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General Introduction

Recently, Anatomic Pathology (AP) has seen the introduction of several tools such as high-resolution histopathological slide scanners and efficient software viewers [1]–[3] for virtual slide technologies. These initiatives created the conditions for a broader adoption of computer-aided diagnosis based on whole slide images (WSI) with the hope of a possible contribution to decreasing inter-observer variability and to more personalized diagnostic and prognostic evaluation.
In particular, there have been decisive advances in terms of recognition rate and accuracy by recent developments [1], [4], [5].

Similarly, in order to reduce inter-observer variability between AP reports of malignant tumors[6], [7], the College of American Pathologists edited 67 organ-specific Cancer Checklists and associated Protocols (CAP-CC&P) [6]. Each checklist includes a set of AP observations that are relevant in the context of a given organ-specific cancer and have to be reported by the pathologist. The associated protocol includes interpretation guidelines for most of the required observations.
All these changes and initiatives bring up a number of scientific challenges, among which the sustainable management of the available semantic resources associated to the diagnostic interpretation of AP images by humans (pathologists) prior to their use by computers (image analysis algorithms). In this context, reference vocabularies and formalization of the associated knowledge are especially needed to annotate histopathology images with labels complying with semantic standards.

Current terminology systems for AP structured reporting (APSR) gather terms of very different granularity [8], [9] and have not yet been compiled in a systematic approach. Moreover, the APSR template designed by the "Integrating the Healthcare Enterprise" initiative (IHE) provides a formal representation of only high-level AP observations resulting from human interpretation of low-level morphological abnormalities. There is still a need to extend the scope of IHE APSR and to integrate in a unique formal representation both high-level AP entities observable by humans and the corresponding low-level morphological abnormalities, especially those that can be quantified using image analysis tools.
In this research work, we present our contribution in this direction. We propose a sustainable way to bridge the content, features, performance and usability gaps [10][11] between histopathology and WSI analysis. Our multi-disciplinary approach covers the histopathology and imaging domains. It is structured as follow:

Histopathology domain:

i. Identify and extract relevant quantifiable observations from the College of American Pathologists (CAP) organ-specific Cancer Checklists and associated Protocols (CC&P)
ii. Identify within the reference biomedical ontologies made accessible by the NCBO Bioportal [12], [13] and within the UMLS metathesaurus [14] the available histopathological formalized knowledge covering the scope of CAP-CC&Ps
iii. Build a sustainable visual representation of this knowledge using the semantic types of the UMLS metathesaurus [15], [16].
iv. Initiate a formal representation of this knowledge under the AP Quantifiable Observation termino-ontology

Imaging domain:

i. Identify effective histopathology imaging methods highlighted by recent Digital Pathology (DP) contests
ii. Identify relevant imaging formalized knowledge within the reference biomedical ontologies in NCBO Bioportal [12], [13] and within the UMLS metathesaurus [14]
iii. Extract the imaging terms and functionalities issued from major biomedical-imaging software (MATLAB, ITK, ImageJ)
iv. Identify the conventional/ common imaging tasks and features in the histopathology imagery surveys
v. Initiate a formal representation by integrating this imaging knowledge (issued from contests biomedical-imaging software's and literature) under the Practical Image Processing Tasks termino-ontology

In both histopathology and imaging approaches, a semi-automatic annotation process was used to label the quantitative parameters and relevant terms with codes from predefined reference semantic resources.
In the histopathology domain, in order to build a terminologic "gold standard", two medical experts independently identified relevant terms corresponding to quantitative parameters observed by pathologists to score or grade malignant tumors. F-measure score were calculated to evaluate concordance between experts.
In the imaging domain, relevant terms and functionalities issued from major biomedical-imaging software were extracted manually. Their hierarchization and integration were then performed with Protégé®.
Based on NCBO Bioportal and UMLS semantic types, the concepts and metadata generated constitute a sustainable vocabulary, dedicated to histopathology, being able to effectively support daily work on WSI. Semantic models and reference terminologies are essential in DP, being generally viewed as able to support the reproducibility and quality of the diagnostic, to assist and standardize anatomopathological reporting, and to enable multi-center clinical collaboration or research, especially in the context of cancer grading[8].

This manuscript contains four main parts organised in six chapters:

1. PART 1 restitutes an "Overview of existing Cancer Grading processes, related standards and recent initiatives"
2. PART 2 covers Histopathology domain knowledge formal representation
3. PART 3 is about Formal representation of the image analysis knowledge issued from different communities.
4. PART 4 gives Concluding remarks and perspectives with Smart'GRADE,[1] an Integration Platform Valorization Prospect.

This research work is a step forward to organized, cross-disciplinary, information-driven collaborations in the histopathological imaging field. Future work should focus on further development toward realizing our longer term goals of advancing interoperability of histopathological imaging systems and performance of computer-assisted diagnosis and prognostic evaluation in histopathology[17], [18].

[1] Smart'GRADE is a valorization project initiated during this doctoral project. It proposes a

List of Publications

1. **Traoré L**, Kergosien Y, Racoceanu D, “Bridging the Semantic Gap Between Diagnostic Histopathology and Image Analysis,” Stud. Health Technol. Inform., vol. 235, pp. 436–440, 2017.

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Acronyms

- Active contour model (ACM)
- Anatomic Pathology Structured Reports (APSRs)
- Anatomic Pathology Workflow (APW)
- AP Information System (APIS)
- Clinical Document Architecture (CDA)
- College of American Pathologists (CAP)
- Computer assisted diagnosis (CAD)
- Convolutional Neural Networks (CNN)
- Digital imaging and communications in medicine (DICOM)
- eCC (electronic Cancer Checklists)
- Fast Healthcare Interoperability Resources (FHIR)
- Health Level Seven (HL7)
- Hematoxylin & Eosin (H&E)
- High power field (HPF)
- Immunohistochemical (IHC)
- Information Object Definition (IOD)
- Integrating the Healthcare Enterprise (IHE)
- International Health Terminology Standards Development Organisation (IHTSDO®)
- International Organization for Standardization (ISO)
- Laboratory Information System (LIS)
- Logical Observation Identifiers Names & Codes (LOINC)
- Nottingham Grading System (NGS)
- Picture Archiving and Communication System (PACS)
- Region of interest (ROI)
- Standard Development Organizations (SDO)
- Support Vector Machine (SVM)
- Systematized Nomenclature of Medicine - Clinical Terms (SNOMED-CT)
- Tissue Micro Array (TMA)
- Unified Code for Units of Measure (UCUM)
- Whole slide image (WSI)

PART 1 Overview of existing Cancer Grading processes, related standards and recent initiatives

1 Overview of existing Cancer Grading processes, related standards and recent initiatives

Main questions

- What is the existing process for cancer grading?
- How could it be improved in the context of digital pathology?

1.1 Predicting cancer prognostic in Anatomy Pathology (AP)

The anatomic pathology (AP) examination enables to establish a diagnosis and to give prognostic indications based on manual evaluation by light microscopy of histological features concerning lesions of tissues or cells derived from specimens. Rules that allow establishing a diagnostic and / or prognostic conclusion from morphological characteristics observed in images are published in the scope of classification systems which, in the context of cancer, address either diagnostic or staging or grading.
Cancer is a disorder of cell life cycle that leads to excessive cell proliferation rates, typically longer cell lifespans and poor differentiation. The histologic tumor *grading*/scoring along with the spread oriented (tumor, nodes, metastases) *staging* are used to evaluate each specific cancer patient, develop their individual treatment strategy, estimate how the cancer might respond to treatment and give a prognosis, which is the expected outcome or course of a disease [19], [20].

1.1.1 Staging and Grading of Cancer

1.1.1.1 What is cancer grading or scoring?

Histologic cancer "grade" or “score” is a way of classifying a tumor based on how different the cancer looks from normal cells and tissue, how quickly and abnormally it is growing and dividing, and how likely it is to spread. As opposed to grading, *staging* relates to the actual extension of the tumor to precise anatomic structures, be it locally or regionally or globally. Different grading systems are used for different types of cancer, e.g., Nottingham for breast cancer and Gleason for prostate cancer. Together with staging systems, they are fundamental to clinical trials (especially for multi-center data collection), prognostic studies, and medical decision-making.

1.1.1.1.1 Generic cancer grading systems

If no specific system is used, the following general grades are most commonly used, and recommended by the American Joint Commission on Cancer (AJCC) and other bodies, following a similar pattern with grades being increasingly malignant over a range of 1 to 2, 3 or 4.
The grade score (numerical: G1 up to G4) increases with the lack of cellular differentiation - it reflects how much the tumor cells differ from the cells of the normal tissue they have originated from. As shown respectively in Table 1-1, Table 1-2 and **Table 1-3**, tumors may be graded on **four-tier**, **three-tier**, or **two-tier** scales, depending on the institution and the tumor type.

Four-tier grading scheme		
Grade 1	Low grade	Well-differentiated

Grade 2	Intermediate grade	Moderately differentiated
Grade 3	High grade	Poorly differentiated
Grade 4	Anaplastic	Anaplastic

Table 1-1 Four-tier scale grading

Three-tier grading scheme		
Grade 1	Low grade	Well-differentiated
Grade 2	Intermediate grade	Moderately differentiated
Grade 3	High grade	Poorly differentiated

Table 1-2: Three-tier scale grading

Two-tier grading scheme		
Grade 1	Low grade	Well-differentiated
Grade 2	High grade	Poorly differentiated

Table 1-3: Two-tier scale grading

The histologic grade can suggest how slow growing (grade I) or aggressive (grade III or IV a tumor is.

- Well-differentiated (low grade or grade I) tumors look more like normal tissue.
- Poorly differentiated (high grade or grade III) tumors look disorganized under the microscope and may behave more aggressively than grade I tumors.
- Those tumors that look neither well differentiated nor poorly differentiated are designated moderately differentiated, or grade II.

1.1.1.1.2 Cancer type-specific grading systems

Breast and prostate cancers are the most common types of cancer that have their own grading systems.

Breast cancer: the Nottingham grading system also called the Elston-Ellis modification of the Scarff-Bloom-Richardson grading system is used for breast cancer grading [21]. This system grades breast tumors based on the following features:

- Tubule formation: how much of the tumor tissue has normal breast duct structures
- Nuclear grade: an evaluation of the size and shape of the nucleus in the tumor cells
- Mitotic rate: how many dividing cells are present, which is a measure of how fast the tumor cells are growing and dividing

Prostate cancer. The Gleason scoring system is used to grade prostate cancer. The Gleason score is based on biopsy samples taken from the prostate [22]. The pathologist checks the samples to see how similar the tumor tissue looks to normal prostate tissue. Both a primary and a secondary pattern of tissue organization are identified. The primary pattern represents the most common tissue pattern seen in the tumor, and the secondary pattern represents the next most common pattern. Each pattern is given a grade from 1 to 5, with 1 looking the most like normal prostate tissue and 5 looking the most abnormal. The two grades are then added to give a Gleason score. The American Joint Committee on Cancer recommends grouping Gleason scores into the following categories [23]:

- Gleason X: Gleason score cannot be determined
- Gleason 2–6: The tumor tissue is well differentiated
- Gleason 7: The tumor tissue is moderately differentiated
- Gleason 8–10: The tumor tissue is poorly differentiated or undifferentiated

Grading in cancer is distinguished from staging, which is a measure of the extent to which the cancer has spread.

1.1.1.2 What is cancer staging?

Grading in cancer is distinguished from staging, which is a measure of the extent to which the cancer has spread

Staging is a way of classifying a cancer based on the extent of tumour in the body[19]. In other words, the stage of a cancer describes its size and if it has spread from where it started to other parts of the body. Stages are based on specific factors for each type of cancer. "*There are different types of staging systems, but the most common and useful staging system is the TNM system."[24].* The TNM system was developed in the 1940's by Pierre Denoix.

- T = Tumor, describes the size of the tumor.
- N = Node involvement, describes whether the cancer has spread to the lymph nodes and which nodes are involved. For example, N0 is no lymph nodes affected. N1 means there are cancer cells in 1–3 of the lymph nodes.
- M = Metastatic spread, describes if the cancer has spread to another part of the body. For example, M0 means the cancer has not metastasized to other parts of the body.

Higher numbers usually mean more extensive disease, larger tumor size, and/or spread of the cancer beyond the organ in which it first developed. It is important to note that once a stage is assigned and treatment given, the stage is never changed. For example, "*if a stage I cancer of the cervix is treated, and 2 years later a metastasis (spread of the same cancer) is found in the lung, it remains a stage I, with recurrence to the lung.*"

1.1.2 Classification systems for AP evaluation of cancer prognosis

1.1.2.1 The College of American Pathologists (CAP)

To reduce inter-observer variability between AP reports of malignant tumors, the College of American Pathologists (CAP) edited organ-specific Cancer Checklists and associated Protocols (CC&P). These guidelines aid the pathologist in collecting the essential data elements - including description of scoring/grading/staging systems - needed in the pathology report for each tissue type. There are currently 67 protocols available covering 20 major organ systems. Since 1986, the CAP Cancer Protocols have served as a resource and reference for complete reporting of malignant tumors, including American Joint Committee on Cancer (AJCC) staging and the World Health Organization (WHO) histologic type standard elements [6]. They are created in printable format (PDF, DOC) by CAP Cancer Committee and transformed to electronic format (XML) with the electronic Cancer Checklist (eCC) [6].

1.1.2.1.1 CAP Cancer Checklists and associated Protocols (CC&P)

Each checklist includes a set of AP observations that are expected to be reported by pathologists in organ-specific AP cancer reports, i.e. Context, Quantifiable Observation, Corresponding value set, Procedures, observable entities, Explanatory notes that unambiguously describe how pathologists should derive a high-level observation from low-level morphological characteristics observed in images [25] .

What is a cancer protocol?

The Cancer Protocols are created by a multidisciplinary team of expert medical professionals, led by the members of the CAP Cancer Committee. A cancer protocol is composed of two parts:

- A case summary (i.e. the 'synoptic report' format)
- Accompanying explanatory notes that provide brief educational material to facilitate accurate completion of the case summary

Included in each case summary are "required data elements" for optimal patient care as well as "optional" elements that may be clinically important but are not yet evidence-based or regularly used in patient management. In some instances, a data element will be noted as conditionally required if present in the specimen (i.e., lymph nodes in invasive carcinoma of the breast) [26]. Figure 1-1 shows an extract of the *"Title page" (Green),* the subsections of the "Case Summary" (Yellow) and accompanying "Explanatory Notes" (Red) of the *"Protocol for the Examination of Specimens From Patients with Invasive Carcinoma of the Breast."* A complete version (33 pages) of the overall protocol can be found in

Appendix 1: ***Overall Protocol for the Examination of Specimens From Patients with Invasive Carcinoma of the Breast***

COLLEGE of AMERICAN PATHOLOGISTS

Protocol for the Examination of Specimens From Patients With Invasive Carcinoma of the Breast

Version: InvasiveBreast 4.0.0.0 **Protocol Posting Date:** June 2017

Includes pTNM requirements from the 8th Edition, AJCC Staging Manual

For accreditation purposes, this protocol should be used for the following procedures AND tumor types:

Procedure	Description
Excision less than total mastectomy	Includes specimens designated excision, segmental resection, lumpectomy, quadrantectomy, and segmental or partial mastectomy, with or without axillary contents
Total Mastectomy	Includes skin-sparing and nipple–sparing mastectomy, with or without axillary contents
Tumor Type	**Description**
Invasive breast carcinoma of any type, with or without ductal carcinoma in situ (DCIS)	Includes microinvasive carcinoma and carcinoma with neuroendocrine features

Extract of a CAP Cancer Checklists and Protocols (CC&P) "Title page" (Green)

Surgical Pathology Cancer Case Summary

Protocol posting date: June 2017

INVASIVE CARCINOMA OF THE BREAST:

Select a single response unless otherwise indicated.

Specimen Identification
The following 3 elements identifying the specimen may be listed separately or on 1 line:

Procedure (Note A)
___ Excision (less than total mastectomy)
___ Total mastectomy (including nipple-sparing and skin-sparing mastectomy)
___ Other (specify): _____
___ Not specified

Histologic Grade (Nottingham Histologic Score) (Note F)

Glandular (Acinar)/Tubular Differentiation
___ Score 1 (>75% of tumor area forming glandular/tubular structures)
___ Score 2 (10% to 75% of tumor area forming glandular/tubular structures)
___ Score 3 (<10% of tumor area forming glandular/tubular structures)
___ Only microinvasion present (not graded)
___ No residual invasive carcinoma
___ Score cannot be determined

Nuclear Pleomorphism
___ Score 1 (nuclei small with little increase in size in comparison with normal breast epithelial cells, regular outlines, uniform nuclear chromatin, little variation in size)
___ Score 2 (cells larger than normal with open vesicular nuclei, visible nucleoli, and moderate variability in both size and shape)
___ Score 3 (vesicular nuclei, often with prominent nucleoli, exhibiting marked variation in size and shape, occasionally with very large and bizarre forms)
___ Only microinvasion present (not graded)
___ No residual invasive carcinoma
___ Score cannot be determined

Mitotic Rate
___ Score 1 (≤3 mitoses per mm^2) (see Table 1)
___ Score 2 (4-7 mitoses per mm^2) (see Table 1)

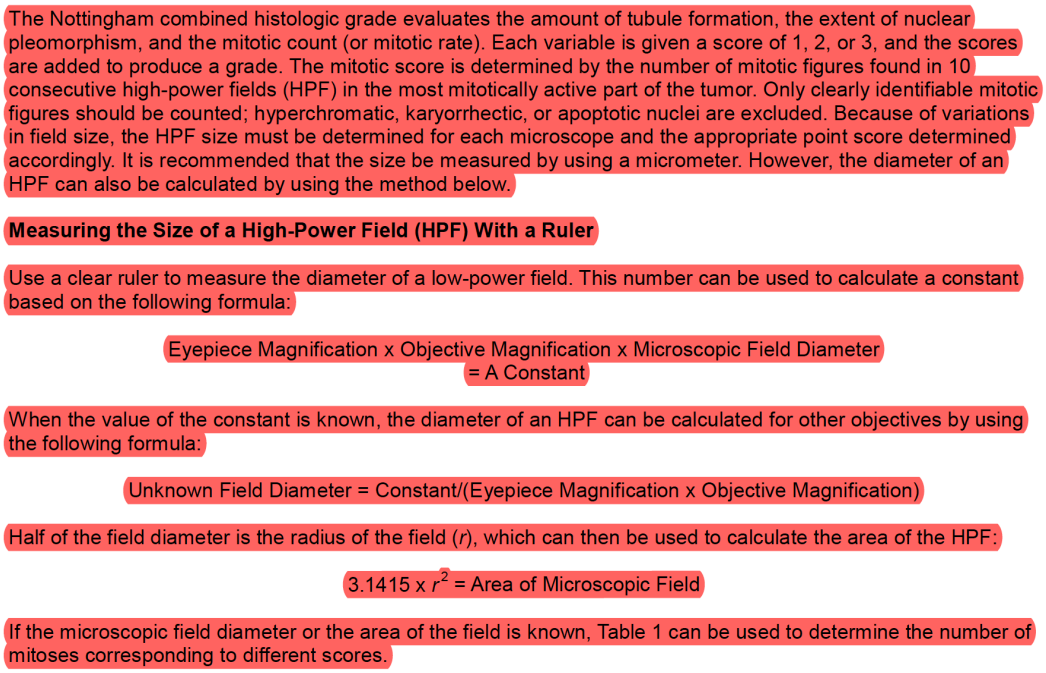
Extract of a CAP Cancer Checklists and Protocols (CC&P) Data elements (i.e. observable entities with corresponding value sets) in "Case Summary" (Yellow)

Explanatory Notes

A. Procedures
The following types of breast specimens and procedures may be reported with the case summary:

Excisions: These procedures resect breast tissue without the intent of removing the entire breast. The nipple is usually not included with excisions. Excisions include specimens designated "partial mastectomies," "lumpectomies," and "quadrantectomies."

F. Histologic Grade
All invasive breast carcinomas should be graded.[12] The Nottingham combined histologic grade (Elston-Ellis modification of Scarff-Bloom-Richardson grading system) should be used for reporting. Within each stage grouping there is a relation between histologic grade and outcome.



The Nottingham combined histologic grade evaluates the amount of tubule formation, the extent of nuclear pleomorphism, and the mitotic count (or mitotic rate). Each variable is given a score of 1, 2, or 3, and the scores are added to produce a grade. The mitotic score is determined by the number of mitotic figures found in 10 consecutive high-power fields (HPF) in the most mitotically active part of the tumor. Only clearly identifiable mitotic figures should be counted; hyperchromatic, karyorrhectic, or apoptotic nuclei are excluded. Because of variations in field size, the HPF size must be determined for each microscope and the appropriate point score determined accordingly. It is recommended that the size be measured by using a micrometer. However, the diameter of an HPF can also be calculated by using the method below.

Measuring the Size of a High-Power Field (HPF) With a Ruler

Use a clear ruler to measure the diameter of a low-power field. This number can be used to calculate a constant based on the following formula:

Eyepiece Magnification x Objective Magnification x Microscopic Field Diameter = A Constant

When the value of the constant is known, the diameter of an HPF can be calculated for other objectives by using the following formula:

Unknown Field Diameter = Constant/(Eyepiece Magnification x Objective Magnification)

Half of the field diameter is the radius of the field (*r*), which can then be used to calculate the area of the HPF:

$3.1415 \times r^2$ = Area of Microscopic Field

If the microscopic field diameter or the area of the field is known, Table 1 can be used to determine the number of mitoses corresponding to different scores.

Extract of a CAP Cancer Checklists and Protocols (CC&P) "Explanatory Notes" (Red)

Figure 1-1: Extract of CAP Cancer Checklists and Protocols (CC&P with the *"Title page"(Green),* the subsections of the "Case Summary" (Yellow) and accompanying "Explanatory Notes" (Red)

1.1.2.1.2 Synoptic reporting for Cancer Cases

In June 2017, the College of American Pathologists (CAP) updated a total of 52 cancer protocols to reflect the AJCC 8th Edition Cancer Staging Manual. With more than 7,000 changes, the summary of the required and conditional elements for each protocol will help pathologists to audit the required elements within his cancer reporting system and reports to ensure that complete reports are provided within clinical colleagues and for patients.

The synoptic report format has two main requirements:

All required cancer data elements from a cancer protocol must be included in the report whether they are applicable or not and

All required data elements must be displayed using the following format:

Required data element followed by response, (e. g. "Tumor size: 5.5 cm)

Data elements (i.e. observable entities with corresponding value sets) are of different types: related to the clinical context, the specimen collection procedure, the AP diagnosis (tumor location and type), the evaluation of the tumor prognosis (grade/score, stage, treatment response, etc.). For most data elements, explanatory notes unambiguously describe how pathologists should derive a high-level observation from low-level morphological characteristics observed in images [6], [25]. Most of the data elements allowing prognostic evaluation are quantifiable observable entities and the corresponding explanatory notes include the description of the quantitative features observed by the pathologist during the interpretation.

Figure 1-2 shows Summary of required elements of the "Protocols for the examination of Specimens from Patients with Invasive Carcinoma of the Breast"

Protocol for the Examination of Specimens from Patients with Invasive Carcinoma of the Breast
2017 Breast v4.0.0.0
Protocolposting date: June 2017
Surgical Pathology Cancer Case Summary
INVASIVE CARCINOMA OF THE BREAST:
Procedure
Specimen Laterality
Tumor Size
Histologic Type
Histologic Grade: Nottingham Histologic Score Glandular (Acinar)/Tubular Differentiation Nuclear Pleomorphism Mitotic Count Overall Grade
Ductal Carcinoma In Situ (DCIS)
Macroscopic and Microscopic Extent of Tumor* Skin Nipple Skeletal Muscle
Margins Invasive Carcinoma* DCIS*
Regional Lymph Nodes Number of Lymph Nodes with Macrometastases (>2 mm): ____* Number of Lymph Nodes with Micrometastases (>0.2 mm to 2 mm and/or >200 cells): ____* Number of Lymph Nodes with Isolated Tumor Cells (≤0.2 mm and ≤200 cells)#: ____* Number of Lymph Nodes Examined* Number of Sentinel Nodes Examined*
Pathologic Stage Classification (pTNM, AJCC 8th Edition) TNM Descriptors* Primary Tumor (pT) Regional Lymph Nodes (pN) Distant Metastasis (pM)*

* If Applicable

Figure 1-2: Required data elements of the "Protocols for the examination of Specimens from Patients with Invasive Carcinoma of the Breast"

The summary of required elements for the 67 protocols can be found in

Appendix 3. This document can also be downloaded from the CAP website by completing the form (name, institution, email, etc.) and answering questions about the method the applicant uses to create a synoptic report. "Insight into how pathologists use the protocols will help the CAP improveA the format and design of the cancer protocols."
Recognizing that there is significant variability in format from institution to institution, the CAP has established a specific format for 'synoptic reporting' within a surgical pathology report on cancer specimens.

Appendix 5 gathers a copy of Invasive breast carcinoma "work aid document" with related filled examples.

1.1.2.1.3 Cancer Biomarker Reporting Template

The Cancer Biomarker Reporting Templates are produced to establish reporting guidance for commonly ordered biomarkers, create stand-alone reporting templates, and improve

consistency and completeness of results reporting to assist tumor registrars and others involved in data collection, exchange, and surveillance. These reporting templates are intended to cover all important data elements for routinely assessed tumor markers and are designed to be incorporated into electronic reporting systems. Completion of the template is the responsibility of the laboratory performing the biomarker testing and/or providing the interpretation [26]. Below is an example of a CAP Biomarker protocol extract of the "*Template for Reporting Results of Biomarker Testing of Specimens from Patients with Carcinoma of the Breast*." It is mainly composed of the CAP Approved Breast Biomarker Reporting Template and the "*Background Documentation"* with "*Explanatory notes*." Subsections of the Breast Cancer Biomarker Reporting Template are summarized in Table 1-4.

CAP Approved Breast Biomarker Reporting Template - RESULTS and METHODS
Results Estrogen Receptor (ER) Status (Note A) Progesterone Receptor (PgR) Status (Note A) HER2 (by immunohistochemistry) (Note B) HER2 (ERBB2) (by in situ hybridization) (Note B) + Ki-67 (Note C) + Multiparameter Gene Expression/Protein Expression Assay (Note D) Cold Ischemia and Fixation Times Methods + Testing Performed on Block Number(s): Fixative Estrogen Receptor (required for US-based laboratories) Progesterone Receptor (required for US-based laboratories) + ER and PgR Scoring System HER2 (by immunohistochemistry) (required for US-based laboratories) HER2 (ERBB2) (by in situ hybridization) (required for US-based laboratories) + Ki-67 + Image Analysis
Background Documentation with Explanatory notes – Explanatory Notes
A. Estrogen Receptor and Progesterone Receptor Testing B. HER2 (ERBB2) Testing HER2 Testing by Immunohistochemistry HER2 Testing by In Situ Hybridization C. Ki-67 Testing D. Multigene Expression Assays

Table 1-4: Subsections of Template for reporting results of Biomarker Testing Specimen from Patients with Carcinoma of the Breast

1.1.3 Inter-expert variability in cancer grading/scoring

Inter-experts variability in AP is widely reported in the literature [27].

"*The medical literature occasionally discusses aspects of the pathologic diagnosis processes, generally departing from the pathologic practice. The lack of a model makes discussions about the subject a matter of preference or personal style. Educational programs are largely based on the apprenticeship model, and the development of specific abilities rests on the personal aspects of both apprentice and mentor."[28]*
For many cancers, qualitative evaluation of well-established histopathology patterns based on existing grading/scoring/staging systems is insufficient for predicting the survival outcomes of patients. This is partly due to the fact that classification systems are numerous and evolving. Thus, identical cases may lead to different prognostic conclusions depending on the classification system on which the pathologist's refers to [29].

1.2 Digital Pathology

1.2.1 Workflow of the AP diagnosis and prognostic process

An AP diagnosis is the result of a complex series of activities, mastered by the pathologist. For any AP exam, the following three (3) main steps can be considered.

- **Input (Order)**: an interrogation or request to which the pathologist tries to find an answer (what is the histologic type of the lesions of tissues or cells observed in the specimens (AP diagnostic)? What is their score/grade ? What is their stage?)
- **Process (Procedure):** the AP expert observes, describes and judges what he sees by using parameters at his disposal (medical context, question, image,) and his expertise (knowledge, reasoning, experience)
- **Output (Report):** AP diagnostic report, which summarize required and significant elements of his judgement.

An important part of this process is based on the analysis of histopathological slides. This is performed by the expert with the identification and description of visual characteristics of form, texture structure, location, etc. The visual analysis of AP images consists of interpreting the features that can be identified and to consider those that are relevant to answer the initial questions.
In the cancer domain, the initial question is twofold: i) establishing the topographic and morphologic diagnostic of the cancer and ii) predicting the survival outcomes of patients, or at least providing relevant elements for decision making by a multidisciplinary meeting. Cancer diagnostic and prognostic evaluation remains largely a tedious human activity with disruptive elements such as fatigue, mood, etc. As shown in Figure 1-3, today, an important part of the histopathological image analysis routine remains subjective.

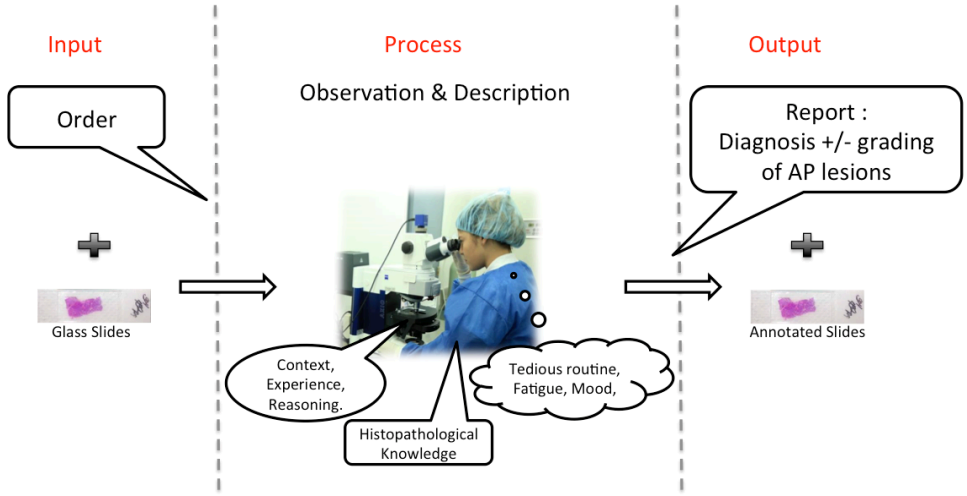


Figure 1-3: Current scenario for the visual analysis by light microscopy of an histopathology slides in AP laboratory

We should note that the knowledge of an expert is partly included in the sum of the images associated with the case reports encountered during his medical practice. This set of cases and associated images constitutes a non-negligible part of the empirical knowledge of the domain [30].

From the literature different approaches [31], [32] attempt to explain the diagnostic process. In this work we decided to consider the diagnostic process as a problem solving strategy. In resolving the problems presented by the case, the pathologist must elaborate an action plan, contemplating four (4) different domains: cognitive, communicative, normative, and medical conduct [28].

At first, the pathologist observes and classifies what he sees, but he « makes no deduction of inferences » in this process. In all « diagnostic » decisions, the pathologist is alleged to classify what he observes [33].
The purpose of this model is to reconstruct the practice of pathology within a structured framework. We categorized actions and operations involved in the diagnostic task within appropriate domains and provided some brief theoretic insight in support of them.
As shown in Figure 1-4, the authors discuss cognitive aspects of diagnosis (referring to pattern recognition, histologic findings, and immunohistochemical markers); communicative aspects (clinical information needed to diagnose soft tissue tumors, the content of the surgical pathology report); normative aspects (rules to classification and grade, rules to handle the resection specimen) and medical conduct aspects (consequences of diagnosis to the management of the case, second opinion request) [28].

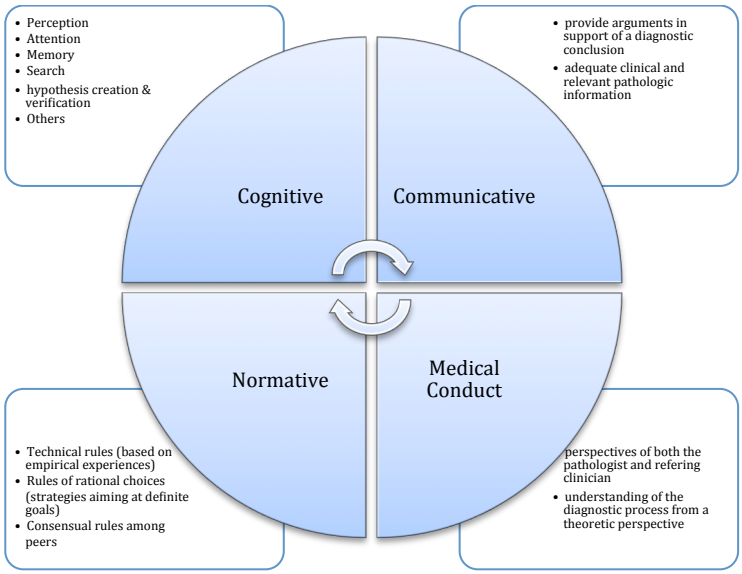


Figure 1-4: Actions and operation involved in the AP diagnostic task as a problem solving strategy

1.2.2 Slide scanners and Whole slide Imaging (WSI)

Significant technologic gains have led to the adoption of innovative digital imaging solutions in pathology. Whole slide imaging (WSI), which refers to scanning of conventional glass slides in order to produce digital slides, is the most recent imaging modality being employed by pathology departments worldwide [34], [35]. WSI are obtained by histopathological slide digitization process as shown in Figure 1-5 by using a high-resolution slide scanner. A single image is generally about 140 000 by 60 000 pixels RGB (3 bytes per pixel), that is about 8 gigapixel. Each image is usually about 1 to 2GB when compressed, and 15 to 25GB when uncompressed. WSI represents an efficient way to store the slides, protecting their critical information from degradation [36]. AP images are key information objects of the collaborative workflow of digital AP and will become an integral component of Electronic Health Records (EHR) as part of AP reports [37], [38].

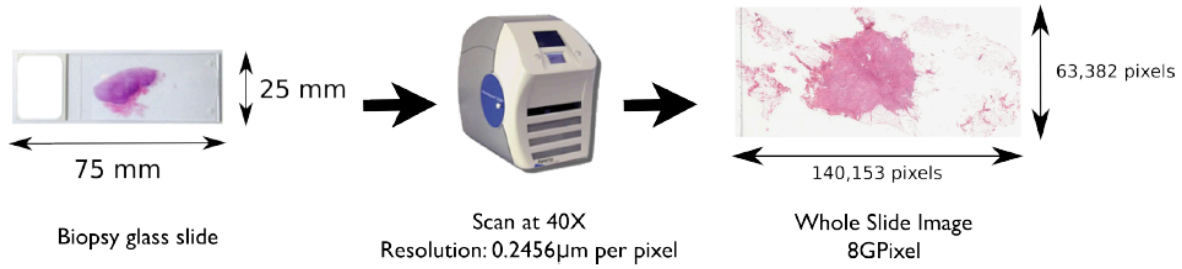


Figure 1-5 Whole slide imaging digitisation process (source [36])

The typical organization of a WSI may be thought of as a "pyramid" of image data. As shown in Figure 1-6, the WSI consists of multiple images at different resolutions. The "altitude" of the pyramid corresponds to the "zoom levels." The base of the pyramid is the highest resolution image data as captured by the instrument. A thumbnail image may be

created which is a low-resolution version of the image to facilitate viewing the entire image at once. One or more intermediate levels of the pyramid may be created, at intermediate resolutions, to facilitate retrieval of image data at arbitrary resolution. Each image in the pyramid may be stored as a series of tiles, to facilitate rapid retrieval or arbitrary sub regions of the image. Figure 4 shows a retrieved image region at an arbitrary resolution level, between the base level and the first intermediate level. The base image and the intermediate level image are "tiled." The shaded areas indicate the image data which must be retrieved from the images to synthesize the desired sub region at the desired resolution [39].

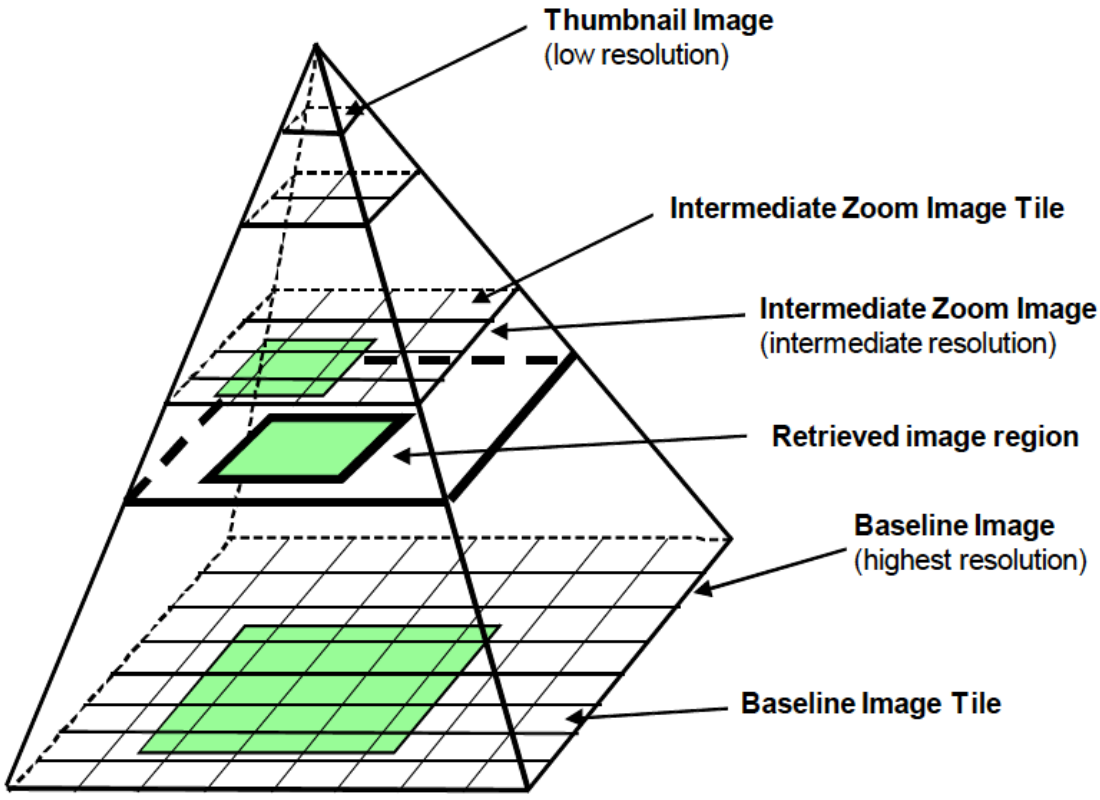


Figure 1-6 Typical "pyramid" organisation of Whole slide imaging (source [39, p. 145])

WSI is challenging the AP domain since it offers new promising perspectives for more efficient collaborative practices and also brings some barriers to overcome. WSI is already being widely used in AP undergraduate teaching, distance learning and continuing medical education, proficiency testing, quality assurance programs, research (tumor banking) and teleconsultation (for second opinion)[40]. Regarding the latter, the use of WSI has been validated for diagnostic applications in surgical pathology, cytopathology, and immunohistochemistry[41], [42]. Since about 5% of surgical procedures require rapid diagnosis [43], the use of WSI in the context of daily practice is expected to increase significantly especially in rapid frozen intraoperative studies, not only in hospitals lacking an AP laboratory but also in small and medium sized AP laboratories which often lack the required subspecialty expertise [8].

It is difficult to foresee the consequences of the full digitalization of AP laboratories in part because short-term issues have arisen. These are challenging standards organizations with regards to the integration of WSI in the collaborative digital AP.

Beyond the technical challenges (scanning speed, quality of image during capture, storage dimension, and viewing processes) the greatest changes involve the workflow organization

and require solutions for automation, integration, and simplification [8]. Simplification allows converting complex, time-consuming tasks into more straightforward ones. For example, integration between **Anatomic Pathology** Information System (APIS) and a Picture Archiving and Communication System (PACS) eliminates the time-consuming steps of manual double data entry of data or manually transferring digital images from one computer to another [44].

The IHE integration profile **Anatomic Pathology Workflow (APW)** describes how WSI management can be closely integrated to the information flow of collaborative digital AP using existing and emerging medical informatics standards like the Digital Imaging and Communications in Medicine (DICOM) upgraded by DICOM supplements 122 and 145 and HL7 [8]. Albeit slow, DICOM are underway to help standardize the use of WSI in pathology [35]. Figure 1-7 shows the APW IHE integration profile, which is part of the Pathology and Laboratory Medicine (PaLM) domain. PaLM merged the former AP and LAB domains since 2016, January 4th.

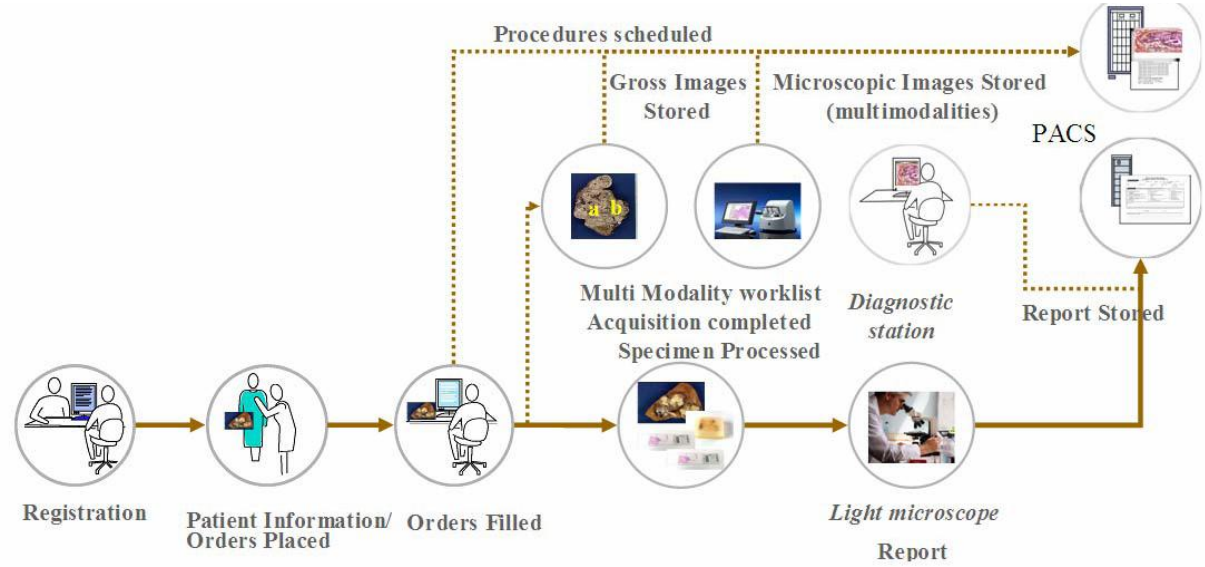


Figure 1-7 IHE AP Workflow (APW) profiles integrating the pathology department in the healthcare institution, and covering these specialties: surgical pathology, clinical autopsy, and cytopathology (source [?, p. ?])

"*APW establishes the continuity and integrity of basic pathology data acquired during examinations ordered for an identified inpatient or outpatient. This profile covers three main aspects of the workflow*" [45]*:*

- **The ordering aspects of the workflow** - APW specifies a number of transactions to maintain the consistency of ordering information and specimen management information.
- **The reporting aspects of the workflow** - APW specifies a number of transactions to create and store observations and reports outside the Pathology department and to maintain the consistency of these results.
- **The imaging aspects of the workflow** - APW specifies a number of transactions to create and store images and to maintain the consistency of these images. Work lists

for image acquisition is generated and can be queried. This Integration Profile also describes evidence creation.

These issues are more detailed in section 1.3.2,.

1.3 Standards for Digital Pathology

According to the International Organization for Standardization (ISO), a standard is « a document, established by consensus and approved by a recognized body, that provides, for a common and repeated use, rules, guidelines or characteristics for activities or their results, aimed at the achievement of the optimum degree of order in a given context » [46], [47]
This is the formal definition of a generic standard, but it applies to informatics standards as well [42].

1.3.1 Standard for WSI - Digital imaging and communications in medicine (DICOM)

Materials in this section mainly refer to recent works of the DICOM Working Group 26 and publications of D. Cluny, B.Smith and B. Gibaud et Al. [17], [48]

In order to be able to make best use of medical images, it is desirable to be able to associate to images the corresponding context including patient demographic information, details on the venue and the AP examination and technical information related to image acquisition. The need to address similar issues with radiology images lead to the creation of the DICOM standard, following the ACR-NEMA standard. DICOM is the most widely used medical imaging standard in the world [49], [50]. It is a high-level communications standard, which facilitates interchange of images and metadata and has been widely adopted in radiology. It allows image acquisition devices from one manufacturer to work smoothly with Picture Archiving and Communication System (PACS) from a different vendor and an image viewer from yet another company [42], [51]. Table 1-5 shows a list of some types of devices or systems which might interact with digital DICOM AP files [42].

Device/System	**Role**
Slide scanner	Creates WSI image data
PACS/Image archive	Stores images and corresponding meta data in a DICOM format
Anatomic Pathology Laboratory Information System (LIS)	Contains workflow and report information, history of specimen and slide preparation and results of pathologic examination
Image viewer for pathologists	Displays AP images/WSI for diagnostic analysis by pathologists. The viewer may be customized for pathologist's needs
Image analysis software	Generates quantitative or qualitative data from images
Image viewer for clinicians	Displays AP images/WSI for consultation by clinicians. current general purpose DICOM viewers will need to be modified to properly display WSI

Table 1-5: Devices or systems which may interact with AP images (source [42])

The DICOM standard is organized in 18 independent parts. Figure 1-8 represents them with Part 2, 3, 4, 5 and 16 being the most important ones [48]:

- "*Part 2 "Conformance" specifies how a manufacturer can claim conformance to the DICOM standard for a particular product or implementation, by writing a document called a "conformance statement." Part 2 explains in detail how this document must be written and the information it must contain.*
- *Part 3 "Information object definitions" provides the specification of the information objects to be exchanged (more than 1000 pages of text), as well as the definition of the semantics of each data element. The main reason why this part is so long and complex is related to the many existing imaging modalities (such as computed tomography, ultrasound, magnetic resonance, positron emission tomography etc.), that require many technical parameters.* Whole slide microscopic image Information Object Definition (IOD) of supplement 145 (page 38) is also specified in this Part.
- *Part 4 "Service Class specifications" defines the services for exchanging information, either images or information that is useful to manage images.*
- *Part 5 "Data structure and encoding" specifies how the information objects specified in Part 3 can be organized into a linear bit stream, in order to be sent over a network connection or stored in a file. All aspects related to image compression are addressed in Part 5.*
- *Part 16 "Content Mapping Resource" addresses the question of terminology, i.e. on the one hand, it defines how existing terminological resources can be used in DICOM (e.g. SNOMED, LOINC, UCUM), and on the other hand, how content items can be grouped together and re-used in DICOM Structured Reporting documents (notion of Templates)."*

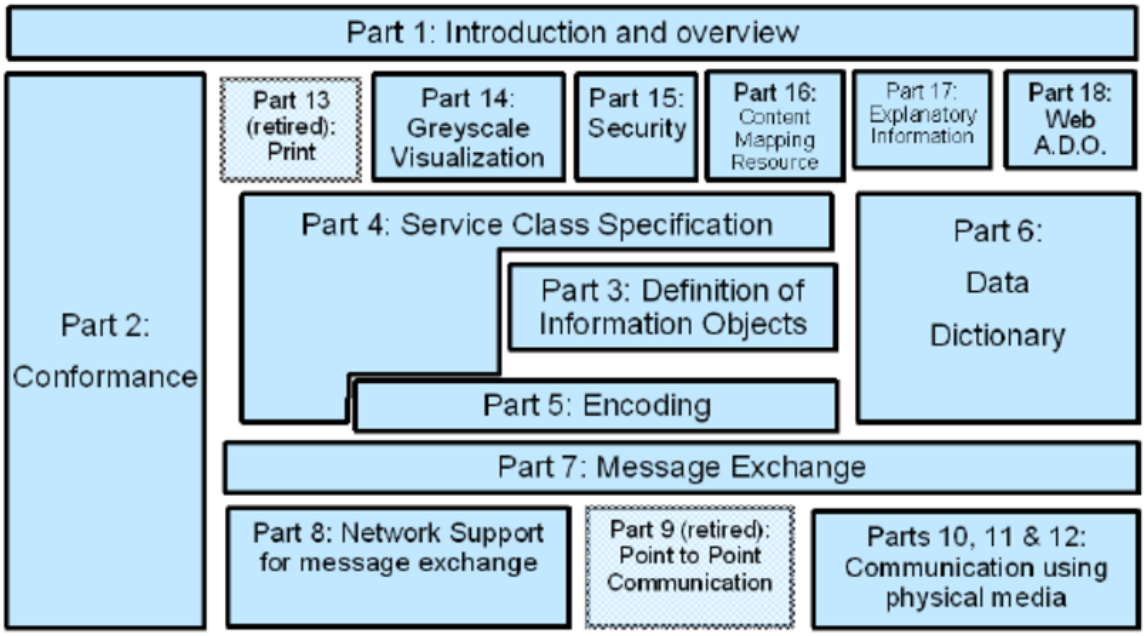


Figure 1-8: The different Numbered Parts of the DICOM standard (source B. Gibaud: The DICOM standard : a brief overview)

« *Despite its widespread adoption, some parts of DICOM still lack consistent semantics, so that different systems can use DICOM to tag similar elements in different ways, which can affect the consistent sharing of data across different applications* » [17] . By refering to the litterature and recent works, one can note an increased interest in developping DICOM standard features for better interoperability, further automate compliance and conformance testing [17]:

- The DICOM standard document, encoded into XML [52] « *represents one step towards transforming the standard into an ontology framework to support development of next-generation image management systems* ».
- The 'DICOM Controlled Terminology' DCM, an OWL resource in Bioportal [53]
- The Semantic DICOM Ontology (SEDI) [54], a more recent effort which aims "*to support the real-time translation of semantic queries into DICOM queries*" while targeting radiotherapy PACS.

1.3.1.1 Standard Committee Working Group 26

The WG-26 was created fall 2005. It gathers pathologists, consultants, researchers and representatives from most major pathology imaging vendors. Some pathology-related image formats do not yet have applicable DICOM Information Object Definitions. Examples include whole slide images, high-order multi-spectral images, flow cytometry, electron microscopy and others. The initial goals of the WG 26 were: i) to extend minimal capabilities to describe specimens in DICOM, ii) to create a mechanism to allow exchange and use of whole slide microscopic images within DICOM. Its longer-term goals are related to other imaging modalities, such as multi-spectral images, electron microscopy, flow cytometry, clinical laboratory images[2]. The following two DICOM supplements were defined by the DICOM WG26 in order to better address the specificity of information object definitions dedicated to whole slide image acquisition, storage and display.

1.3.1.1.1 Supplement 122: Specimen Module and Revised Pathology SOP[3] Classes [8]
The DICOM supplement 122 defines formal DICOM attributes for the identification and description of specimens to support the imaging workflow in the pathology department [55]. Specimen attributes include attributes that (1) identify the specimen (within a given institution and across institutions); (2) identify and describe the container in which the specimen resides; (3) describe specimen collection, sampling, and processing; and (4) describe the specimen or its ancestors when these descriptions help with the interpretation of the image.

[2] « DICOM and the Pathology Community Experience », Bruce Beckwith, MD, Chairman of Pathology at North Shore Medical Center, DICOM Website : www.medical.nema.org
[3] Service Object Pair (SOP)

1.3.1.1.2 Supplement 145 : Whole Slide Microscopic Image IOD[4] and SOP Classes[5] [8]

The DICOM supplement 145 [39] defines DICOM Information Object Definition (IOD) applicable to Whole Slide Images. Whole Slide Images are different from traditional microphotographs in multiple ways. They are considerably larger, and therefore, for performance reasons, Whole Slide Images are usually accessed remotely using an image browser which only loads a small portion of the overall image pixel data. In addition, the need for displaying these images at multiple different "magnifications" is another technical and architectural challenge. The DICOM supplement 145 provides the maximum amount of flexibility to image acquisition, storage and display devices and software. For a variety of reasons, the proposal introduces the concept of tiling as shown in Figure 1-9 (breaking down the full image into multiple smaller images which can be handled separately) for storage of Whole Slide Images. However, images which are smaller than the current image size limits in DICOM can also be stored as JPEG2000[6] images [56] and accessed via the JPIP[7] protocol [57], both of which are supported by DICOM already. In addition, the proposed Information Object Definition has provisions for handling multi-spectral images, multiple focal planes and other necessary features, as well as allowing for detailed descriptions of the optical components used to create the image (Figure 1-9). A system compliant with Supplement 145 will be able to store WSI directly on a PACS, while a compliant viewer will be able to retrieve WSI directly from a PACS.

[4] Information Object Definition (IOD)

[5] A Service-Object Pair (SOP) Class is defined by the union of an Information Object Definition (IOD) and a DICOM Service Elements (DIMSE). The SOP Class definition contains the rules and semantics which may restrict the use of the services in the DIMSE Service Group or the Attributes of the IOD. Examples of Service Elements are Store, Get, Find, Move, etc. Examples of Objects are CT images, MR images, but also include schedule lists, print queues, etc.

[6] JPEG 2000 (JP2) is an image compression standard and coding system. It was created by the Joint Photographic Experts Group committee in 2000 with the intention of superseding their original discrete cosine transform-based JPEG standard (created in 1992) with a newly designed, wavelet-based method.

[7] JPIP (JPEG 2000 Interactive Protocol) is a compression streamlining protocol that works with JPEG 2000 to produce an image using the least bandwidth required.

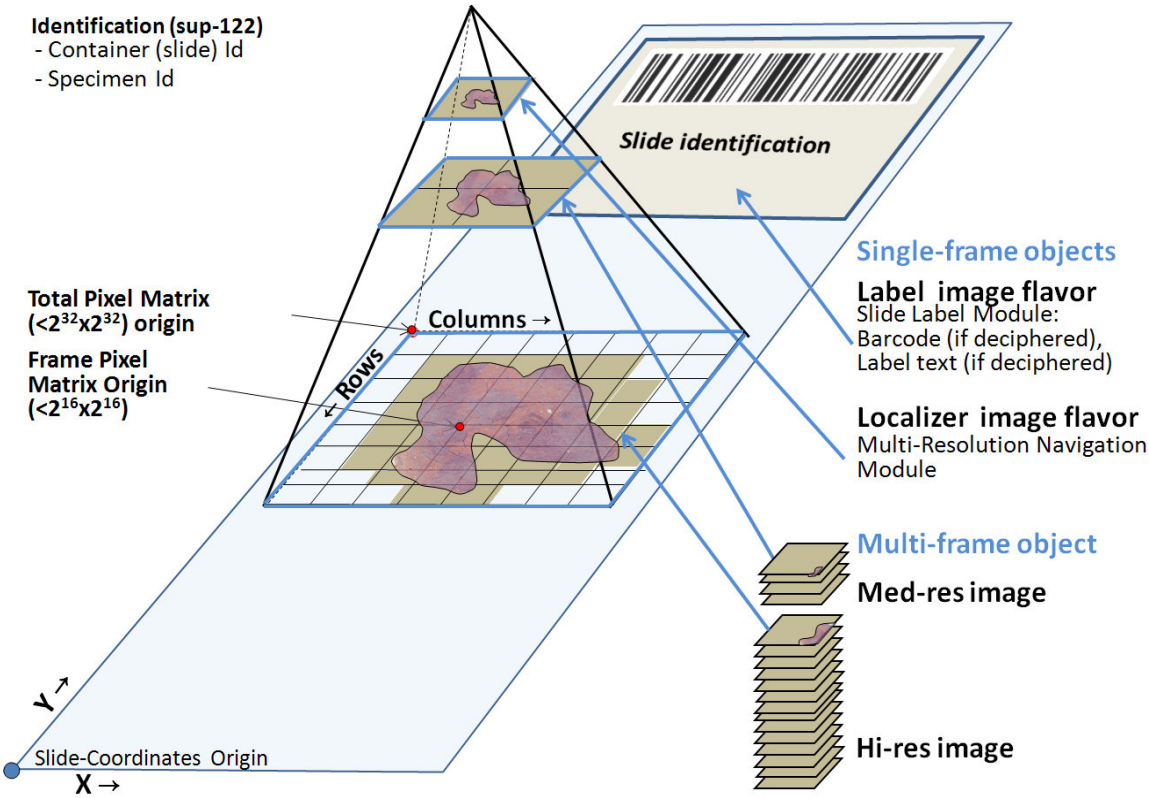


Figure 1-9: Whole Slide Image Information Object Definition (WSI IOD) from DICOM supplement 145 proposes storing tiles from a multi resolution hierarchy in multi-frame object(s). Each tile is stored in a frame and is located within a $2^{32}x2^{32}$ total pixel matrix. Specific Z planes or/and optical paths may be specified at the frame level (Source [39])

1.3.1.2 Standards for AP reports

1.3.1.3 CAP electronic Cancer Checklist (eCC)

The College of American Pathologists (CAP) eCC (electronic Cancer Checklists) enables pathologists to use the CAP Cancer Protocols directly within their laboratory information system (LIS) workflow and to ensure that each report is completed with the necessary required elements. Most anatomic pathology (AP)-LIS vendors offer a CAP eCC synoptic module for reporting on surgical cancer resections and selected biopsies [58]. The CAP eCC are interoperable (platform independent), portable, exchangeable format (XML). It is customizable for individual lab practices and contains structured data elements in a logical workflow. Its XML format is endorsed by main standards organizations (HL7, IHE, IHTSDO, etc.) [59].

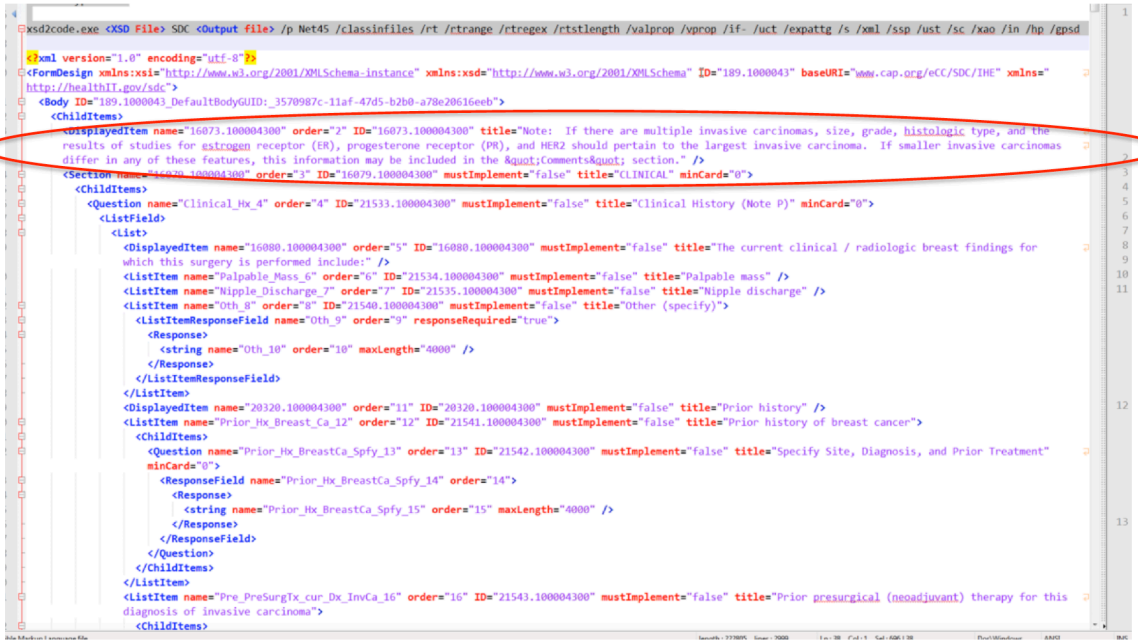


Figure 1-10: XML format CAP electronic Cancer Checklist (eCC)

1.3.2 Health Level Seven (HL7) CDA based Anatomic Pathology Structured Reports (APSRs)

1.3.2.1 Health Level Seven (HL7)

Founded in 1987, Health Level Seven (HL7) International is a not-for-profit, ANSI-accredited standards developing organization dedicated to providing a comprehensive framework and related standards for the exchange, integration, sharing, and retrieval of electronic health information that supports clinical practice and the management, delivery and evaluation of health services. Its mission is to provide standards that empower global health data interoperability. "Level Seven" refers to the seventh level of the International Organization for Standardization (ISO) seven-layer communications model for Open Systems Interconnection (OSI) - the application level.
Meanwhile, the HL7 Anatomic Pathology WG was established to investigate the complex relationships between specimens, observations, images and documents in AP. Joint meetings between the IHE AP and HL7 AP have also been regularly conducted [60], [61]. Generally, such meeting are organised in accordance with the main Digital Pathology community events such as the European Congress on Digital Pathology (ECDP) which was lastly on 2016, may 25th-28thin Berlin[8].

1.3.2.2 Clinical Document Architecture (CDA)

The HL7 Clinical Document Architecture (CDA) is an XML-based mark-up standard intended to specify the encoding, structure and semantics of clinical documents for exchange. CDA is an ANSI-certified standard from Health Level Seven International (HL7.org).

[8] http://www.digitalpathology2016.org

CDA specifies the syntax and supplies a framework for specifying the full semantics of a clinical document. It defines a clinical document as having the following six characteristics in Table 1-6 [62].

Number (#)	Characteristic	Definition
1	Persistence	remaining in use for a long period
2	Stewardship	maintained by a trusted organization, e.g. a hospital using CDA
3	Potential for authentication	legal attestation that the clinical information is accurate
4	Context	a default context to the record, such as the patient identity and who created the document
5	Wholeness	the full document, not just parts of it, can be authenticated
6	Human readability	a person can read the material on a browser or mobile device

Table 1-6: Clinical Document Architecture characteristic definitions

A CDA can contain any type of clinical notes. Typical CDA document types include Discharge Summary, Imaging Report, History & Physical, and Pathology Report [63]. An XML element in a CDA supports unstructured text, as well as links to composite documents encoded in pdf, docx, or rtf, as well as image formats like jpg and png [64].
To represent health concepts, CDA uses HL7's Reference Information Model (RIM), which puts data in a clinical or administrative context and expresses how pieces of data are connected. CDA also takes advantage of coding systems such as SNOMED-CT (Systematized Nomenclature of Medicine-Clinical Terms) and LOINC (Logical Observation Identifiers Names and Codes). With the HL7 format using XML and RIM, Clinical Document Architecture allows EHRs and other health IT systems to process documents while also letting people easily read them on Web browsers and mobile devices.
Standardizing and computerizing AP reports is necessary to improve the quality of reporting and the exchange of AP information [65]. Several studies provide recommendations that delineate the required, preferred, and optional elements, which should be included in any AP report, regardless of report types (e.g. reporting guidelines in [66], [67]. Several national initiatives intend to define standard clinical models for generic AP Structured Reports (APSRs) (e.g. in Germany, the Netherlands or Australasia). Other initiatives focus on specific types of APRs, mainly in the cancer domain. In France, the French Society of Pathology (SFP) has published minimum data sets for 28 cancer locations [68]. In Australasia, the Royal College of Pathologists Australasia (RCPA) has published 6 organ specific cancer templates [69]. In some cases implementation guides for these APSR models based on information technology standards (e.g. XML) or healthcare information technology standards (e.g. HL7 CDA or CEN archetypes) are also provided.

Based on the CAP-CC&Ps, a joint IHE and Health Level 7 (HL7) AP initiative defined a formal information model for AP Structured Report (APSR) based on HL7 Clinical

Document Architecture (CDA) that was published in March 2011[9]. The objective of the IHE/HL7 APSR template was to make it interoperable so that different healthcare facilities could collect, exchange and mine AP information at an international level. The current scope of the IHE APSR content profile addresses all fields of AP (inflammatory diseases as well as cancer). In the cancer domain, the IHE APSR value set appendix provides a list of organ-specific AP observations derived from the CAP-CC&Ps of the 20 most frequent cancers (Appendix 6). The clinical content of APSR was encoded using reference terminologies (LOINC, SNOMED CT (including items from TNM UICC, 7th edition), ICD-O and PathLex).

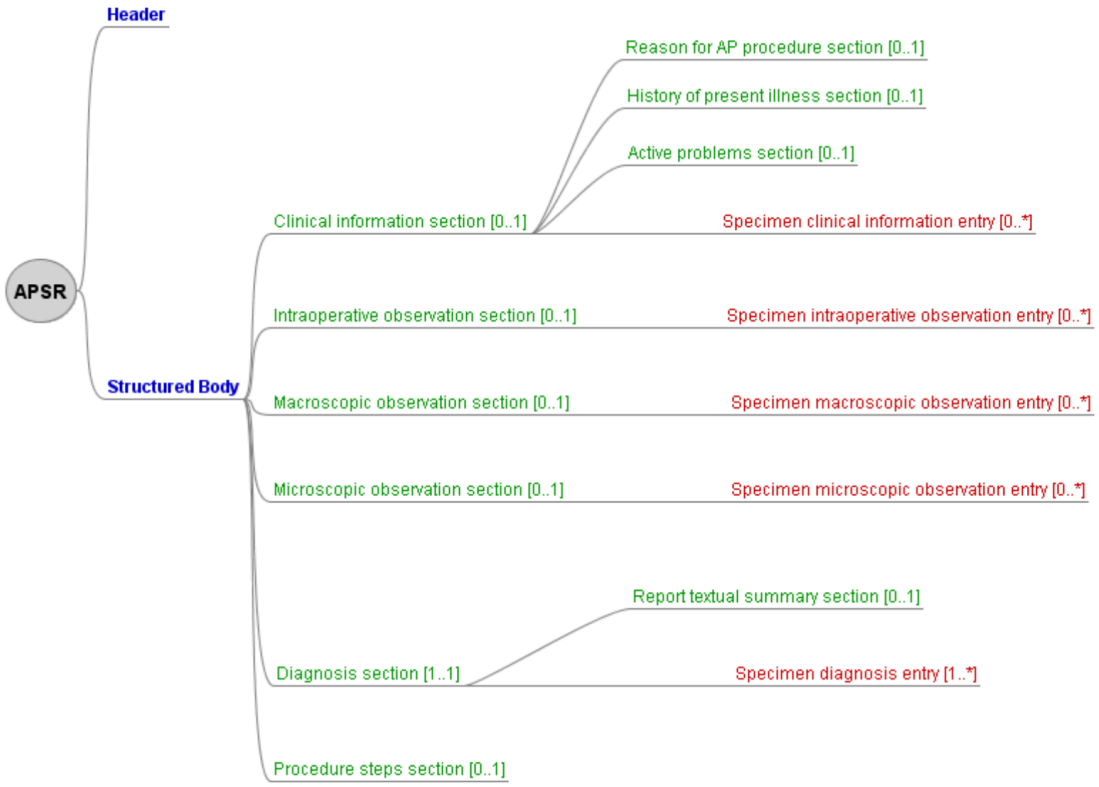


Figure 1-11: Common hierarchy for all AP Structured Reports (APSRs) document content modules (source [8])

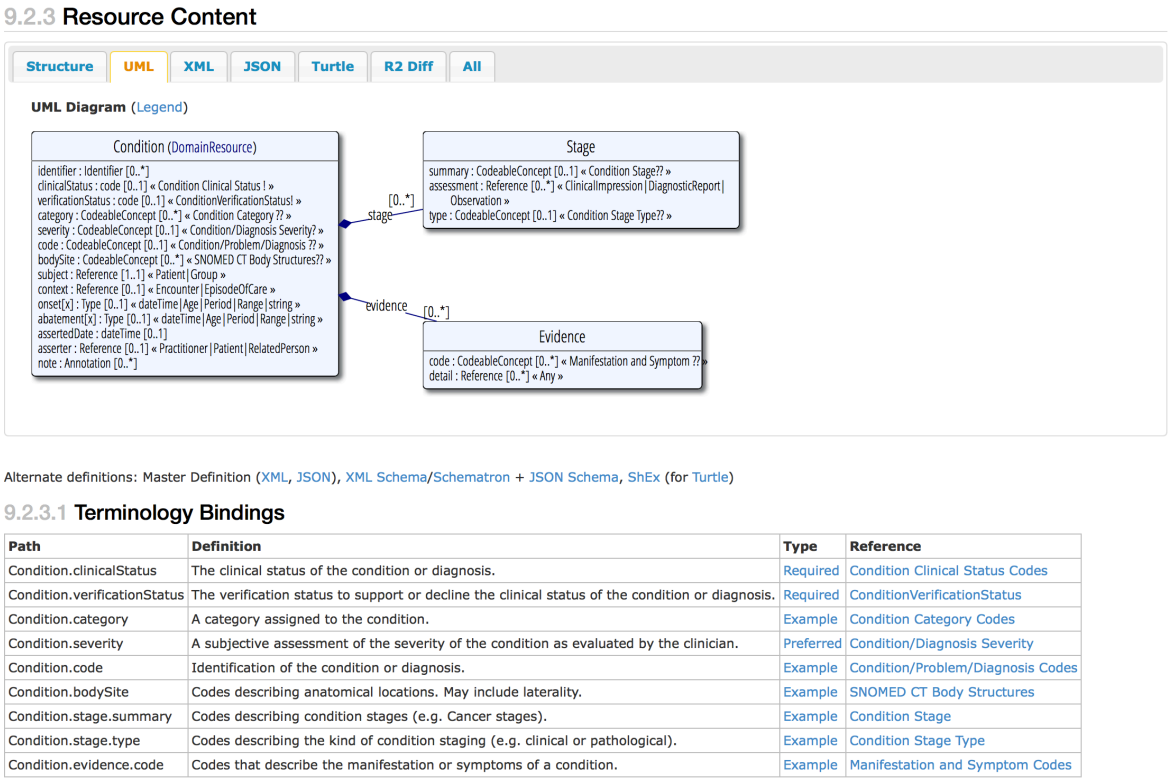
1.3.3 Fast Healthcare Interoperability Resources (FHIR)

Fast Healthcare Interoperability Resources (FHIR, pronounced, "fire") is a draft standard describing data formats and elements (known as "resources") and an Application Programming Interface (API) for exchanging Electronic health records. The standard was created by the Health Level Seven International (HL7) healthcare standards organization [70].

Healthcare records are increasingly becoming digitized. As patients move around the healthcare ecosystem, their electronic health records must be available, discoverable, and

[9] Daniel C, Macary F, Rojo MG, Klossa J, Laurinavičius A, Beckwith BA, et al. Recent advances in standards for Collaborative Digital AP. Diagn Pathol. 2011; 6 Suppl 1:S17: www.ihe.net/Technical_Frameworks/#anatomic

understandable. Further, to support automated clinical decision support and other machine-based processing, the data must also be structured and standardized.
The philosophy behind FHIR is to build a base set of resources that, either by themselves or when combined, satisfy the majority of common use cases. FHIR resources aim to define the information contents and structure for the core information set that is shared by most implementations. There is a built-in extension mechanism[10] to cover the remaining content as needed[71]. For example, in the scope of AP knowledge formalization, the FHIR resources of interest are "Condition" (to represent diagnostic information) and "Observation" (to represent scores and grades). The link between the two resources can be established via the attribute "Stage"[11].



Path	Definition	Type	Reference
Condition.clinicalStatus	The clinical status of the condition or diagnosis.	Required	Condition Clinical Status Codes
Condition.verificationStatus	The verification status to support or decline the clinical status of the condition or diagnosis.	Required	ConditionVerificationStatus
Condition.category	A category assigned to the condition.	Example	Condition Category Codes
Condition.severity	A subjective assessment of the severity of the condition as evaluated by the clinician.	Preferred	Condition/Diagnosis Severity
Condition.code	Identification of the condition or diagnosis.	Example	Condition/Problem/Diagnosis Codes
Condition.bodySite	Codes describing anatomical locations. May include laterality.	Example	SNOMED CT Body Structures
Condition.stage.summary	Codes describing condition stages (e.g. Cancer stages).	Example	Condition Stage
Condition.stage.type	Codes describing the kind of condition staging (e.g. clinical or pathological).	Example	Condition Stage Type
Condition.evidence.code	Codes that describe the manifestation or symptoms of a condition.	Example	Manifestation and Symptom Codes

Figure 1-12: Screenshot of the Content and Terminology Bindings of "Condition" FHIR resource in UML format

FHIR can be used as a stand-alone data exchange standard, but can and will also be used in partnership with existing widely used standards. FHIR aims to simplify implementation without sacrificing information integrity. It leverages existing logical and theoretical models to provide a consistent, easy to implement, and rigorous mechanism for exchanging data between healthcare applications. FHIR has built-in mechanisms for traceability to the HL7 RIM and other important content models. This ensures alignment to HL7's previously defined patterns and best practices without requiring the implementer to have intimate knowledge of the RIM or any HL7 v3 derivations [70]. Figure 1-13 summarizes the FHIR resources with their five (5) organisation levels.

[10] https://www.hl7.org/FHIR/extensibility.html
[11] Refer to : http://build.fhir.org/condition.html

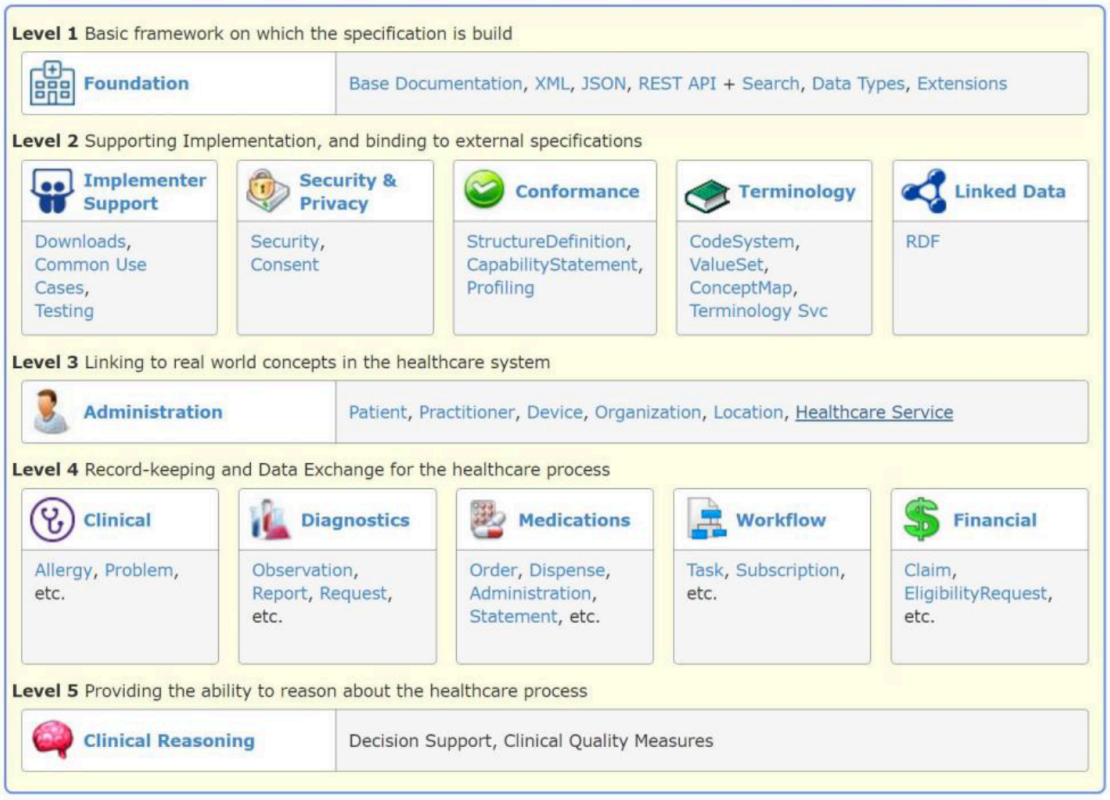


Figure 1-13: FHIR resources with the five (5) levels organisation (à revoir dans le texte)

1.3.4 International Organization for Standardization (ISO) and other related efforts

International Organization for Standardization (ISO) ensures that products and services are safe, reliable and of good quality. It has over 21000 International Standards covering almost all aspects of technology and business [72]. When considering medical informatics standards, it is useful to think about the « activities or their results » as being processes which generate medical information. This information may be in the form of electronic documents or images, such as pathology specimen images, which are annotated with demographic and medical information regarding the source patient [42].

Apart from main standards stated above, there are other projects, which are relevant to AP and Digital Pathology standards.

First, between 2004 and 2006 the Association of Pathology Informatics sponsored a project called the Laboratory Digital Imaging Project which had a goal of creating an Extensible Mark-up Language (XML) based specification for pathology image objects which would provide a common format for file interchange and be compatible with DICOM. This was an ambitious effort but unfortunately it was not able to gain sufficient traction and it foundered [9].

Another effort to mention is the Open Microscopy Environment[12]. This project is aimed at research biological microscopy, not at medical application, but it has created a data model in XML, which is expandable, and self-describing [10]. They also provide open source software through a collaborative development model. It may be of interest to those using digital microscope imaging for research purposes.

[12] www.openmicroscopy.org

A third project which is aimed at addressing the lack of a universal data format for a whole slide images is OpenSlide[13], which has created a vendor neutral C library for viewing and manipulating whole slide images in a variety of different vendor formats [73]. While not being a standard, this project provides a useful tool to help bridge the gap until digital pathology standards are widely adopted in practice. Given the long time frame that is typical of standards adoption, this project may be relevant and useful for a long time into the future.

1.3.5 Integrating the Healthcare Enterprise (IHE)

The Integrating the Healthcare Enterprise (IHE) initiative, which has been developed in North America, Europe, and Asia, aims at defining and promoting the best use of medical informatics standards. It works precisely to specify how medical informatics standards should be implemented to meet specific health care needs and making systems integration more efficient and less expensive [40].

The IHE process is based on working groups that include both health care providers who define precise users' needs and information systems vendors in charge of defining domain-specific Integration Profiles, i.e. standard-based exchange of information in real-world situations. Integration Profiles describe informatics transactions leveraging and constraining established industry standards such as DICOM or HL7. The annual definition cycle of new profiles by users and suppliers ends in the organization of international platforms of the IHE initiative interoperability tests (called ''connectathons'') that confer the unique efficiency. Participation of European researchers in IHE Anatomic Pathology has been fostered and partly coordinated by the COST action IC0604 "Euro-telepath", funded by the European commission [74].

The early sponsors of the IHE initiative in the AP domain (ADICAP[14], SEAP[15], SEIS[16], CAP[17]) solicited practicing pathologists and haematologists; information technology professionals; and vendors from France, Spain, Italy, Germany, Japan and the United States to work on the IHE AP technical framework. The IHE AP working group conducted working sessions approximately one meeting every three months. If errors in existing standards or the need for extensions are identified, IHE's policy is to report them to the appropriate standards bodies (HL7, DICOM or IHTSDO) for resolution within their conformance and standards evolution strategy. American, European, and Japanese groups agreed that, although specific DICOM objects were defined for AP digital images, modification and/or extension were necessary for two main reasons.

First, the DICOM model did not initially describe specimens in sufficient detail or associate images with specimens with enough precision for the complexity of AP practice; and second, some pathology-related image formats (Whole Slide Images, multispectral images, flow cytometry, etc.) did not have applicable DICOM information object definitions.

To address these issues, a specific DICOM pathology working group (WG26) was created in December 2005 and several IHE AP–DICOM WG26 joint working sessions have been organized [75].

The IHE AP Domain, established in 2005, managed the AP Technical Framework. This activity consisted in describing the workflow of collaborative digital anatomic pathology, identifying the IHE actors (i.e. functional information technology components and their application roles), and defining the standard-based transactions between them. This

[13] www.openslide.org

[14] Association pour le Développement de l'Informatique en Cytologie et Anatomie Pathologiques, France.

[15] Spanish Society of Pathology, Spain.

[16] Spanish Society of Health Informatics, Spain.

[17] College of American Pathologists, USA.

description is organized into functional units called integration or content profiles that highlight the capacity of IHE actors to address specific clinical needs.

1.3.5.1 The IHE Pathology and Laboratory Medicine (PaLM)

The PaLM domain of IHE merges and supersedes the Laboratory (LAB) and AP (AP) fields respectively launched in 2003 and 2006. The fundamental purpose behind this merger was the acknowledgment of a lot of similarities within the scopes of both fields, and a long routine with regards to reuse their advantages related to content modules, exchanges and common thinking from each other.

"*The decision for this merger has been prepared collectively along year 2015 by the LAB and AP leaderships, common secretariat and memberships, and was approved by the Board of IHE International on November 12, 2015. It becomes active on January 2016*." Table 1-7 summarizes the IHE PaLM laboratory specialty and sub-specialties

Laboratory specialty	Sub-specialties
AP specialties	surgical pathology, autopsy, cytopathology, image cytometry, immunohistochemistry
clinical pathology specialties	clinical chemistry, haematology, coagulation, blood gas, microbiology, immunology (allergy, auto-immunity, serology), transfusion medicine (blood bank testing), transplant compatibility testing (HLA), fertility, assisted medical procreation, cytogenetic (karyotype, molecular cytogenetic), drug monitoring and toxicology, flow cytometry
molecular pathology specialties	gene mutations detection in tumor cells, genetic identification and characterization of infectious agents, diagnostic of genetic disorders

Table 1-7: Summary of IHE PaLM laboratory specialty and sub-specialties

1.4 Innovative initiatives on "Integrated" Digital Pathology Platforms

Recently, the introduction of several tools such as slide scanners and WSI technologies created the conditions for a broader adoption of computer aided diagnosis based on WSI with the hope of a possible contribution to decreasing inter-observer variability in AP and improving diagnostic and prognostic evaluation. Semantic models are formal representations of knowledge in a given domain that allow both human users and software applications to consistently and accurately interpret domain knowledge [76], [77]. The formalisms used to represent meaning and the protocols to interact with semantic stores permits the community to create and accumulate semantic data in a form that both machines and humans could use and reuse. Following initiatives are related to either or both of these domains.

1.4.1 Academic & research platforms

1.4.1.1 Cognitive MIcroscope (MiCo) Project

Being part of a long-term process, initiated by the experience of the industrial and university partners (Figure 1-14), the Cognitive MIcroscope (MICO) aims at radically modifying the medical practices by proposing a new cognitive medical imaging environment able to improve reliability of decision-making in histopathology [78], [79]. "*Its goal is to realize a generic, open-ended, semantic digital histology platform including a cognitive dimension*." MICO combines visual perception, context, cognition and experience to reinforce a visual diagnosis assistance following an approach centred on user behaviour.

Figure 1-14: Cognitive MIcroscope project consortium members

A major issue plaguing current computer-aided diagnosis systems is the "*opaqueness*" of their mechanism, making the users untrustworthy and the results hard to validate. MICO aims at moving a step forward in medical practice by proposing the foundations of a new confluent cognitive medical imaging technology using context in an intelligent way.
MICO behaves in a cognitively consistent way with standard medical practice, following a uniform representation of image analysis, reasoning and context elements. This transparency throughout the whole process is exploited in order to permit confluence between the user and the platform, and medical validation of the results through technologies of content-based semantic image retrieval. In order to demonstrate the viability and relevance of the system, breast cancer grading was developed, tested and validated in the frame of project MICO.

1.4.1.2 FlexMim

The FlexMIm consortium includes 27 pathology laboratories in the Paris region (coordinated by Assistance Publique-Hôpitaux de Paris), research laboratories from University Pierre et Marie Curie (UPMC Univ Paris 06) and University Paris Diderot, as well as 3 companies: TRIBVN, PERTIMM and Orange (project coordinator). Based on a cloud architecture, the project embeds a dedicated Whole Slide Image (WSI) database and visualisation support [80].
In addition to developing new tools for helping WSI analysis by pathologists, the cooperative research project FlexMIm aims at setting up a shared platform allowing further technological improvements to be tested and evaluated online by a community of pathologists [80]. FlexMIm treats the user needs, expressed by anatomo-pathologists, in a context of decrease of their demography and increase of the number of medical acts. It provides the pathologists with tools increasing their cooperative (initial tele-diagnostic, tele-expertise, e-learning) and collaborative capabilities (medical protocols including slides cross reading), based on whole slide imaging (WSI) technologies [81]. FlexMIm also worked to issues [81] related to :

- Develop and setup cognitive algorithms, driven by medical knowledge models (image exploration and cancer grading rules, annotation procedures, valid medical ontologies), to identify specific regions of interest for pathological analysis/grading.
- Provide innovative, effective solutions to manage and manipulate WSI according to the used devices and networks. Provide intelligent algorithms allowing fluid data

sharing and exchange via telecommunication network in the «Télépathologie Ile de France» cluster.

1.4.1.3 Planuca

The PLANUCA[18] (Digital Platform of Pathology for the Management of Cancers) project is a cross-disciplinary collaboration between mathematicians, computer scientists, technicians, pathologists, academics and industrial actors who have previuosly collaborated for research programs. It aims to develop a digital platform of tools available to pathologists to help with screening, diagnosis, prognosis and teaching in pathology tumor [82]. As shown in Figure 1-15, project's scientific and industrial partners are: DATEXIM for the development of softwares; The GREYC (Research Group in Computer Science, Image, Automation and Instrumentation of Caen) for the development of algorithms; The AP service of the Cotentin Public Hospital Center and the University Hospital of Caen for the supply of medical dataset, expertise and tests in real conditions.

Figure 1-15: Planuca project logo representing different partners of the consortium

1.4.2 Industrial R&D platforms

1.4.2.1 TissueGnostic[19]

TissueGnostics (TG) was established in 2003 after nearly a decade of basic research. It is an Austrian company with subsidiaries in EU, USA and China, specialized in integrated solutions for high content and/or high throughput scanning and analysis of digital slides and images of tissue sections, Tissue Microarrays (TMA), cell culture monolayers, smears, etc. Imaging and analysis in micro well plates, petri dishes and culture flasks are made very easy in TG integrated workflows. TG provides a dedicated workflow for FISH, CISH and dot structure analysis in all its cytometry systems [83]

[18] http://planuca.datexim.com
[19] http://www.tissuegnostics.com/en/

It is one of the expert in microscope automation, image analysis, cell analysis, tissue analysis and blood analysis and automated digital cell morphology. Their list of products ranges from microscopy workstations to stand alone software for image analysis, cell analysis, tissue analysis and systems that can identify and pre-classify white blood cells for the haematology field [84]. Since 2004, TG starts TissueQuest and HemoFAXS software-development.
TissueQuest 1.0, automated identification and functional characterization of single cells in tissue sections. In contrast to morphometry, which provides values referring to the metric dimensions of cells, the term "tissue cytometry" refers to quantification of molecular parameters. Though methodically different, tissue cytometry exhibits a functional similarity to flow cytometry. While flow cytometer is restricted to cells in suspension (e.g. blood) and cannot be applied to solid tissue, tissue cytometry refers to the cytometric analysis (as opposed to morphometric analysis) of histological sections. TissueGnostics has been the first manufacturer of tissue cytometers offering a flow cytometry-like workflow (but applied to tissue sections) [85].
HemoFAXS is a CE-IVD ISO 13 485 conform complete solution for clinical routine haematology. It offers fully automated classification of leukocytes and erythrocytes in peripheral blood and body fluid. A bone marrow application as well as an application for veterinary medicine is also available. FDA approval is in progress [85].

1.4.2.2 Definiens

Definiens Tissue Studio is a digital pathology image analysis software application based on Cognition Network Technology (CNT)[86]. The intended use of Definiens Tissue Studio is for biomarker translational research in formalin-fixed, paraffin-embedded tissue samples which have been treated with immunohistochemical staining assays, or haematoxylin and eosin (H&E) [87]. The central concept behind Definiens Tissue Studio is a user interface that facilitates machine learning from example digital histopathology images in order to derive an image analysis solution suitable for the measurement of biomarkers and/or histological features within pre-defined regions of interest on a cell-by-cell basis, and within sub-cellular compartments [87]. The derived image analysis solution is then automatically applied to subsequent digital images in order to objectively measure defined sets of multiparametric image features. These data sets are used for further understanding the underlying biological processes that drive cancer and other diseases. Image processing and data analysis [86] are performed either on a local desktop computer workstation, or on a server grid [87]. To emulate the human mind's cognitive powers, Definiens used patented image segmentation and classification processes, and developed a method to render knowledge in a semantic network. CNT examines pixels not in isolation, but in context. It builds up a picture iteratively, recognizing groups of pixels as objects. It uses the colour, shape, texture and size of objects as well as their context and relationships to draw conclusions and inferences, similar to a human analyst.

1.4.2.3 Tribvn

TRIBVN Healthcare designs and provides solutions to acquire, manage, process and share images for cell and tissue diagnosis. Its wide range of products and services provides solutions to assist doctors and researchers in their diagnostic decision-making and their scientific evaluation on behalf of patients. TRIBVN Healthcare brings its know-how for a better diagnostic efficiency in the field of cancer, neurological diseases and dermatology [88]. The solution is based on the implementation of virtual slide modality (slide scanner or

motorized microscope) and CaloPix[20] [89] software. This solution offers a high efficiency in data management thanks to the possibility of database centralization on a server and thanks to the implementation of standardized and automated analysis routines[90]. In the field of analysis, TRIBVN Healthcare has many macros to automate counting tasks and tiresome and time-consuming measures: TMA, IHC, Fibrosis, Neuro, Dermato. The user can handle large sets of slides or complete regions of interest

1.4.2.4 DATEXIM[21]

Datexim is an innovative company founded in 2011, specialized in medical imaging. Inspired by imaging technology in fighting cancer, they are committed to improve pathologist practice by making disease detection faster, simpler and more precise. Datexim offers:
CytoProcessor™ [91] automatic screening system for cervical cancer « *A cost-efficient automatic screening system for cervical cancer with a sensitivity of 97% or higher* »[91]. CytoProcessor™ empowers cytologists with virtual microscopy tools that emulate their natural working environment, but with more systematic cell screening and significant time saving. Inspired by the need for reliable, rapid screening, Datexim designed its automated system to detect, analyze, and classify each cell in the sample. As a result, CytoProcessor™ will provide precisely the information the pathologist needs to make a decision in seconds.
LinkedPath[92] digital pathology solution « *Makes your AP laboratory workflow faster and more cost efficient using the latest innovations in digital pathology* ». It retrieves your data in real time from any computer or tablet without importing images, saving or storing them. Datexim ensures the security of the user connection and the confidentiality of medical data. Their full web application makes examining slides as easy as browsing an Internet site[92].
VirtualMultihead™ [93] real-time collective pathology review solution « Perform Collective Pathology Reviews from anywhere with any device ». Whether for diagnostic purposes or for training, VirtualMultihead™ empowers the pathologist to perform collective reviews from anywhere. Datexim secured connection via the web ensures instantaneous availability of whole slide images for review, and confidentiality of patient data. VirtualMultihead™ offers an unparalleled virtual microscopy experience. Zoom in and scan through the slide as easily as with a conventional microscope. With VirtualMultihead™ , colleagues on the other side of the world see the same perfectly synchronized image. Each participant in the session can point out areas of interest in the slide using Datexim proprietary pointer tool. The voice or videoconference can be done in parallel using any modern technology or a simple telephone.

Apart from these platforms, there exist other key actors in the global digital pathology domain like Philips, GE Healthcare, Leica Biosystems, Hamamatsu Photonics, etc. Table 1-8 shows leading companies in the global digital pathology sector.

#	**Company**	**Country**
1	Leica Biosystems Nussloch GMBH	Germany
2	Ventana Medical Systems, Inc.	U.S.
3	Hamamatsu Photonics K.K	Japan

[20]Calopix is an in-vitro diagnostic medical device for general use, and a regulated health care product, which carries the CE mark. CaloPix is a registered class II medical instrumet in Canada. CaloPix is a software solution for the management of all gross and microscopic images generated in a pathological, heamatological or histological laboratory. Wetht for research or for diagnosis, CaloPix permits the browsin, indexing, retrieval, analysis and sharing of departmental images (source: www.tribvn.com).

[21] http://www.datexim.com/fr/

4	3DHISTECH Ltd.	Hungary
5	Philips Healthcare	Netherlands
6	Apollo Enterprise Imaging Corp.	U.S.
7	XIFIN, Inc.	U.S.
8	Definiens AG	Germany
9	Visiopharm A/S	Denmark
10	Omnyx, LLC	U.S.
11	Corista LLC	U.S.

Table 1-8: Leading industrial actors in the global digital pathology market

1.5 Relevance and limits of existing approaches

1.5.1 WSI technology adoption and limits

WSI technology has matured enormously. Whole slide images have offered the AP community novel clinical, nonclinical, and research image-related applications. WSI platforms have the potential to improve diagnostic accuracy, increase workflow efficiency, balance workloads, better integrate images with information systems, and financially enhance return on investment. However, the adoption of WSI by pathologists worldwide has been slow for several reasons, including Limiting technology, image quality, shortcomings to scan all materials (e.g., cytology, microbiology), the cost of these systems and digital slide storage, their inability to handle high-throughput routine work, regulatory barriers in certain countries, user-unfriendly ergonomics, and pathologists' reluctance to use WSI [35].
As more image analysis algorithms and computer-assisted diagnosis tools get developed and validated for clinical use, they will empower pathologists to become more efficient, precise, and reproducible at quantifying prognostic features/parameters.

1.5.2 Use of standard and publicly available knowledge

Semantic models and reference terminologies are important in optical microscopy based diagnostic histopathology to improve reproducibility and quality, to assist and standardize reporting, and to enable multi-center clinical collaboration or research, especially in the context of cancer grading [8]. Reference vocabularies and ontologies are especially needed for the annotation of histopathology images with labels complying with semantic standards.
The MiCo project achieved a prototype system to perform some histopathology diagnosis related tasks on WSI where elementary imaging processes were combined by a logic engine, which could use formalized knowledge available as a set of rules. These rules, however, had been elaborated through local collaboration between pathologists and image scientists whereas sustainability calls for the use of publicly available knowledge gathered in standard formats from collaborative multi-centric efforts and constantly updated.
Current terminology systems for AP structured reporting gather terms of very different granularity and have not yet been compiled in a systematic approach. Moreover, the IHE APSR template provides a formal representation of only high-level AP observations resulting from human interpretation of low-level morphological abnormalities. There is still a need to extend the scope of IHE APSR and to integrate in a unique formal representation both high-level AP entities observable by humans and the corresponding low-level morphological abnormalities, especially those that can be quantified using image analysis tools

The availability of digital tools in pathology, especially WSI and the possibility to perform on them some image analysis tasks, call for an extension of semantic modelling to the realm of image processing and its integration with clinical semantics. Bridging the semantic gap between diagnostic histopathology and image analysis is needed for a broader use of image analysis in routine pathology.

1.5.3 Collaboration and interoperability issues

Building a usable and useful interoperability framework requires close collaboration with both healthcare professionals and their association(s)'and industrial stakeholders. Integration and content profiles need to be clinically driven and health information system vendors need to be involved since their systems will need to adopt semantic interoperability resources or will utilize clinical data that conforms to such resources. This real-world approach is needed to ensure that beyond theoretical and technical issues prioritized clinical needs are addressed.
At the technical level, we want to highlight the need of collaborative platform providing a scalable solution for regularly extending the scope of the semantic interoperability framework such as editing tools for managing templates and their binding to reference terminologies.
At the organizational level, special emphasis is put on the need of education strategy to enable wider clinical and patient/citizen acceptance and use of knowledge-rich healthcare information systems. A key success factor is when professional organizations (clinical and public health) and patient groups who drive the needs educate their members.

1.5.4 Modelling and standardizing

Modelling and standardizing the semantics of AP diagnostic interpretation requires a major input from AP experts and tools are welcome to partly relieve them from the burdens of identifying and integrating concepts from a complex and rapidly evolving domain. Extending the scope of such resource would benefit from the involvement of an international consortium of pathologists provided with supportive tools enabling community members to contribute terminological content and provide feedback on existing classes and properties. To go towards these achievements, the following scientific challenges need to be considered:

- The sustainable management of the semantic resources associated to the diagnostic and prognostic interpretation of AP images by both humans (pathologists) and computers (image analysis algorithms).
- A visual representation summarizing the current state of the concepts available in existing biomedical ontologies in the scope of the AP of tumors.
- The integration of quantitative image analysis in routine AP workflow with associated histopathology semantics.

1.6 Conclusion

As we move towards an era where digital pathology becomes more commonplace in clinical practice it has become clear that simply being able to exchange an image file is insufficient to fulfil the needs of practicing pathologist. The scanning of a slide and the viewing of the resulting image are key steps, but in addition, the associated metadata needs to be combined with the image file so that the images can be considered in combination with the clinical information. In addition, there are multiple systems which are involved with creating, storing, viewing and annotating pathology images and the entire workflow must be considered if digital pathology images are going to be seamlessly integrated into the work of practicing

pathologists [34]. We believe that the management of available semantic resources associated to histopathological image files semantics standard tools of the image analysis communities will ease the integration of WSI in clinical routine and support new generation diagnostic/prognostic protocols in Digital Pathology. At last, with the emergence of big data and machine learning technologies applied in AP, premiminary studies are showing that automated image analysis of quantitative features in AP images is useful in predicting patient prognosis in several cancers and pre-cancerous lesions [Yu + ref 31,32,34 et 51 de Yu]. In order to enhance cancer grading/scoring in the context of digital pathology, there is a need of formal models for each specific score/grade system and its details to allow cross-studies comparison of survival prediction methods as well as supporting interoperability purposes among different applications.

1.7 Summary

1.7.1 What was already known on the topic?

- Score/grade systems are specific to cancer types
- For some cancer types, different Score/grade systems are used or the same score/grade system uses different values to represent the results making it difficult cross-studies comparison of survival prediction method

- For many cancer types, grade/score and stage alone only have limited predictive values in stratifying survival outcomes of patients

- WSI are more used in the context of digital pathology.
- Standard-based model of image meta data and AP reports are used with integration profiles to seamlessly support the workflow of AP diagnostic and prognostic evaluation
- Automated image analysis of quantitative features in AP images can predict cancer prognosis

1.7.2 *What this study added to our knowledge?*

- There is a need of formal models for each specific score/grade system and its details to allow cross-studies comparison of survival prediction methods as well as supporting interoperability purposes among different applications.

- Appropriate codes must be assigned to both, the score systems and each of their details including the quantitative features that could be involved in survival prediction methods based on image analysis

PART2

Histopathology (CAP Cancer Protocols) domain knowledge formal representation

2 A sustainable visual representation of available histopathological digital knowledge for breast cancer grading

Main questions

- What are the existing formal models for representing AP Observable entities (APO) and AP Quantifiable Features (APQF)?
- How to build a terminological system for AP Quantifiable Features taking into account the existing formal models?

2.1 Background

Beside the development of digital pathology, AP computer aided diagnosis systems are emerging. Such systems, without replacing the work of the pathologist, can provide decision support and accelerate the interpretation of images by automating the evaluation of certain quantifiable features. They could also contribute to improve inter-observer diagnostic reproducibility by making quantitative assessment more objective.

Such computer systems use image processing and machine learning techniques to define models for diagnostic or prognostic evaluation [94] [95], [96]. Their development and validation require that i) sets of quality images of the **tumor pathology of interest** be constituted and ii) relevant **quantifiable parameters** in each context of specific tumor pathology are defined. It is therefore important to formalize both the concepts characterizing the types of tumor pathology (anatomical and pathological diagnoses) and the quantifiable parameters having a potential prognostic value for each type of tumor pathology.

In this context, **reference vocabularies and formalization of the associated knowledge** are especially needed to annotate histopathology images with labels complying with semantic standards or to automatically compute diagnostic or prognostic information from images.

2.1.1 Semantic models

Semantic models are formal representations of knowledge in a given domain that allow both human users and software applications to consistently and accurately interpret domain terminology [76], [77]. The formalisms used to represent meaning and the protocols to interact with semantic stores have permitted humans to create and accumulate semantic data in a form that both machines and humans could use and reuse. Coming after technologies like semantic networks – UMLS still uses them -- ontologies are nowadays the preferred way to formalize semantic knowledge and to convert it into a standard storable form (e.g. using triples at a lower level). According to Gruber [97], an ontology is « an explicit specification of a conceptualization », where « conceptualization » means an « abstract, simplified view of the world that we wish to represent for some purpose ».
Another requirement is that such a specification should be shared, e.g., published. Most available ontologies use Description Logics (DL), often hidden under specialized languages like OWL, a standard of the World Wide Web Consortium (W3C), to describe pieces of reality – domains -- and to control the complexity of query processing, e.g., to forbid asking for undecidable questions. Tools like Protégé [98] enable humans to create, check, and query ontologies. Portals like BioPortal are servers, which make ontologies available for queries by

humans and machines alike, either through human-oriented Graphics User Interfaces (GUIs) that execute in browsers, or Application Programming Interfaces (APIs) that programmers can use to set up client machines. Portals also play the role of publishers as they accept ontologies to be uploaded by authors, which entails an important service related to concept identification: each author is only responsible for uniquely identifying each concept within her proposed ontology, the portal providing a unique identifier for each ontology it publishes and also its own unique portal identifier (the concatenation of the three identifiers results in a universal resource identifier (URI) for each concept in each ontology).

2.1.2 Existing efforts for representing AP observable entities

Recently, the introduction of several tools such as slide scanners and virtual slide technologies created the conditions for a broader adoption of computer aided diagnosis based on whole slide images (WSI) with the hope of a possible contribution to decreasing inter-observer variability in Anatomic Pathology (AP) and enhancing the capability of pathologists to provide accurate diagnoses and prognostic evaluations. These changes bring up a number of scientific challenges such as the sustainable management of the available semantic resources associated to the diagnostic interpretation of AP images by both humans (pathologists) and computers (image analysis algorithms).

In order to reduce inter-observer variability between AP reports of malignant tumors, the College of American Pathologists edited more than 60 organ-specific Cancer Checklists and associated Protocols (CAP-CC&P) [6]. Each checklist includes a set of AP observations that are relevant in the context of a given organ-specific cancer and have to be reported by the pathologist. The associated protocol includes interpretation guidelines for most of the required observations.

Based on the CAP-CC&Ps, a joint IHE and Health Level 7 (HL7) AP initiative defined a formal information model for Anatomic Pathology Structured Report (APSR) (detailed in section 1.3.2.2, page 40). The clinical content of APSR was encoded using reference terminologies (LOINC[2], SNOMED-CT[3] (including items from TNM UICC[4], 7th edition), ICD-O[5] and PathLex[6]).

Current terminology systems for AP structured reporting gather terms of very different granularity [8], [9] and have not yet been compiled in a systematic approach. Moreover, the IHE APSR template provides a formal representation of only high-level AP observations resulting from human interpretation of low-level morphological abnormalities. There is still a need to extend the scope of IHE APSR and to integrate in a unique formal representation both high-level AP entities observable by humans and the corresponding low-level morphological abnormalities, especially those that can be quantified using image analysis tools.

2.1.3 Existing efforts for representing AP quantitative features

The modelling of prognostic evaluation systems in medicine has been the subject of work proposing a generic model of medical scores or grades [21], [99]–[101]. This model is

adapted to represent the result of the grade evaluation by a human observer but does not allow to effectively represent a set of quantifiable parameters from the grade evaluation.

From recent works on the formalization of histopathology knowledge:

The work proposed by Zillner et al. [102], [103] uses medical image annotation and reasoning technologies in spatio-anatomical reasoning [104] context to automatically classify patients with lymphoma.

Kurtz et al. propose ePAD [105], [106] a radiological tool for image retrieval based on semantic annotations. This automated approach provides real time support for radiologists, showing them images associated with similar diagnoses. Although systems, such as ePAD, enable the creation of image annotations (in the AIM format), they do not represent them in a format that is directly suitable for reasoning.

On another hand, Fouad et al, present an ontological perspective in histological and histopathological imaging. This on-going work focuses on the quantitative and algorithmic analysis of digitised images of cells and tissues [107].

Luque et al. focus on helping cancer specialists in automatic patient classification (staging) using semantic annotations in images [108]. The classification is made by semantic reasoning on annotations encoded in AIM, these annotations, made by radiologists, describe lesions in images.

In Racoceanu et al. [36], the authors describe a prototype that controls an entire histological image analysis protocol developed in MICO 3 in order to improve the Whole Slide Image (WSI) analysis protocol and become a reliable assessment for breast cancer classification.
In Benmarouf et al., « Interpretation breast cancer imaging by using ontology » [109], the authors propose a methodology to improve the clinical model that performs the score of breast cancer, based on the Nottingham Grading System (NGS). They designed OWL-DL ontology and SWRL rules based on histopathological images annotations in WFML2.

Marquet et al. presented an OWL ontology for automated TMN classification [110]. However, they did not use it to do classification based on image annotations.

Smith et al. propose to develop ontology to represent imaging data and methods used in pathological imaging and analysis. The ontology is named as « Quantitative Histopathological Imaging Ontology – QHIO ». It is under construction and aims to foster organized, cross-disciplinary, information-driven collaborations in the pathological imaging field [18].

Still in this direction, a preliminarily work has been recently published by our team [25] proposing the use of the CAP organ-specific CC&P. Based on NCBO BioPortal and UMLS semantic types, the metadata and semantic information generated represent a sustainable vocabulary, dedicated to histopathology, being able to effectively support daily work on Whole Slide Images, in Digital Pathology.

2.2 Problem, hypothesis and objectives

A major aspect of ontology design is the effort to rely as much as possible on existing semantics by referring to available ontologies for concepts already modelled. That emphasis on collaboration backed by web standards is probably the main reason for the breakthrough of ontologies compared to former technologies. This entails investing time to explore how the domain of interest relates to existing semantic knowledge. Modelling and standardizing the semantics of AP diagnostic interpretation requires a major input from AP experts and tools are welcome to partly relieve them from the burden of identifying and integrating concepts from a complex and rapidly evolving domain.

Our hypothesis is that it is possible to provide AP experts with a visual representation summarizing at any time the current state of the concepts available in existing biomedical ontologies in the scope of the AP of tumors. In particular, such tool is intended to support the development of a future AP Observation Ontology (APOO) including both observable entities (APO) reported by humans (pathologists) and quantifiable entities (APQF) automatically computed by machines.

Our objectives were:

i) to identify within the reference biomedical ontologies made accessible by the NCBO BioPortal [12], [13] and within the UMLS metathesaurus [14] the available histopathological formalized knowledge covering the scope of breast cancer CAP-CC&Ps

ii) to build a sustainable visual representation of this knowledge using the semantic types of the UMLS metathesaurus [16], [111].

2.3 Materials and methods

We propose a methodology and some tools to build a sustainable visual representation of standard-based AP knowledge about AP observations. Our approach consists in two steps:

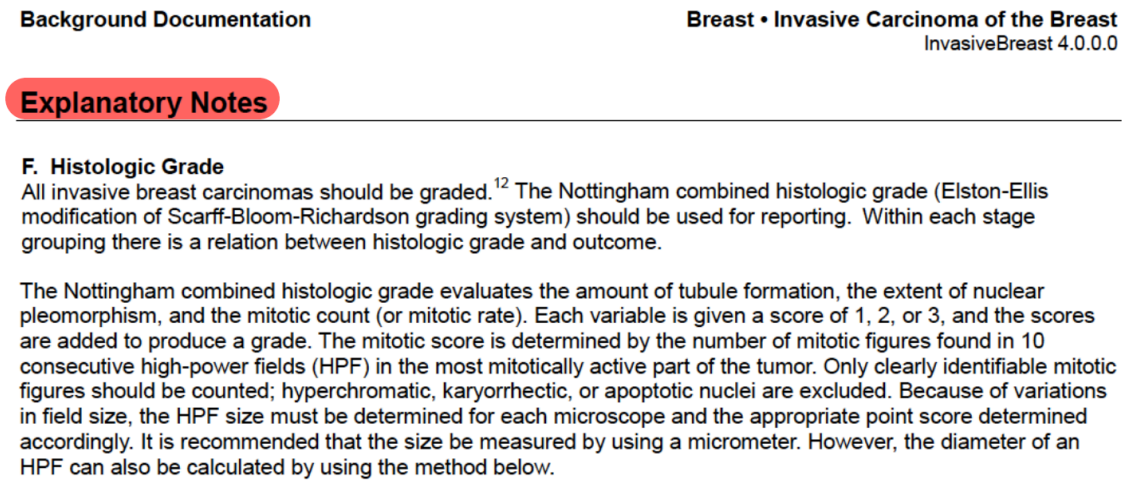
i. identifying the set of reference biomedical ontologies that are most relevant for semantic annotation of low-level morphological abnormalities;
ii. annotating CAP-CC&Ps notes using these reference ontologies and building for each high level observable entity an integrative visual representation of the concepts corresponding to relevant low-level morphological abnormalities.

We first evaluated the methodology in the limited scope of the two CAP-CC&Ps dedicated to **invasive carcinoma (IC)** and **ductal carcinoma in situ (DCIS)** of the breast.

2.3.1 Step 1: defining the set of reference biomedical ontologies that are the most relevant for semantic annotation of low-level morphological abnormalities.

We selected from the two CAP-CC&Ps a subset of five quantifiable AP observations - i.e. *observable entities that could be computed by image analysis tools* - and the corresponding

notes in the protocols (4 notes from IC, 1 note from DCIS). All five notes can be found in Appendix 2.



Background Documentation **Breast • Invasive Carcinoma of the Breast**
InvasiveBreast 4.0.0.0

Explanatory Notes

F. Histologic Grade
All invasive breast carcinomas should be graded.[12] The Nottingham combined histologic grade (Elston-Ellis modification of Scarff-Bloom-Richardson grading system) should be used for reporting. Within each stage grouping there is a relation between histologic grade and outcome.

The Nottingham combined histologic grade evaluates the amount of tubule formation, the extent of nuclear pleomorphism, and the mitotic count (or mitotic rate). Each variable is given a score of 1, 2, or 3, and the scores are added to produce a grade. The mitotic score is determined by the number of mitotic figures found in 10 consecutive high-power fields (HPF) in the most mitotically active part of the tumor. Only clearly identifiable mitotic figures should be counted; hyperchromatic, karyorrhectic, or apoptotic nuclei are excluded. Because of variations in field size, the HPF size must be determined for each microscope and the appropriate point score determined accordingly. It is recommended that the size be measured by using a micrometer. However, the diameter of an HPF can also be calculated by using the method below.

Figure 2-1: Extract of an explanatory note example from Breast Invasive Carcinoma corresponding to the Observable entity "Histologic Grade"

Two senior pathologists independently identified in each note a list of key concepts that unambiguously describe how pathologists should derive a high-level observation from low-level morphological characteristics observed in images. The union of the lists provided by the pathologists was considered as a "gold standard" (Appendix 4)

The NCBO platform provides Recommender [112], [113], a web service that proposes a selection of ontologies found to be relevant to a text. The ontology-ranking algorithm used by Recommender evaluates the relevance of each ontology to the input using a combination of the following four evaluation criteria: **coverage, acceptance, detail of knowledge, and specialization**. For each of these four criteria, a score is computed, then the scores obtained are weighted and aggregated into a final score for each ontology [114]. The weights are modifiable by users, with default values: Coverage =0.55, Acceptance=0,15, Knowledge detail = 0.15 and Specialization = 0,15. We tested Recommender with the full notes (Table 2-1) and the gold standard (Table 2-2) using in each case either the default set of weights for the four criteria used by Recommender or giving full weight to the coverage coefficient.

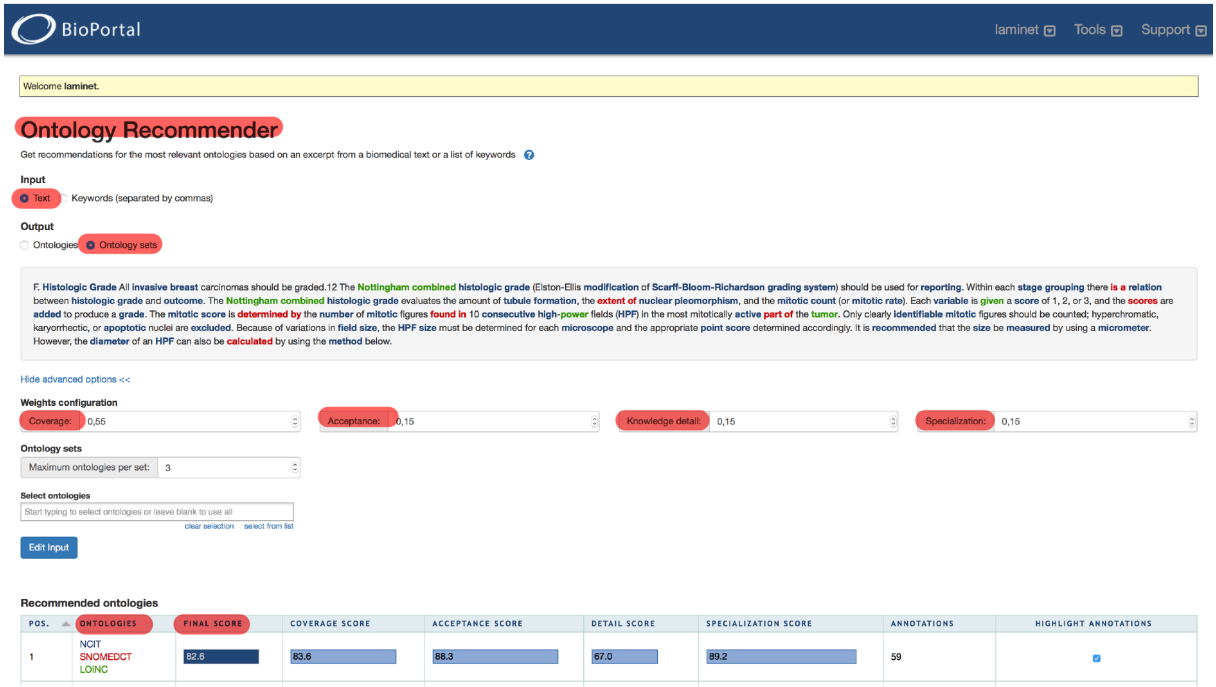


Figure 2-2: NCBO BioPortal Recommender service User Interface

NCBO however does not make public the explicit definition of the computed criteria (their authors were contacted), so we decided to implement our own method for ranking ontologies.

We used NCBO Annotator [115], [116], a tool supporting the biomedical community in tagging raw texts automatically with concepts from the biomedical ontologies and terminologies hosted by BioPortal (an option is provided to annotate from a user defined subset of ontologies).

We automatically annotated the notes by NCBO Annotator using the 508[7] ontologies available on the NCBO platform. For each note i and each ontology j, we kept only one occurrence of each of the terms annotated by ontology j in note i (some terms appear several times in the same note and Annotator returns one hit per occurrence to permit contextual studies), getting a set $S_{i,j}$ with $n_{i,j}$ elements. For each note i we computed the set of terms annotated in note i by any ontology (merging hits from all ontologies and removing multiple occurrences) getting the set $S_{i,tot}$ with $n_{i,tot}$ elements. We then computed ratios $n_{i,j} / n_{i,tot}$ which we call « coverage ratios » since an ontology that would hit every term in a note would get a ratio of 100% for that note.

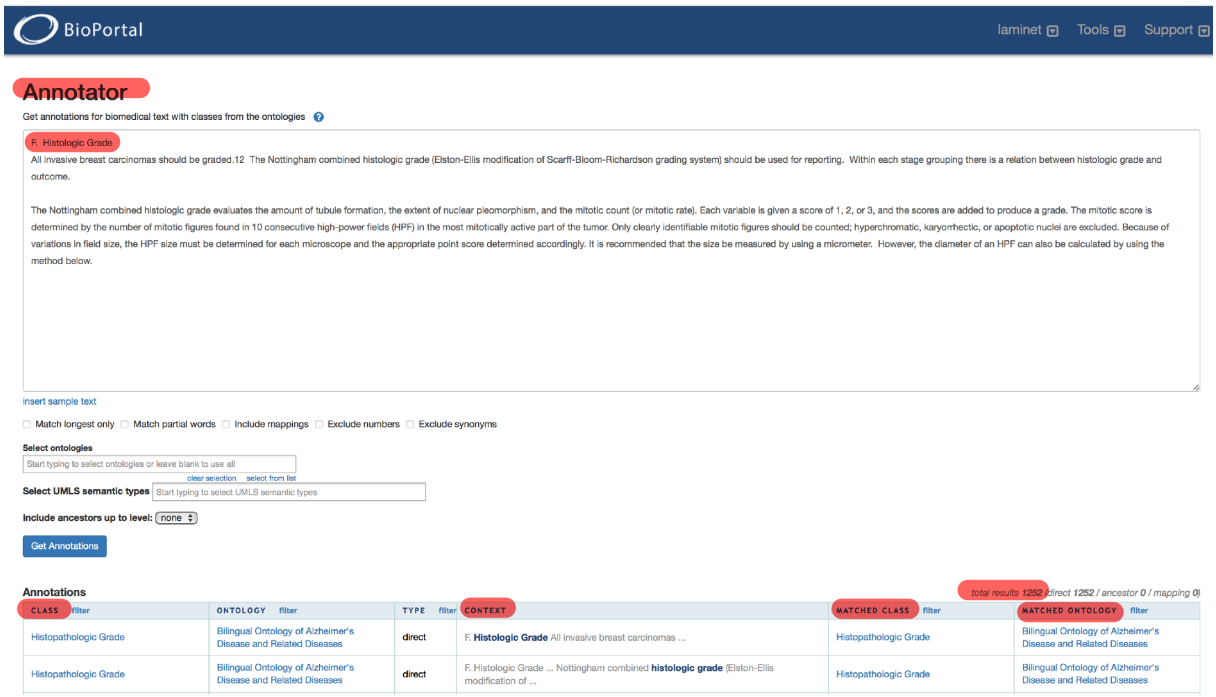


Figure 2-3: NCBO BioPortal Annotator service User Interface

Ordering the ontologies in decreasing order of their coverage ratios led to the selection of the 5 best ontologies for each note. We also computed coverage rates averaged over the set of 5 notes to summarize our results and ease discussion. The results are presented in Table 2-1.

Then the same procedure was followed for annotating the "gold standard". The results are presented in Table 2-2.

2.3.2 Step 2: building for each high level observable entity an integrative representation of the concepts representing the corresponding low-level morphological abnormalities.

By using the Terminology Services REST APIs of the Unified Medical Language System (UMLS) [117] we queried the UMLS metathesaurus to recognize in the text of our 5 notes concepts belonging to the UMLS. Then, we identified their Concept Unique Identifiers (CUIs), and their semantic types as modelled in the UMLS semantic network, which is a semantic formalism different from ontologies.

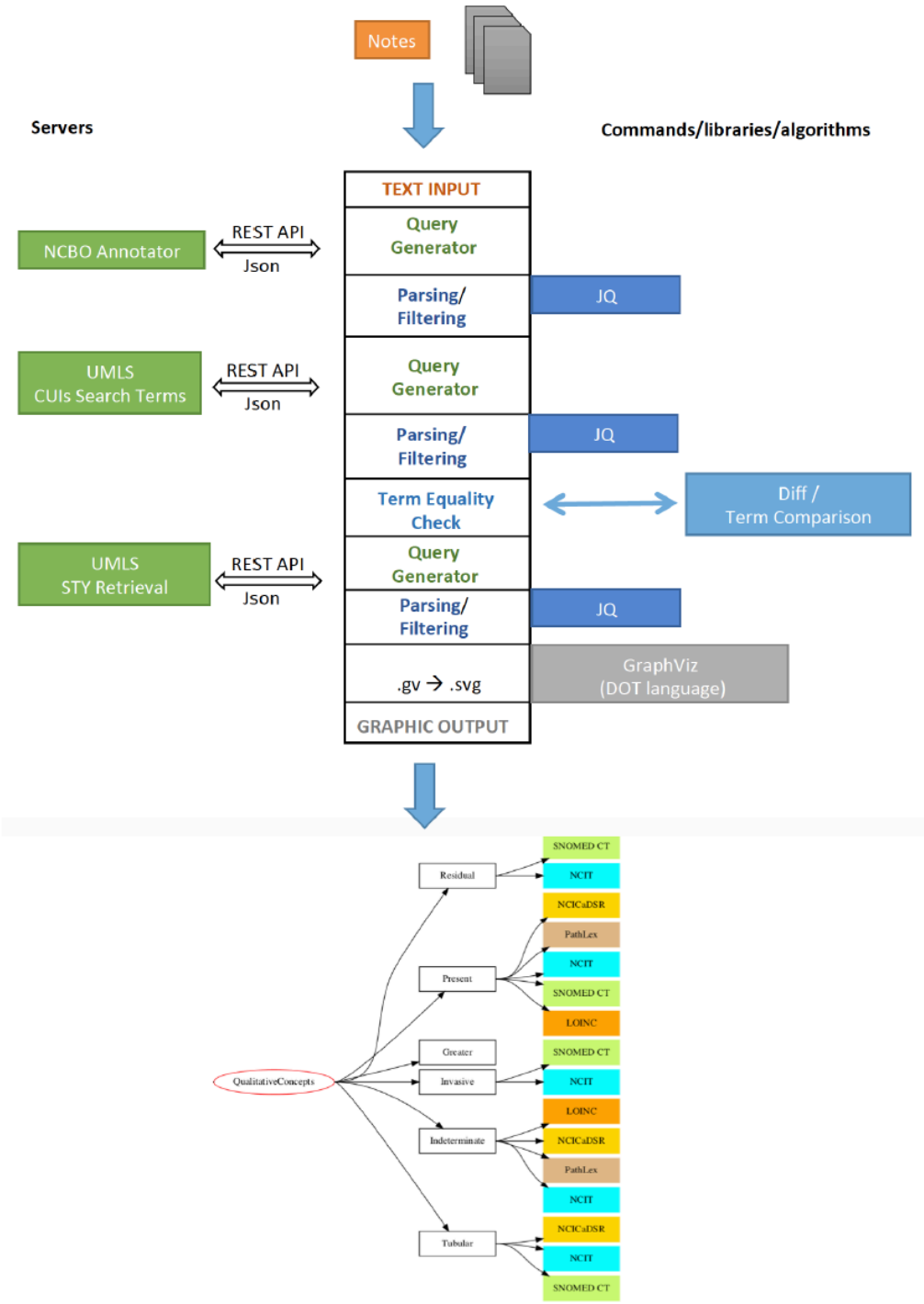


Figure 2-4: Automated workflow for the identification of available histopathological formalized knowledge from NCBO BioPortal and UMLS metathesaurus and building of the sustainable visual representation in the scope of the CAP-CC&P

To explore possible presentations, a first graphical visualization of the semantics associated to the notes was manually built from the lists of extracted concepts using the free version of the commercial visualization application MindMaple [118]. We then used the Python programming language [119], the JQ tool [120], and GraphViz [121] to automate each step of the workflow (Figure 2-4) from text input to visual display, replacing MindMaple by GraphViz and producing a slightly different visualization.

2.4 Results

Our first result is the selection of the subset, at the time of writing, of the most relevant biomedical ontologies to be used for annotation of CAP-CC&P: SNOMED-CT, LOINC, NCIT, NCI CaDSR Value Sets and PathLex were found as the most appropriate reference ontologies in the context of the two notes related to breast cancer grading methods. For individual note annotation, the set of ontologies changes: NCIT and SNOMEDCT remains for all 5 Notes, LOINC for 4 notes, NCI caDSR for 3 notes, PATHLEX and CTV3 for 2 notes. However if we take the union of the first 5 ontologies in the annotation of each individual and Rank them, the order is as follow: SNOMEDCT, LOINC, NCIT, NCI CaDSR with PathLex and CTV3 ex æquo at the 5th position. Table 2-1 shows as percentages the coverage of the concepts of each note by the annotations of the 5 reference ontologies. That these percentages can add to more than 100 for a single note reflects the possible overlap in ontologies coverage.

Notes	Number of Concepts	Number of annotations	NCIT		SNOMED-CT		LOINC		NCI Value set		PATHLEX	
			Annotations	%	Annotations	%	Annotations	%	Annotations	%	Annotations	%
Note#1	23	37	15	41	10	27	6	16	5	14	1	3
Note#2	102	154	68	44	40	26	18	12	25	16	3	2
Note#3	58	92	31	34	28	30	12	13	15	16	6	7
Note#4	47	70	33	47	17	24	11	16	6	9	3	4
Note#5	103	146	71	49	31	21	12	8	29	20	3	2
Average Rate of Concepts		499	218	44%	126	25%	59	12%	80	16%	16	3%

Table 2-1: Number of concepts and coverages of the reference ontologies in the annotation of observation notes of CAP-CC&P

Table 2-2 uses the same format when only concepts from the gold standards are counted to quantify annotations. Minor changes in the average coverages of the 5 ontologies can be observed resulting -- besides an unsurprising tie considering low counts of the gold standards -- in one swap between LOINC and SNOMED-CT in the ordered sequence. Overall the automated process reported in Table 2-1 captured well the quality scale deduced from the manually extracted gold standard in Table 2-2.

Gold standard		Ontologies									
		NCIT		LOINC		NCI Value set		SNOMED-CT		PATHLEX	
Notes	Number of Concepts	Concepts	Coverage (%)	Concepts	Coverage (%)	Concepts	Coverage (%)	Concepts	Coverage (%)	Concepts	Coverage (%)
Note#1	2	2	100	2	100	0	0	0	0	0	0
Note#2	11	4	36	0	0	0	0	0	0	0	0
Note#3	16	7	44	1	6	0	0	1	6	1	6
Note#4	8	4	50	3	38	1	13	1	13	1	13
Note#5	29	10	34	1	3	6	21	4	14	1	3
	Total	Total	Average coverage (%)	Total	Average coverage (%)	Total	Average coverage (%)	Total	Average coverage (%)	Total	Average coverage (%)
	66	27	41	7	11	7	11	6	9	3	5

Table 2-2:Number of concepts and coverages of the reference ontologies using the gold standard

For each note, NCBO Recommender gave either a score for each ontology or set of 4 preferred ontologies, in both cases with adjustable weights for sub criteria. In Table 2-3, column 2 shows the 6 best-scored ontologies with the default weights, while column 3 shows the best 6 ontologies for all the weight put on the coverage criterion. Surprisingly, that second result is not exactly equal to our former procedure where we computed coverage after a query to Annotator. Indeed we could not find in the documentation or literature a precise formula for the coverage computed by Recommender. Columns 4 and 5 report the results for best sets

of 4 of ontologies (4 being the maximal size of sets to be recommended). For the two longest notes, no answer (NA) came from the server after 30 minutes of multiple tries, after which we decided that no answer was available for such input size.

Notes	Input: text, Weights configuration: default, Output: ranked ontologies	Input: text, Weights configuration: Coverage =1, Output: ranked ontologies	Input: text, Weights configuration: default, Output: ontology sets (4 per set)	Input: text, Weights configuration: Coverage =1, Output: ontology sets (4 per set)
Note#1	1. PATHLEX (64.4)	1. PATHLEX (74.8)	PATHLEX	PATHLEX
	2. NCIT (60.8)	2. NCIT (48.0)	NCIT	SNOMEDCT
	3. SNOMEDCT (51.9)	3. SNOMEDCT (41.2)	SNOMEDCT	LOINC
	4. LOINC (40.9)	4. LOINC (35.2)	LOINC	CADSRVS
	5. RCD (33.6)	5. CADSRVS (25.2)	Final Score=78.2	Final Score=98,7
	6. CADSRVS (33.1)	6. SNMI (23.0)		
Note#2	1. NCIT (75.4)	1. NCIT (71.2)	NA	NA
	2. SNOMEDCT (58.5)	2. SNOMEDCT (50.9)		
	3. LOINC (45.4)	3. LOINC (41.1)		
	4. RCD (41.0)	4. RCD (33.0)		
	5. CADSRVS (38.3)	5. CADSRVS (31.5)		
	6. MESH (30.9)	6. SWEET (26.8)		
Note#3	1. PATHLEX (71.0)	1. PATHLEX (85.8)	PATHLEX	PATHLEX
	2. NCIT (55.3)	2. NCIT (41.9)	NCIT	SNOMEDCT
	3. SNOMEDCT (47.0)	3. SNOMEDCT (35.0)	SNOMEDCT	LOINC
	4. LOINC (39.6)	4. LOINC (31.9)	LOINC	CADSRVS
	5. RCD (35.9)	5. RCD (25.9)	Final Score=78.9	Final Score=100
	6. CADSRVS (31.2)	6. SNMI (18.6)		
Note#4	1. NCIT (78.7)	1. NCIT (74.1)	NCIT	NCIT
	2. SNOMEDCT (55.5)	2. LOINC (48.5)	SNOMEDCT	SNOMEDCT
	3. LOINC (47.5)	3. SNOMEDCT (48.4)	LOINC	LOINC
	4. RCD (40.5)	4. CADSRVS (34.7)	CADSRVS	CADSRVS
	5. CADSRVS (38.5)	5. RCD (33.4)	Final Score=85.5	Final Score=95.1
	6. ONTOAD (30.5)	6. PATHLEX (26.2)		
Note#5	1. NCIT (79.8)	1. NCIT (76.2)	NA	NA
	2. SNOMEDCT (63.8)	2. LOINC (61.9)		
	3. LOINC (54.3)	3. SNOMEDCT (61.9)		
	4. CADSRVS (51.7)	4. CADSRVS (49.2)		
	5. RCD (49.2)	5. RCD (48.4)		
	6. NIFSTD (41.5)	6. NIFSTD (43.8)		

Table 2-3: NCBO Recommender results for Note#1 to Note#5 processed as text, with ontology ranking or set of ontologies as output

Notes	Input: text, Weights configuration: default, Output: ranked ontologies	Input: text, Weights configuration: Coverage =1, Output: ranked ontologies	Input: text, Weights configuration: default, Output: ontology sets (4 per set)	Input: text, Weights configuration: Coverage =1, Output: ontology sets (4 per set)
Note#1	1. NCIT (88.8)	1. LOINC (100)	LOINC	NP
	2. LOINC (81.9)	2. NCIT (90.4)	Final score = 81.9	
	3. SNOMEDCT (43.2)	3. AURA (38.5)		
	4. ONTOAD (43.2)	4. NPO (38.5)		
	5. NPO (40.9)	5. OntoVIP (38.5)		
	6. EFO (38.3)	6. CMPO (28.8)		
Note#2	1. NCIT (65.5)	1. NCIT (74.6)	NCIT	NCIT
	2. SNOMEDCT (48.7)	2. HL7 (50.7)	SNOMEDCT	PATO
	3. OMIM (36.3)	3. PATO (42.0)	OMIM	SNOMEDCT
	4. PATO (35.5)	4. SNOMEDCT (41.6)	SOPHARM	SOPHARM
	5. MESH (34.8)	5. NIFSTD (40.2)	Final score = 80.6	Final score=79.5
	6. LOINC (34.6)	6. FB-CV (40.2)		
Note#3	1. NCIT (71.8)	1. NCIT (64.1)	NCIT	NCIT
	2. SNOMEDCT (59.7)	2. SNOMEDCT (54.0)	SNOMEDCT	RCD
	3. LOINC (51.3)	3. LOINC (48.4)	MESH	SOPHARM
	4. RCD (47.9)	4. RCD (44.2)	OMIM	FYPO
	5. NIFSTD (40.7)	5. NIFSTD (37.1)	Final score=77.2	Final score=80.6
	6. CADSRVS (39.5)	6. BIOMODELS (33.4)		
Note#4	1. NCIT (89.9)	1. NCIT (93.7)	NCIT	NCIT
	2. LOINC (56.7)	2. LOINC (66.7)	LOINC	LOINC
	3. SNOMEDCT (49.1)	3. SNOMEDCT (39.9)	Final score=91.7	Final score=100
	4. CADSRVS (35.6)	4. CSEO (33.9)		
	5. RCD (33.5)	5. CADSRVS (27.9)		
	6. ONTOAD (32.9)	6. ROO (24.1)		
Note#5	1. NCIT (77.6)	1. NCIT (73.1)	NCIT	NA for set 4, Answer with set 3
	2. SNOMEDCT (67.4)	2. SNOMEDCT (65.2)	SNOMEDCT	NCIT
	3. NIFSTD (49.6)	3. LOINC (52.2)	MESH	SNOMEDCT
	4. LOINC (49.1)	4. BIOMODELS (49.8)	CADSRVS	CADSRVS
	5. MESH (47.3)	5. NIFSTD (47.7)	Final score=84.1	Final score=89.5
	6. RCD (44.2)	6. EP (45.4)		

Table 2-4: NCBO Recommender results for Gold Standard terms from Note#1 to Note#5 processed as text, with ontology ranking or set of ontologies as output

Similar results from Recommender for the gold standards as inputs are presented in Table 2-4. Not Present (NP) means that there are no ontology sets recommended for the input provided. Even if one recognizes ontologies selected with annotator at the first positions of Recommender rankings, differences appear soon, even more so for the gold standard results. Emphasizing coverage in the coefficients leads to unexpected ontologies.

Now addressing visualization, we decided to rely on the display of some computed labelled graphs. For each note, the corresponding high-level observable entity term in the title was associated to a central node and peripheral nodes represented all the terms annotated in the note. Peripheral nodes were first linked to nodes representing their UMLS semantic types [16], these semantic nodes in turn linked to the central node. Clickable icons on links near term nodes permitted to pop up windows either to display the text of the notes where they appeared highlighted with the context of analysis and the concerned CAP-CC&P document, or to signal and open each ontology annotating them with the corresponding NCBO or UMLS resources.

The visual report built using Mindmaple is shown on Figure 2-5 for the note on Glandular/tubular differentiation. It federates semantic knowledge from different sources, either from BioPortal's ontologies or from the UMLS metathesaurus and semantic network. The layout proposed by MindMaple after some manual interaction is satisfactory. Notice how the semantic types provide some hierarchical organizations of the peripheral concepts (e.g., the nodes linked to QualitativeConcepts). Figure 2-6 and Figure 2-7 show how popup

windows provide complementary information from source ontologies or the context where annotated terms appear in the notes with the title, ID, version and exact pages of the concerned CAP&CCP.

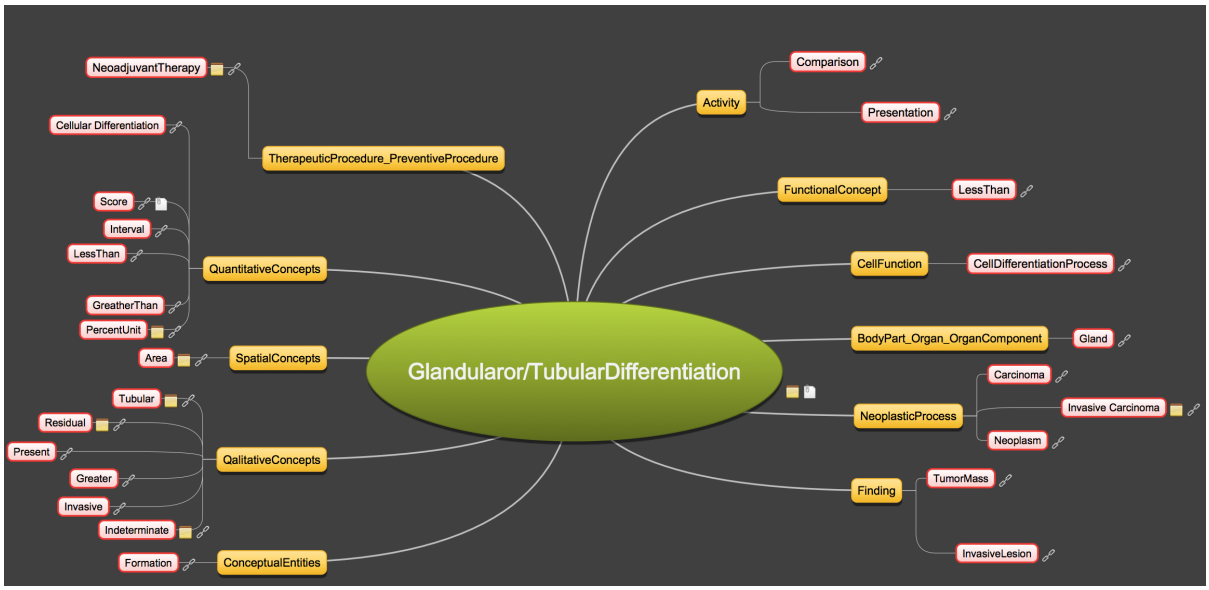


Figure 2-5: Graphical view of the sustainable semantic modelling approach in the context of Glandular/Tubular differentiation

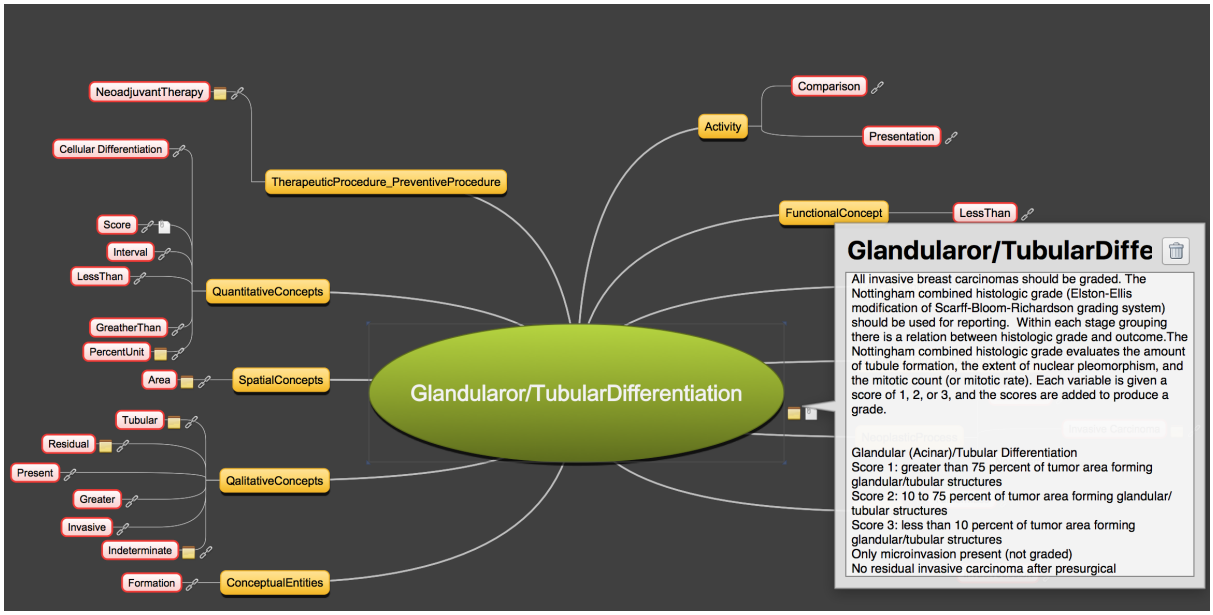


Figure 2-6: The popup window permits reading the term in text context within the note modelled here.

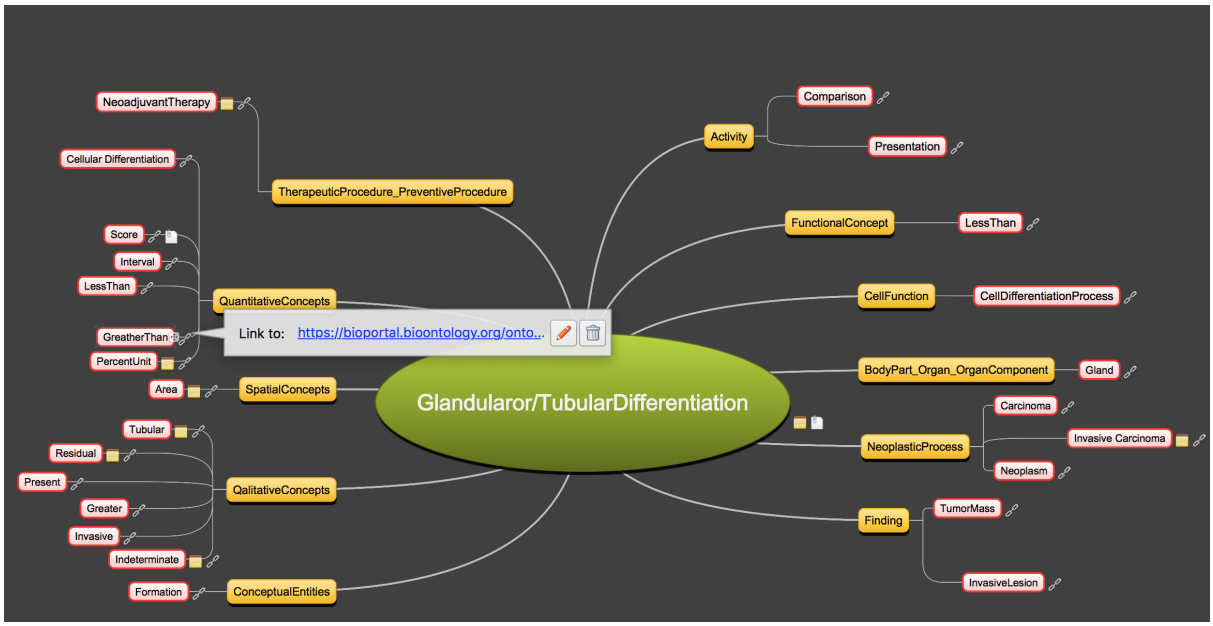


Figure 2-7: access to source ontologies is readily available for further exploration of the semantic modelling of concepts annotated in this note.

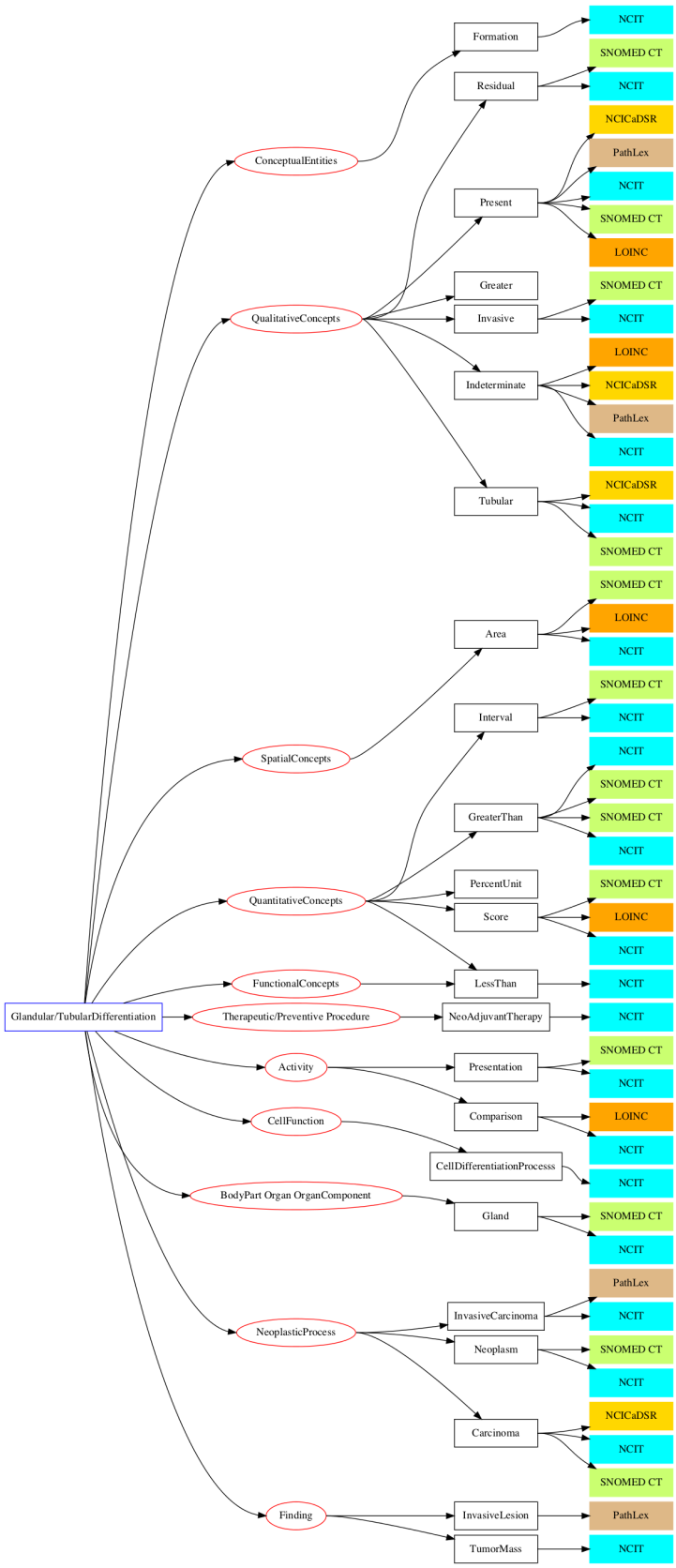


Figure 2-8: Graphical view of the sustainable semantic modelling approach in the context of Glandular/Tubular differentiation obtained with GraphViz

Automating the whole workflow from text input to visual display of the graphical representation was shown to be possible. We addressed the very similar APIs provided by BioPortal and ULMS (both use a REST architecture) and used the Python scripts provided in their documentations to automate all the necessary queries from the note input, obtaining answers in the common JSON file format. The JQ tool was used to parse the results and extract the data we needed to build the graphical representation. GraphViz, and especially its « dot » program were used to produce the graphical representation (shown in Figure 2-8) from a text file which can be automatically written from the two outputs of JQ using by a python program.

2.5 Discussion

The novelty of this approach is the federation of the knowledge issued from different ontologies (and even different semantic formalisms), and the sustainable management that automation eases. This formal representation is based on the UMLS semantic types of the concepts and will refer to source ontologies for future maintenance. Figure 2-5 shows the proposed semantic modelling in the context of glandular/tubular differentiation. For each concept we have information related to its Concept Unique Identifier (CUI), semantic type, source ontology, semantic relation and links to related metadata. These preliminary results open the prospect of building an Anatomic Pathology Observation ontology that will allow an accurate representation of AP reports understandable by both human and software applications.

Our objectives of sustainability address robustness to resource updates and domain extensibility. Updating the visual report to follow the evolution of the source ontologies or the UMLS metathesaurus and semantic network is addressed by simply rebuilding the visual report often enough. The workflow proposed here is compatible with complete automation. Each query we first performed manually has an API counterpart, using standard formats such as JavaScript Object Notation (JSON) or Extensible Mark-up Language (XML) for data exchange. We chose JSON[122] for simplicity. To build a graphical representation, data from lists of annotated terms (Annotator output) and semantic types (UMLS output) had first to be correlated. In particular one had to check that the preferred term of the UMLS file returned for a term already known as a hit for Annotator was equal to that term. A visual alert could be triggered only if the equality test fails, but one could also exploit the other terms, such as synonyms, that UMLS returns, and build complementary visualizations in further work.

Once the results of Annotator and UMLS were integrated in a common data structure, writing the GraphViz source file was straightforward and would only require a simple algorithm. That file was converted by the « dot » utility of Graphviz into an svg file displayable in standard browsers. The svg format was chosen because of its simplicity for inserting hyperlinks from Graphviz. The visualization presented can be extended in many ways, for instance to replace the role of UMLS semantic types by an ontology specific semantic object. Even at this basic stage we found the presentation quite informative in our quest for links to image processing tasks. The current proposed model includes relevant terms corresponding to the various features defining the grades and scores of breast tumors. It provides a sustainable formal representation of the knowledge involved during the AP diagnostic process.

Extending the scope of such resource would benefit from the involvement of an international consortium of pathologists provided with supportive tools enabling community members to contribute terminological content and provide feedback on existing classes and properties.

2.6 Conclusion

This study proposed a formal representation of histopathological knowledge related to breast cancer grading, underpinning AP-focused informatics tools for patient care and clinical research. We described the role of this semantic approach in bridging the gap between the CAP-CC&Ps data elements, NCBO ontologies, the UMLS Metathesaurus and the UMLS Semantic Network. Greater participation of the AP community is needed in the development, adoption, and maintenance of such a source in a sustainable manner. The proposed approach and tools, based on the CAP-CC&Ps, aim at supporting AP experts in building a standard-based representation of low-level morphological abnormalities observed in cancer that can be quantified using image analysis tools. This effort is complementary to the Integrating the Healthcare Enterprise (IHE) initiative building a standard-based representation of high-level AP observations required in cancer AP reports. Additional efforts are needed to achieve a workable standard-based formal representation of histopathological knowledge integrating both observable entities reported by humans (pathologists) and quantifiable entities automatically computed by machines.

Providing such unique formal representation facilitates the way for a more efficient use of computer aided diagnosis in AP. Sustainable management of the explicit and unambiguous semantics associated to the diagnostic interpretation of AP image by both humans (pathologists) and computers grading process, (image analysis algorithms) will support a better use of existing image analysis algorithms such as the ones elaborated in the MICO[8] [36] and their adaptation to other contexts (same type of cancer but different organs, e.g., from breast to prostate, or same organ but different types of cancer).

3 Proposal of an Anatomo-Pathology Quantifiable Features (APQF) Formal representation for grading malignant tumors

3.1 Problem, hypothesis and objectives

In this section, the main issue is related to the management of the semantics associated to the diagnostic and prognostic interpretation of histopathology images. Based on the results of the previous work (see **section 56**), our objective is to "integrate" existing biomedical semantic resources in NCBO BioPortal and UMLS with relevant quantifiable features extracted from the whole scope of the 67 CAP-CC&P.

3.2 Materials and methods

[Beyond the State of the Art enumeration, please note that all the tools and resources described in this section were used during this work]

3.2.1 Existing terminologies and semantic resources for AP Diagnosis & prognostic Observation

3.2.1.1 Terminologies for AP diagnosis coding

In cancer centres, pathologists use the International Classification of Diseases for Oncology (ICD-O) to code AP diagnoses. Pathologists use the coded AP data locally as well as regionally or internationally for epidemiological purposes[9]. In France, under the direction of the Association for Developing Informatics in Cytology and Anatomic Pathology (ADICAP), the use of interoperable repositories was developed in Anatomic Pathology. For more than 30 years, Pathologists include in their AP reports topographic and morphological codes by using the ADICAP terminology. ADICAP codes are integrated into AP Laboratory Information Systems (LIS) or image acquisition modalities (e.g. slide scanners).

3.2.1.1.1 International Classification of Diseases ICD-O

Based on the tenth revision of the International Classification of Diseases (ICD-10) and published by the World Health Organization (WHO), this bi-axial classification applies only to neoplastic tumor pathology. It contains an alphanumeric topography code (4 characters with a separator) and an alphanumeric morphology code (5 characters with a separator). A 6th optional character can complete the morphology code, allowing specifying the grade of the tumor. Originally published in English in 2000, ICD-O-3 has been available in French since November 2008 [123], [124]. The ICD-O site provides access to a prioritized list of 3,616 ACP diagnoses of tumor pathology corresponding to all relevant precoordinated combinations of topography and morphology codes.

By agreement with the College of American Pathologists, the morphology section of ICD-O is incorporated into the Systematized Nomenclature of Medicine (SNOMED)[10,11] classification as the neoplasm section of the morphology field.

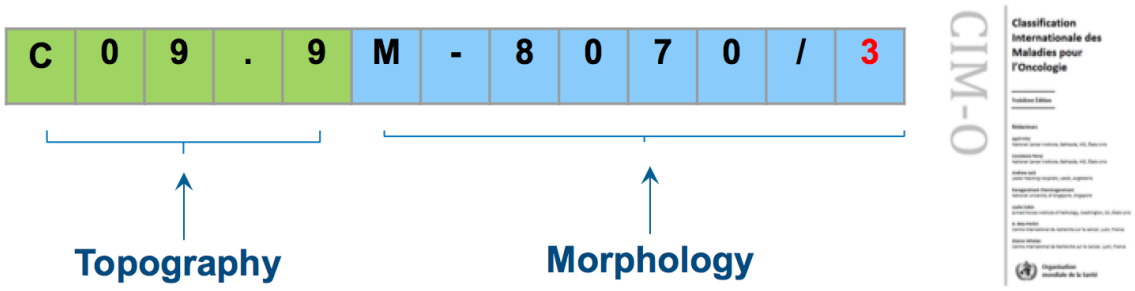


Figure 3-1: Organisation of the International Classification of Diseases (ICD-O) coding system

Limitations of the ICD-O coding system: i) Partial coverage of AP diagnoses (tumor pathology only) ii) Pre-coordinated definition of non-formalized diagnosis of tumor pathology (ICD-O).

3.2.1.1.2 The Association for Developing Informatics in Cytology and Anatomic Pathology (ADICAP)

The coding system for AP lesions developed by the ADICAP [125], [126] is a thesaurus in French, covering all fields of pathology. Each AP diagnosis is associated to a mnemonic alphanumeric code of 15 characters based on 8 Dictionaries: D1 Methods of sampling, D2 Types of technique, D3 organs and regions, D4 General non-tumoral pathology, D5 Tumor pathology, D6 Specific organ lesions, D7 Cytopathology, D8 Topography specifying D3.

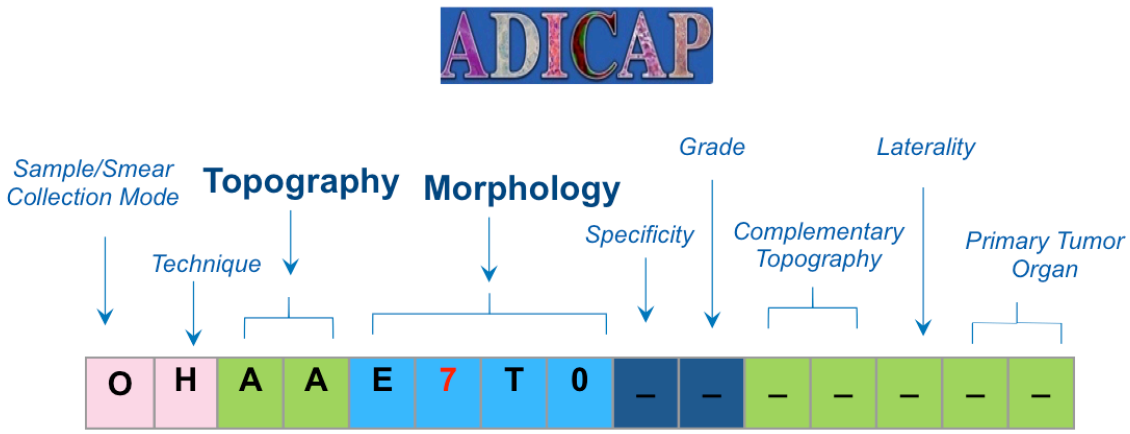


Figure 3-2: Organisation of the Association for Developing Informatics in Cytology and Anatomic Pathology (ADICAP) coding system with the 15 characters and 8 dictionaries

The fields of ADICAP are organized in two zones:

- a mandatory 8-character field, which allows coding the mode of sampling of the received sample, the type of technique, applied to this sample and the AP diagnosis consisting of the topography (2 characters) and the morphology (4 characters) of the lesion.
- an optional 7-character field for pathologists wishing to provide additional precision, particularly in tumor pathology (grade, precise topography, laterality, location of the primary tumor in the case of metastasis).

	Characters number (#)	Code mnemonic alphanumeric	Dictionary Name
Mandatory	1	A	Sample/smear collection mode
	2	A	Technic type
	3-4	A A	Organ
	5-6-7-8	N N N N	Non-tumor lesion
	5-6-7-8	A N A N	Tumor lesion
	5-6-7-8	0 A N N	Cytopathology
Optional	9	A	Free field
	10	A	Grading of tumors
	11-12	A A	Topography
	13	A	Laterality
	14-15	A A	Primary tumor organ

Table 3-1: ADICAP 15 characters coding of lesions : mandatory zone (field 1 to 8) and optional zone (field 9 to 15)

3.2.1.2 Reference biomedical semantic resource for AP Observation

3.2.1.2.1 Integrating Healthcare Enterprise (IHE) AP Observation

Below is an extract of the Breast AP Observations from the IHE Anatomic Pathology Technical Framework Supplement. The overall list of 20 organs can be found in Appendix 6 (IHE_PAT_elementTemplates Tab).

Document template name	Element template id2014		Element fully specified name	Concept domain	Value set or code	Rim source	Element type	Value set (for CD element)
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.439	Estrogen receptor	Breast-Infiltrating malignant neoplasm-Estrogen rec	anatomic pathology observation	432	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.31
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.415	Extensive intraductal co	Breast-Infiltrating malignant neoplasm-Extensive int	anatomic pathology observation	17	observation.value	BL	
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.416	HER2/neu (FISH method	Breast-Infiltrating malignant neoplasm-HER2/neu (F	anatomic pathology observation	433	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.32
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.417	HER2/neu (immunoperd	Breast-Infiltrating malignant neoplasm-HER2/neu (ir	anatomic pathology observation	434	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.33
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.525	Histologic grade	Breast-Infiltrating malignant neoplasm-Histologic gr	anatomic pathology observation	418	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.17
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.525	Histologic grade	Breast-Infiltrating malignant neoplasm-Histologic gr	anatomic pathology observation	419	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.19
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.525	Histologic grade	Breast-Infiltrating malignant neoplasm-Histologic gr	anatomic pathology observation	420	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.18
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.525	Histologic grade	Breast-Infiltrating malignant neoplasm-Histologic gr	anatomic pathology observation	421	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.16
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.525	Histologic grade	Breast-In situ neoplasm-Histologic grade of ductal ca	anatomic pathology observation	417	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.23
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.173	Histologic type	Breast-In situ neoplasm-Histologic type	anatomic pathology observation	436	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.254
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.173	Histologic type	Breast-Infiltrating malignant neoplasm-Histologic ty	anatomic pathology observation	437	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.15
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.173	Histologic type	Breast-Lesion-Histologic type	anatomic pathology observation	1904	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.388
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.440	Microinvasion (DCIS)	Breast-Infiltrating malignant neoplasm-Microinvasio	anatomic pathology observation	98	observation.value	BL	
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.429	Necrosis of ductal carcin	Breast-In situ neoplasm-Necrosis of ductal carcinom	anatomic pathology observation	1901	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.253
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.158	pM	Breast-Infiltrating malignant neoplasm-pM	anatomic pathology observation	1903	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.30
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.159	pN	Breast-Infiltrating malignant neoplasm-pN	anatomic pathology observation	428	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.29
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.438	Progesterone receptor	Breast-Infiltrating malignant neoplasm-Progesteron	anatomic pathology observation	431	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.34
Breast APSR	1.3.6.1.4.1.19376.1.8.1.4.161	pT	Breast-Infiltrating malignant neoplasm-pT	anatomic pathology observation	430	observation.value	CD	1.3.6.1.4.1.19376.1.8.5.27

Table 3-2 : Breast AP Observations from the IHE Anatomic Pathology Technical Framework Supplement

3.2.1.2.2 NCBO BioPortal Ontology Repository

The National Center for Biomedical Ontology (NCBO) is one of the National Centers for Biomedical Computing funded under the NIH Roadmap Initiative. Contributing to the national computing infrastructure, NCBO has developed BioPortal, a web portal that provides access to a library of biomedical ontologies and terminologies [127]. Via the NCBO Web services [128], BioPortal enables community participation in the evaluation and evolution of ontology content by providing features to add mappings between terms, to add comments linked to specific ontology terms and to provide ontology reviews [13]. As well as the evaluation of ontologies based on criteria such as usability, domain coverage, content quality,

support and documentation. Currently[12], BioPortal contains 566 ontologies with 8 152 116 classes.

One can browse the NCBO library of biomedical ontologies and submit one's ontology through the BioPortal Web site. Detailed summary information is available on the ontology summary page for each ontology. Table 3-3 summarize the BioPortal statistics.

Ontologies	566
Classes	8,152,116
Resources Indexed	48
Indexed Records	39,537,360
Direct Annotations	95,468,433,792
Direct Plus Expanded Annotations	144,789,582,932

Table 3-3: NCBO BioPortal semantic resources content statistics

Ontologies from a number of different groups are published in BioPortal, including Biodiversity Information Standards (BIS), The Consultative Group on International Agricultural Research (CGIAR), Clinical and Translational Science Awards (CTSA), OBO Foundry, Proteomics Standards Initiative (PSI), Unified Medical Language System (UMLS), World Health Organization-Family of International Classifications (WHO-FIC) and Cancer Biomedical Informatics Grid (caBIG). One can also narrow the list of ontologies shown by selecting one of the following "format" (a domain) OBO (107), OWL (374), SKOS (2) and UMLS (32) to which the ontology represented.

3.2.1.2.3 UMLS Metathesaurus & Semantic Network

The Unified Medical Language System (UMLS) is a set of files and software applications that brings together many health and biomedical vocabularies and standards to enable interoperability between computer systems. The UMLS integrates and distributes key terminology, classification and coding standards, and associated resources to promote creation of more effective and interoperable biomedical information systems and services, including electronic health records [117].

UMLS metathesaurus is a large, multi-purpose thesaurus that contains biomedical and health related concepts, synonyms and concept relationships arranged as a semantic network. It is used in documenting patient care and further in billing, statistical work, research and indexing [14], [117]. In this study, semantic groups were based on quantifiable concepts associated to most relevant anatomy, concepts and ideas, objects and procedures, etc.

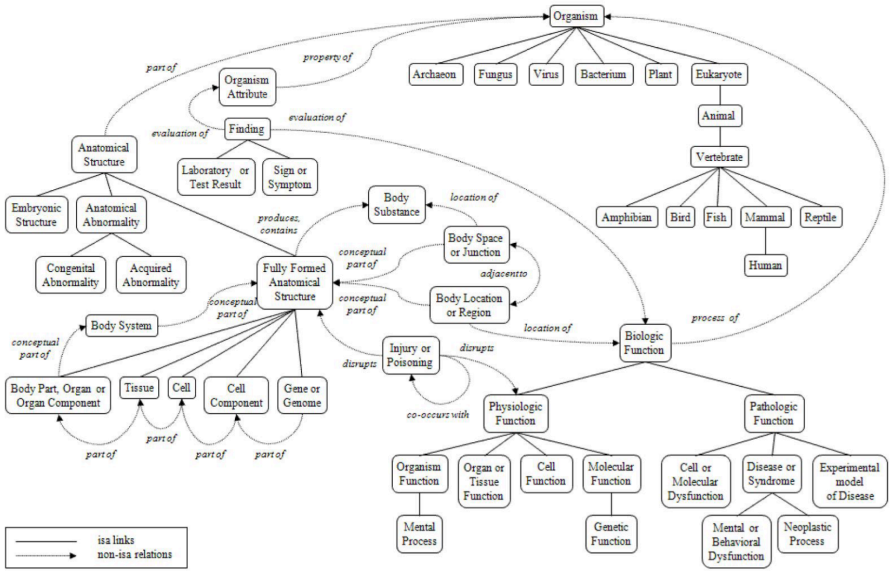


Figure 3-3: A Portion of the UMLS Semantic Network

The UMLS Terminology Services (UTS) provides three ways to access the UMLS: Web browsers, local installation and Web services APIs. In this work, we used the Semantic Network Browser, to view the names, definitions, and hierarchical structure of the Semantic Network. Figure 3-3 above, shows a portion of the UMLS Semantic Network [129], [130].

3.2.1.2.4 Open Biomedical Ontologies (OBO) Foundry

Developed by a US consortium, the Open Biomedical Ontologies (OBO) team is pursuing a strategy to overcome the « proliferation of ontologies, which itself creates obstacles to integration. » [131]. By misuse of language, OBO designates a format but it is at the origin a project of creation of controlled vocabularies shared by the fields of medicine and biology. Existing OBO ontologies, including the Gene Ontology, are undergoing coordinated reform, and new ontologies are being created on the basis of an evolving set of shared principles governing ontology development. The result is an expanding family of ontologies designed to be interoperable and logically well formed and to incorporate accurate representations of biological reality.

3.2.2 Terminology and information model editors

3.2.2.1 Protégé [132]

Protégé is a free, open-source ontology (knowledge-based) editor and framework for building intelligent systems. It allows users to create ontologies via the Protégé-Frames and Protégé-OWL editors. Protégé ontologies can be exported into a variety of formats including RDF, RDFS, OWL, and XML Schema. With Protégé, one can [133]:

- Import, edit and save existing ontologies written in OWL or RDF
- Create new ontologies.
- Save ontologies in several formats, including XML expressions of RDF and OWL

- Visualize ontologies in graphical form, showing the functional relationships between classes.
- Populate ontologies with concrete instances of classes.
- Execute reasoners that can perform inferences on an ontology (i.e. classify instances based on their properties)

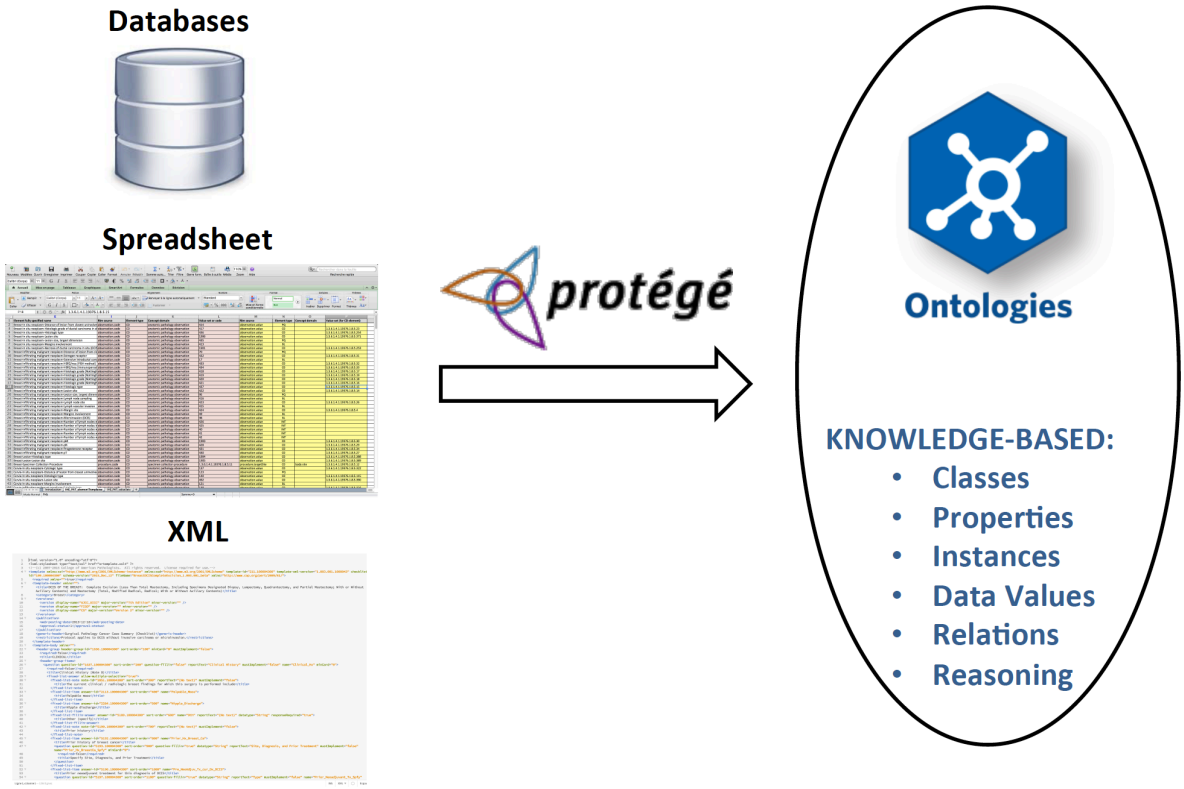


Figure 3-4: Overview of properties and functionalities of Protégé Ontology Editor and Framework

Protégé is built on plugin architecture. The main plugin used in this work was OwlViz [134] that provides a graphical representation of the ontology class hierarchy. Developers can integrate the output of Protégé with rule systems or other problem solvers to construct a wide range of intelligent systems. Protégé fully supports the latest OWL 2 Web Ontology Language and RDF specifications from the World Wide Web Consortium (W3C). Protégé is based on Java, extensible, and provides a plug-and-play environment that makes it a flexible base for rapid prototyping and application development. It is supported by a strong community of academic, government, and corporate users[132].

3.2.2.2 SKOSi [135]

SKOSi is a Simple Knowledge Organization System (SKOS) implementation tool providing a World Wide Web Consortium (W3C) SKOS entities & relations back-end solution. It is a Java RESTful web application that manages SKOS-based resources (vocabularies, terminologies, etc.). SKOSi integrates a SKOS mapping solution. It creates (from GUI), manages and shares mappings between different resources. SKOSi is not a SKOS editor. One cannot create manually concepts and hierarchy inside SKOSI dashboard yet but one can create some data inside a spreadsheet application and import inside SKOSi using given TSV column template editor. Also SKOSi provides a semantic mapping GUI interface to create

mapping (alignment) between SKOS concepts. Figure 3-5 shows an ADICAP-CIM-O alignment tables in SKOSi environment.

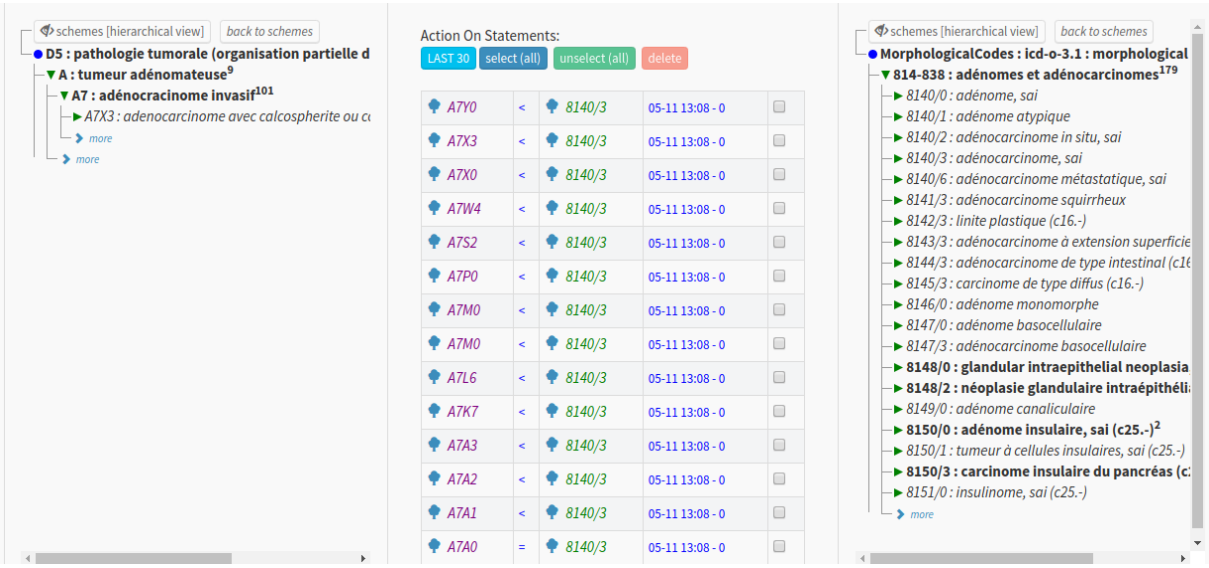


Figure 3-5: ADICAP-CIM-O alignment tables in SKOSi environment

```
<adicap:d3d8#CP> a skos:Concept ;
    skos:altLabel "pericarde"@fr , "péricarde"@fr ;
    skos:broader <adicap:d3d8#CZ> ;
    skos:definition "CP"^^xsd:string ;
    skos:inScheme <adicap:d3d8> ;
    skos:notation "CP"^^<ADICAP:D3D8:CODE> ;
    skos:prefLabel "pericardium"@en , "PERICARDE"@fr .

<adicap:d3d8#LO!S4> a skos:Concept ;
    skos:altLabel "quatrieme metatarsien"@fr , "quatrième métatarsien"@fr ;
    skos:broader <adicap:d3d8#LO> ;
    skos:definition "LO"^^xsd:string ;
    skos:historyNote "S4"^^xsd:string ;
    skos:inScheme <adicap:d3d8> ;
    skos:notation "LO!S4"^^<ADICAP:D3D8:CODE> ;
    skos:prefLabel "fourth metatarsal"@en , "QUATRIEME METATARSIEN"@fr .
```

Figure 3-6: SKOS Format of ADICAP coding system

In this work, we used SKOSi by creating data inside a spreadsheet application then import it inside using tsv column template editor. For installation, one needs Java, Tomcat and a free directory. One of the capabilities of SKOSI is to visualize datasets. After importing a file, one can navigate and search through the data. It offers a SPARQL endpoint allowing queries.

3.2.3 Anatomo-Pathology Quantifiable Features (APQF) Formalisation

Figure 3-7 shows the steps for Anatomo-Pathology Quantifiable Features (APQF) Formal representation

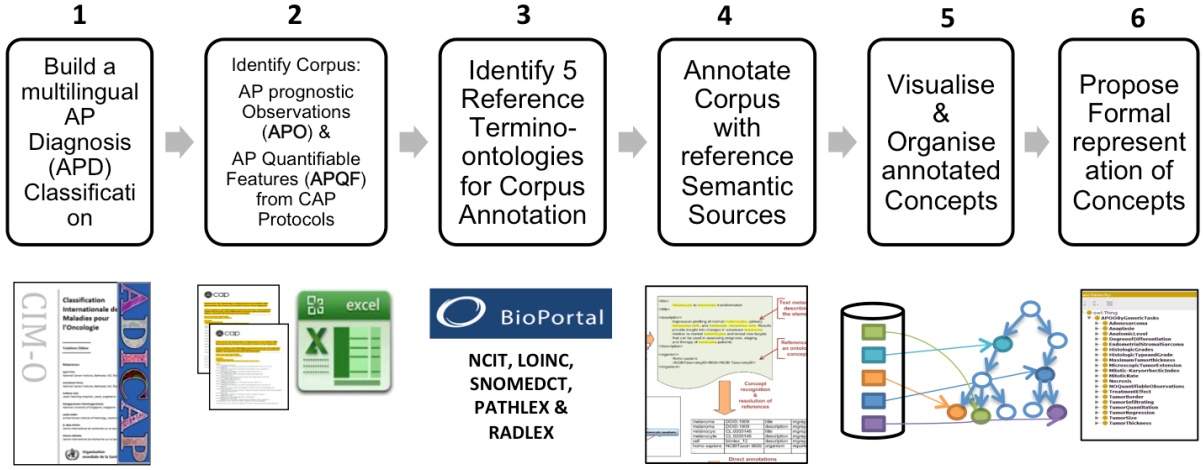


Figure 3-7: Steps for Anatomo-Pathology (AP) quantifiable parameters Formal representation

3.2.3.1 Step 1: Building a multilingual classification of AP Diagnosis (APD) of tumor pathology

At first, ADICAP's D3 (Devices, Organs, Regions) and D5 (Tumor Pathology) dictionaries were organized in an Excel spread sheet, under the supervision of Dr Christel Daniel (CD) a pathologist at the Paris Rothschild hospital (AP-HP). One of the limitations of the ADICAP coding system is related to an inconsistency of the D5 dictionary with respect to the histogenetic classification (1st character). For example: under the character B « Tumeur basocellulaire » and « Tumeur blastemateuse » are classified together whereas they belong to different histogenetic classes. As illustrated Figure 3-7, this type of inconsistency is found in B, D, M, R, S and T subclasses. For more details, please refer to Appendice 7

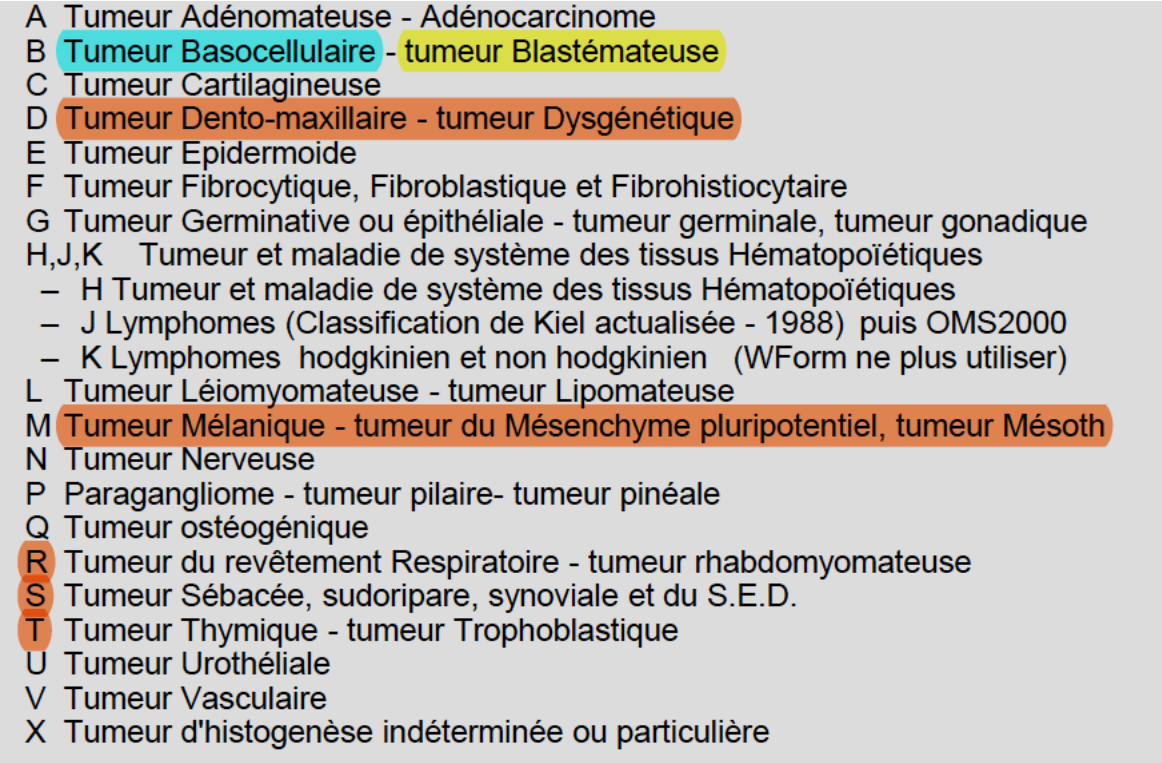


Figure 3-8: ADICAP coding system histogenetic classification

These inconsistency corrections were proposed by Dr CD. The new dictionaries were imported into SKOSi. On another hand, we used the Pubcan resource which consists of the set of AP diagnostics of tumor pathology encoded in ICD-O was used. The axes Morphology and Topography of the ICD-O correspond respectively to the dictionary D3 and D5 of ADICAP (Figure 3-1). The ADICAP-CIM-O alignment table and Pubcan resource have been formalized in SKOS and used to generate pre-coordinated ADICAP codes relevant to the field of tumor pathology. This initial step was performed in collaboration with the AP-HP INnovation Data Web Department (WIND) team. Our contribution is limited to the formalisation of the resource with SKOSi. A Java program allows creating the new resource of the relevant hierarchical codes from the resources in SKOS format.

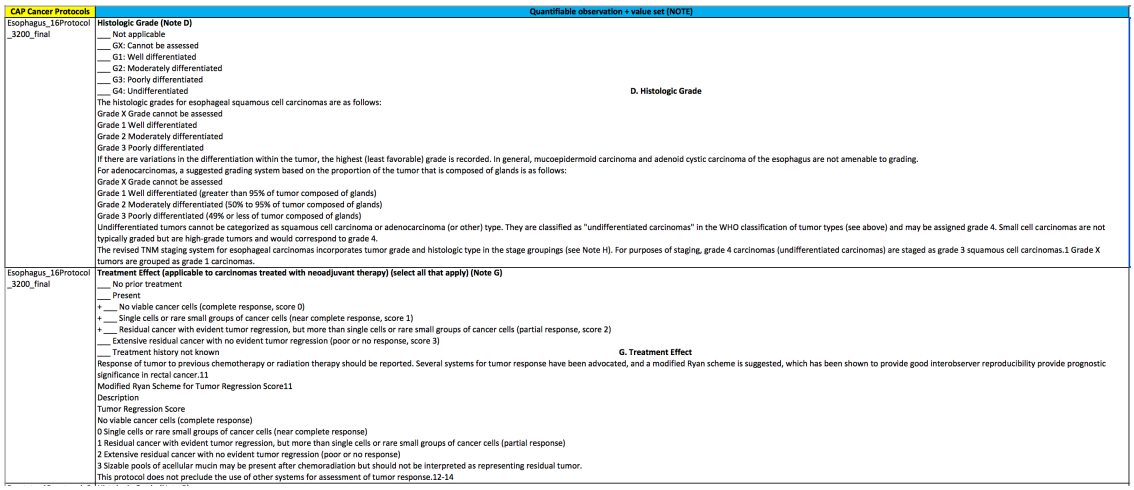
3.2.3.2 Step 2: Identification of relevant quantifiable parameters - AP prognostic Observations (APO) and AP Quantifiable Features (APQF) - from CAP protocols

3.2.3.2.1 Corpus definition:

Each CAP protocol includes a list of required cancer data elements - corresponding to AP observations – with accompanying explanatory notes (

Appendix 1.). To meet the objectives of the study, Dr CD constructed a corpus from the CAP Anatomic Pathology observations. In these protocols, we selected only those AP prognostic Observations that contributed to the evaluation of malignant tumors or the evaluation of the treatment effect and that could be the subject of a semi-automatic Quantifiable analysis and "explanatory notes" associated to these observations. These observations are called "quantifiable" AP prognostic Observations. Figure 3-9 shows an extract of the corpus, for the overall extracted corpus please refer to

Appendix 8 (“Corpus” Tab)



CAP Cancer Protocols	Quantifiable observation + value set (NOTE)
Esophagus_16Protocol_3200_final	**Histologic Grade (Note D)** ___ Not applicable ___ GX: Cannot be assessed ___ G1: Well differentiated ___ G2: Moderately differentiated ___ G3: Poorly differentiated ___ G4: Undifferentiated **D. Histologic Grade** The histologic grades for esophageal squamous cell carcinomas are as follows: Grade X Grade cannot be assessed Grade 1 Well differentiated Grade 2 Moderately differentiated Grade 3 Poorly differentiated If there are variations in the differentiation within the tumor, the highest (least favorable) grade is recorded. In general, mucoepidermoid carcinoma and adenoid cystic carcinoma of the esophagus are not amenable to grading. For adenocarcinomas, a suggested grading system based on the proportion of the tumor that is composed of glands is as follows: Grade X Grade cannot be assessed Grade 1 Well differentiated (greater than 95% of tumor composed of glands) Grade 2 Moderately differentiated (50% to 95% of tumor composed of glands) Grade 3 Poorly differentiated (49% or less of tumor composed of glands) Undifferentiated tumors cannot be categorized as squamous cell carcinoma or adenocarcinoma (or other) type. They are classified as "undifferentiated carcinomas" in the WHO classification of tumor types (see above) and may be assigned grade 4. Small cell carcinomas are not typically graded but are high-grade tumors and would correspond to grade 4. The revised TNM staging system for esophageal carcinomas incorporates tumor grade and histologic type in the stage groupings (see Note H). For purposes of staging, grade 4 carcinomas (undifferentiated carcinomas) are staged as grade 3 squamous cell carcinomas.1 Grade X tumors are grouped as grade 1 carcinomas.
Esophagus_16Protocol_3200_final	**Treatment Effect (applicable to carcinomas treated with neoadjuvant therapy) (select all that apply) (Note G)** ___ No prior treatment ___ Present +___ No viable cancer cells (complete response, score 0) +___ Single cells or rare small groups of cancer cells (near complete response, score 1) +___ Residual cancer with evident tumor regression, but more than single cells or rare small groups of cancer cells (partial response, score 2) ___ Extensive residual cancer with no evident tumor regression (poor or no response, score 3) ___ Treatment history not known **G. Treatment Effect** Response of tumor to previous chemotherapy or radiation therapy should be reported. Several systems for tumor response have been advocated, and a modified Ryan scheme is suggested, which has been shown to provide good interobserver reproducibility provide prognostic significance in rectal cancer.11 Modified Ryan Scheme for Tumor Regression Score11 Description Tumor Regression Score No viable cancer cells (complete response) 0 Single cells or rare small groups of cancer cells (near complete response) 1 Residual cancer with evident tumor regression, but more than single cells or rare small groups of cancer cells (partial response) 2 Extensive residual cancer with no evident tumor regression (poor or no response) 3 Sizable pools of acellular mucin may be present after chemoradiation but should not be interpreted as representing residual tumor. This protocol does not preclude the use of other systems for assessment of tumor response.12-14

Figure 3-9: AP Prognostic Observation Corpus extract

3.2.3.2.2 Description of the experts annotation process and the inter-experts agreement:

From the previously constructed corpus, two medical experts were asked to identify separately ("double-blind"), relevant terms corresponding to quantifiable parameters with a potential prognostic value. An annotation guideline (detailed in

Appendix 8, "Guide d'annotation" Tab) has been established and inter-expert meetings have been organized in order to clarify the role and scope of the resource to be build and to define the rules enabling the experts to identify within the corpus the terms or groups of terms corresponding to the quantifiable parameters within the corpus. A pre-filled Excel file has been created in

Appendix 8 In this file, the "Annotation-Expert" spreadsheet is structured into four main sections:

- The first column contains the title of the CAP protocols specific to each organ (context),
- The second column contains the corpus, which consists of the Quantifiable AP prognostic Observations (grade, score or treatment effects) + the corresponding note.
- The third column is intended to the expert's Term identification and includes for each expert the columns: "Observation quantifiable by the human", "Parameter quantifiable by the machine" and "Rule of quantification".

Then, the quantifiable parameters identified by the experts were reported in an Excel spreadsheet "Audit contenu" to analyse the results of both experts. When conducting annotation studies, inter-annotator agreement is used to evaluate how well or similarly annotators choose a semantic category or to determine reliability of the annotation results[7]. We first tried to find the classic Kappa (k) value. With respect to our readings [136], [137] and after discussion with an expert in this domain, we decided that (k) cannot be calculated in our case since we do not know Negative case count. Negative cases correspond to non-relevant terms/group of terms that were not annotated. The inter-annotator agreement was then assessed through an F-Measure **calculation 135], [136] according to** a, b, c **and** d **parameters presented in**

Table 3-4.

		Expert 2 annotation	
		Positive	Negative
Expert 1 annotation	Positive	a	b

	Negative	c	d(unknown)

Table 3-4: Inter-annotator agreement parameters

Since the total number of quantifiable parameters in the annotation corpus is undefined, "d" represents the number of relevant terms that none of the two experts identified and remains unknown. Thus, by temporarily considering Expert 2's results as a subject and Expert's 1 response as a gold standard, ***Precision*** can be calculated as **a / (a + b)** and ***Recall*** as **a / (a + c)**, so the F-measure will be calculated according to the following formula ***(1)*** and ***(2)***

$$F = \frac{2[a/(a+b)][a/(a+c)]}{a/(a+b)+a/(a+c)} \qquad (1)$$

$$= 2a/(2a+b+c) \qquad (2)$$

After calculating the F-measure, the answers of the two experts were merged and grouped according to the category of the quantifiable prognostic parameter, to create a gold standard, which will be used in step 4.

3.2.3.3 Step 3: Identification of reference semantic resources

3.2.3.3.1 NCBO Recommender REST APIs

In order to annotate the quantifiable parameters, we have first identified the relevant reference ontologies in the field of tumour pathology, which covers quantifiable prognostic parameters from CAP cancer protocols by considering the following reference contexts: oesophagus, prostate, melanoma, colon and breast. The NCBO platform provides Recommender [113], a service that proposes a selection of ontologies found to be relevant to a text (More details given in section 2.3, page 59]

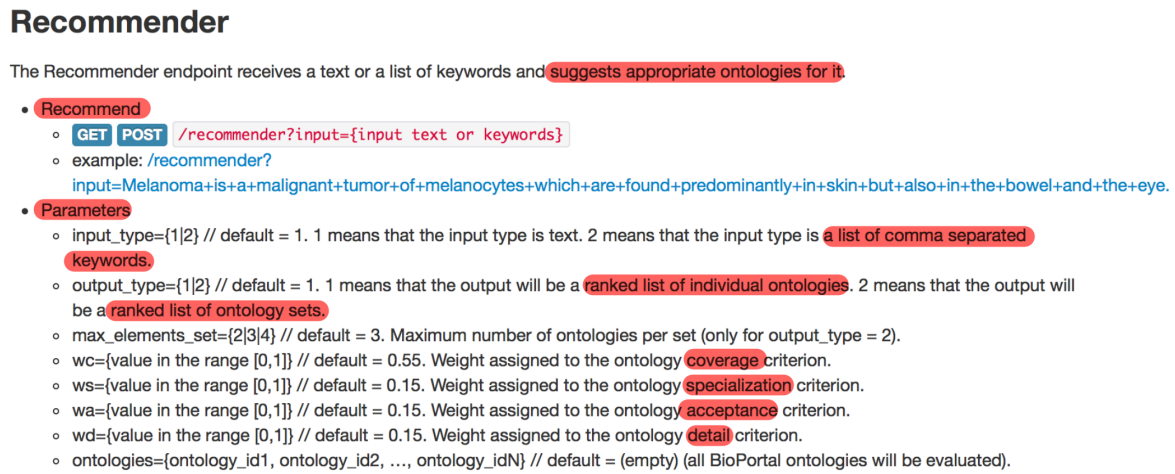


Figure 3-10: Recommender endpoint I/O specifications for a web service query

3.2.3.4 Step 4: Annotation of quantifiable parameters with existing semantic resources (BioPortal ontologies and semantic types of the UMLS)

3.2.3.4.1 NCBO Annotator REST APIs

Annotator supports the biomedical community in tagging raw texts automatically with concepts from the biomedical ontologies and terminologies hosted by BioPortal (an option is provided to annotate from a user defined subset of ontologies). The annotator web service consists of two main steps. First, direct annotations are created from raw text according to a dictionary that uses terms from a set of ontologies. Second, different components expand the first set of annotations using ontology semantics.

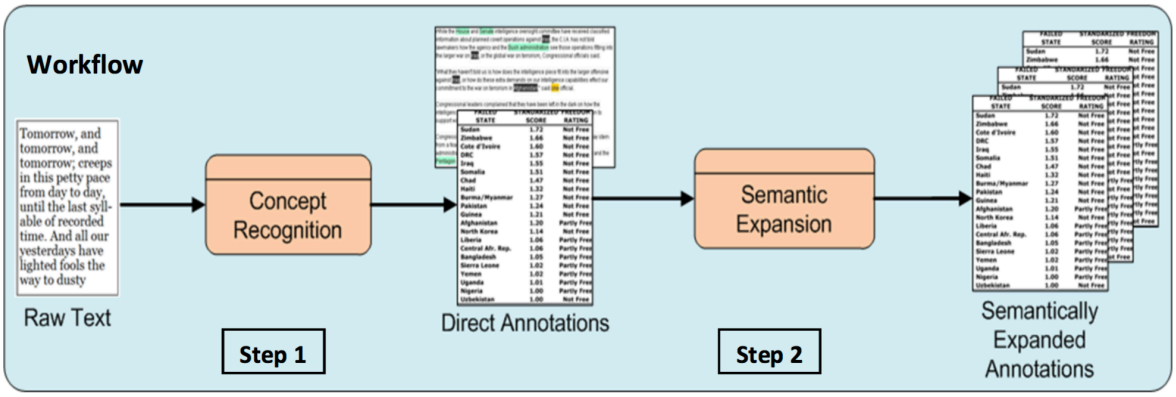


Figure 3-11: Annotator web service workflow, source [15]

A list of concepts are returned representing terms from referenced ontologies. Information related to the Class, Ontology, Type, Context, Matched Class, Matched Ontology are given. The annotated concepts can be obtained in XML and JSON format. By clicking on a concept, one can get the details of the annotation with this concept by following the link back to the concept in BioPortal. Here depending on the reference ontology, information related to the Preferred name, CUI, Synonyms, Definition, SubClass, etc. can be obtained.

3.2.3.4.2 UMLS Terminology Services

UMLS Terminology Services (UTS) provide both web interfaces as well as Web Services to search and retrieve UMLS data. In this work we used Metamap and the Metathesaurus service. Figure shows the UTS web user interface with Tab menu of associated services.

Figure 3-12: UMLS Terminology Service web user interface

3.2.3.4.3 MetaMap

MetaMap is a tool for recognizing UMLS Concepts in a text by referring to UMLS vocabulary sources. It is a highly configurable program developed at the National Library of Medicine (NLM) to map biomedical text to the UMLS Metathesaurus. We used MetaMap to complete semantic information (CUIs and STYs) of concepts that were not found in BioPortal by using the UMLS Semantic Network with its 133 semantic types and 54 semantic relationships. With MetaMap, complementary semantic information (STY, CUI, Relations, Vocabulary Sources, Definitions, etc.) needed to build our semantic model could be extracted. One should note that the STY is not available for of all concepts annotated in BioPortal. MetaMap was developed with respect to UMLS Metathesaurus and UMLS Semantic network. For more info, please refer to this Link[22]. Figure below shows the Metamap User Interface with a screenshot of the Interactive MetaMap Results of CAP note.

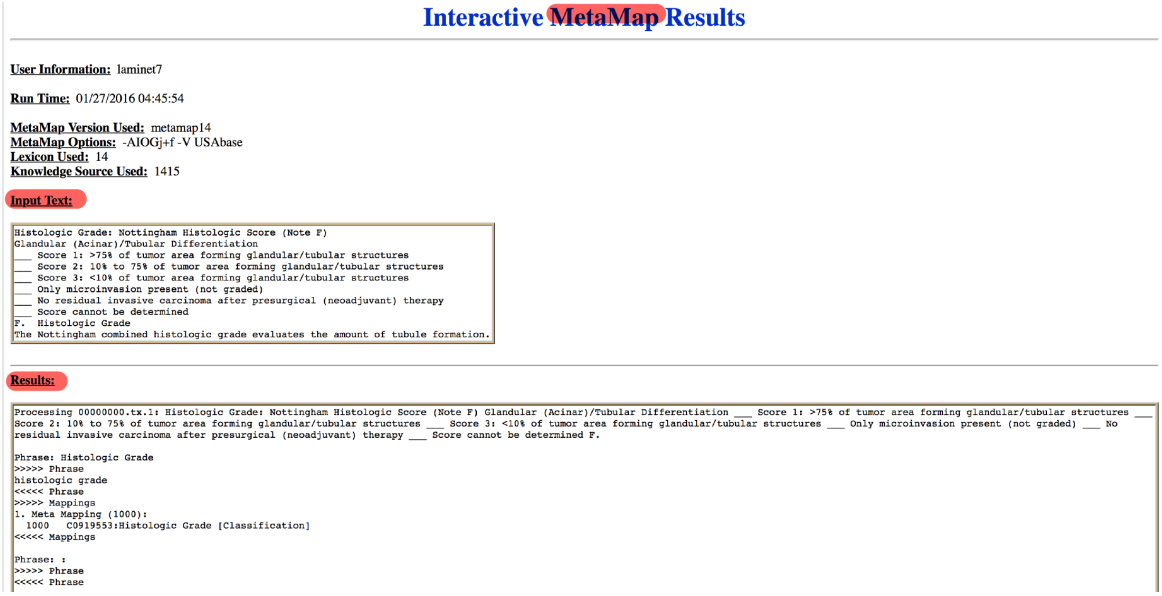


Figure 3-13: Interactive MetaMap Results of a CAP Breast cancer note example

3.2.3.4.4 Metathesaurus Browser Service for term STY, CUI retrieval

22 https://metamap.nlm.nih.gov/

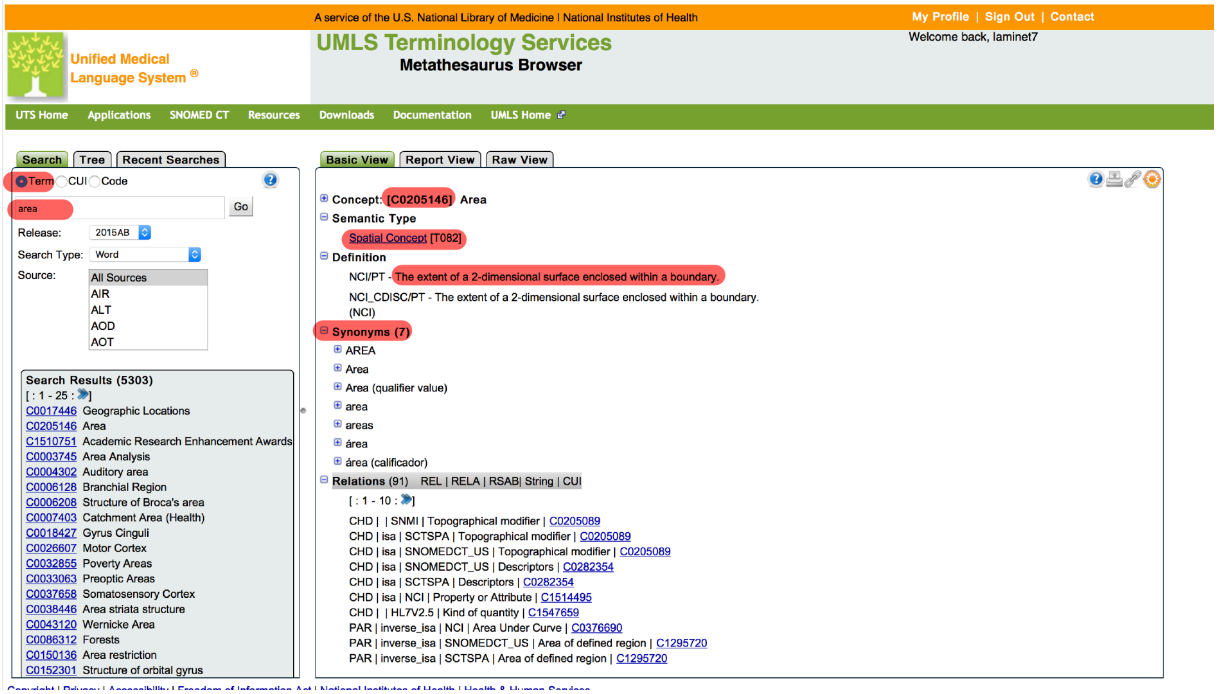


Figure 3-14: UMLS Terminology Service User Interface with result of the semantic knowledge associated to the concept « AREA »

3.2.3.4.5 Bio-YODIE

Bio-YODIE[23][138] is named entity recognition and disambiguation system that identifies various types of biomedical named entities in text and attempts to link them to the most appropriate concept label in the UMLS. Beyond the semantic annotation, Bio-Yodie handles the hierarchical disambiguation between concepts and refers to UMLS to construct updated reference resources. It is based on the General Architecture for Text Engineering (GATE) platform and offers a wide range of Output format for the metadata of annotated concepts. Features with semantic details related to each annotated concept can be obtained by clicking on it (coloured). The Figure below shows « Nuclear Pleomorphism » Features. More details can be obtained by clicking on inst_full (for the concept unique identifier CUI) and tui_full (for the semantic network)

[23] PV a LIMICS collaborator presented his work related to Bio-Yodie for the exploitation of "big data" by extraction of terms in texts.

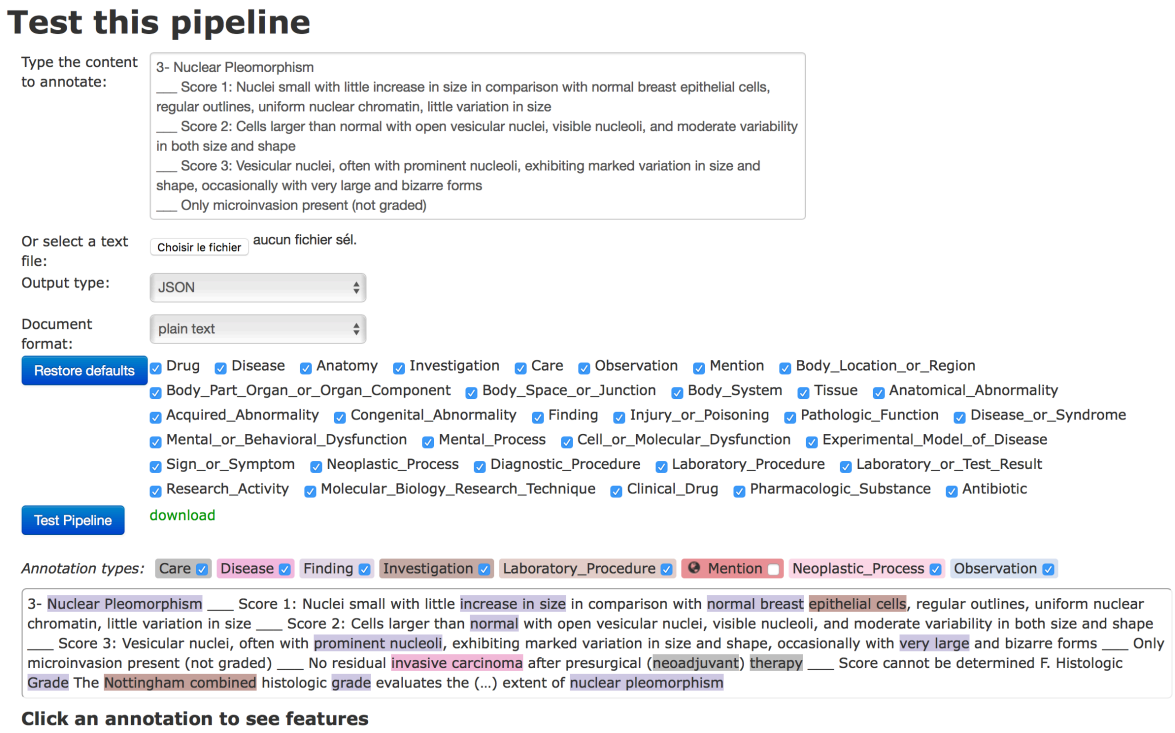


Figure 3-15: Bio-YODIE User Interface with an example of input text and possible I/O parameters

In the current implementation, since the reference ontology is DBpedia[24], the number of annotations for CAP protocols is not significant compared to Annotator and UMLS tools (MetaMap and UTS). If we have the possibility to include reference ontologies specific to our domain, we may expect more relevant annotations. For more information on the entire Bio-YODIE system refer to [138] or see the Demo[25].

- **Conceptualisation**

In this step, we transformed the relevant terms - quantifiable AP prognostic parameters - into concepts and deduced their relations (subClasse) based on the conceptual formalized organization within the reference termino-ontologies previously identified in BioPortal and UMLS.

For the extraction process, we used REST APIs of BioPortal Annotator and UMLS. We queried these web services using a Python source code. The results are in JSON format, then we used JQ to retrieve specific metadata: CUI, Semantic Type, Definition, PrefLabel, Property, Subclass.. The full program to generate annotations from REST APIs is accessible in

[24] www.wiki.dbpedia.org/services-resources/ontology

[25] http://services.gate.ac.uk/yodie/

Appendix 9

3.2.3.5 Step5: Visualization of annotated concepts and associated semantic knowledge

A graphical visualization of the semantics associated with the concepts obtained in the previous step was build using MindMaple. A conceptual model was obtained for the concepts related to the histo-prognostic evaluation of malignant tumor pathologies divided into three conceptual domains:

- AP Diagnosis (APD),
- AP prognostic Observations (APO) with parameters associated to grades, prognostic scores and treatment effect assessment.
- AP Quantifiable Features (APQF).

3.2.3.6 Step 6: Formalization of annotated concepts and associated semantic knowledge under the AP Quantifiable Features termino-ontology

The Anatomic Pathology Quantifiable Features termino-ontology integrates AP Quantifiable Features understandable by both human and software applications with all associated metadata from source ontologies. The Quantifiable Features are associated to their context of use in routine pathology as defined by the CAP protocols: specific grading/scoring/staging system currently used by pathologists in the context of specific AP diagnoses: specific tumor type (morphology) of specific organ (topography).

The objective of this step is to propose a formal representation of the obtained concepts. For this purpose, we used Protégé to organise extracted concepts within classes related to the observable entities of the prognostic potential tumor pathology (APD-diagnosis, APO-prognostic observation and APQF-quantifiable features). At last we proposed "relations" that can exist between the created classes.

3.3 Results

3.3.1 AP Diagnosis (APD) of tumor pathology

A multilingual resource of AP diagnostics in tumor pathology was constructed. This resource contains 9867 pre-coordinated ADICAP concepts (couple D3 & D5) that have been reorganized by proposing inconsistency corrections in the ADICAP D5 dictionary. A subset of 3616 concepts includes an alignment with the corresponding CIM-O pre-

coordinated **concepts.**

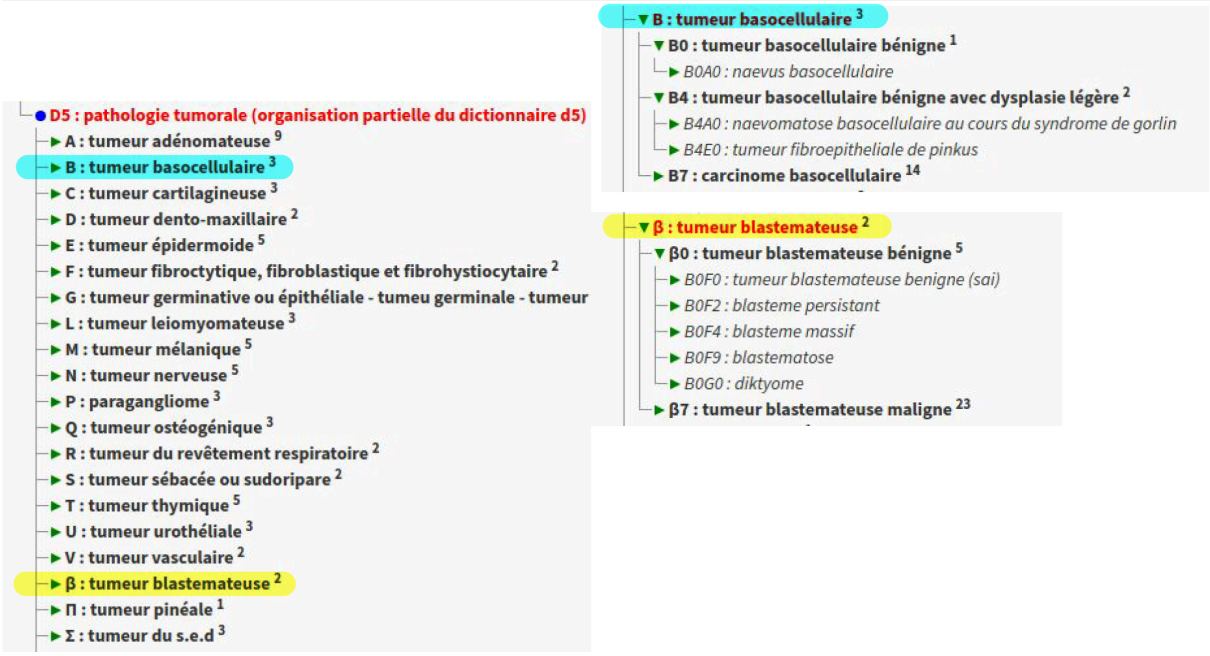


Figure 3-16 shows an example with the difference between 2 types of tumors beginning with B: « Tumeur basocellulaire » and « Tumeur blastemateuse ». This permits to classify AP diagnoses (in ADICAP) within health data warehouse to facilitate construction of queries on AP data. Figure 3-17 is a screenshot of the obtained AP Diagnosis ontology which is integrated in the AP-HP I2B2 (Informatics for Integrating Biology and the Bedside) data warehouse.

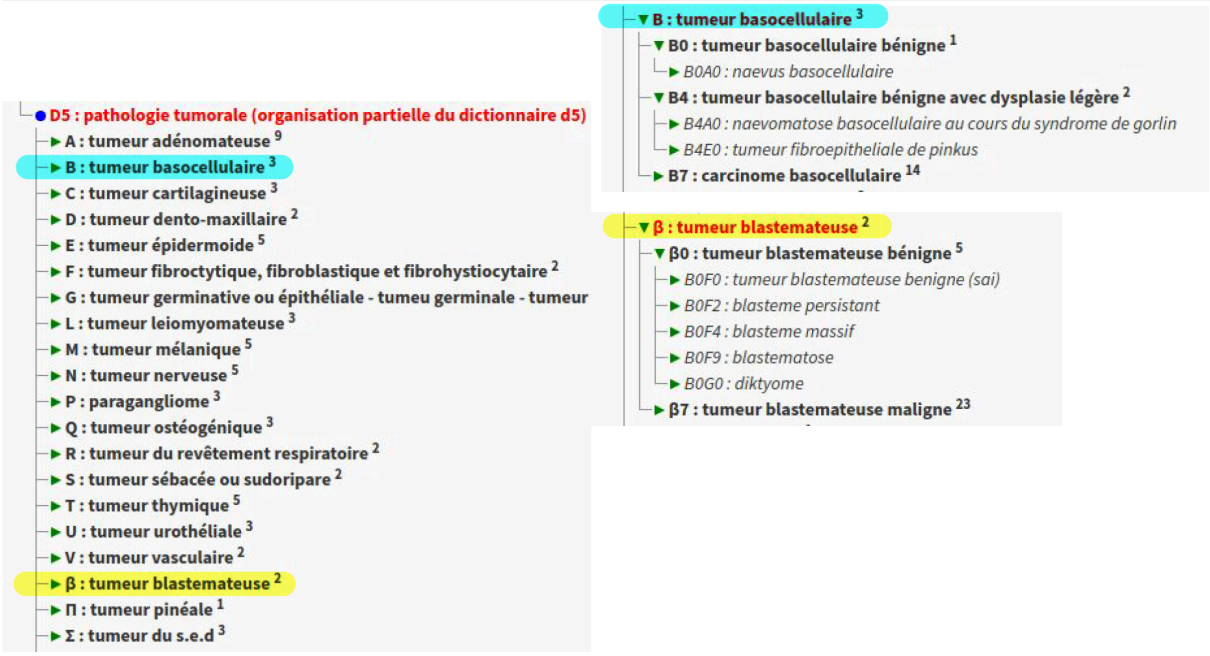


Figure 3-16: Example of an AP Diagnostics resource in tumor pathology constructed with an ADICAP CIM-O

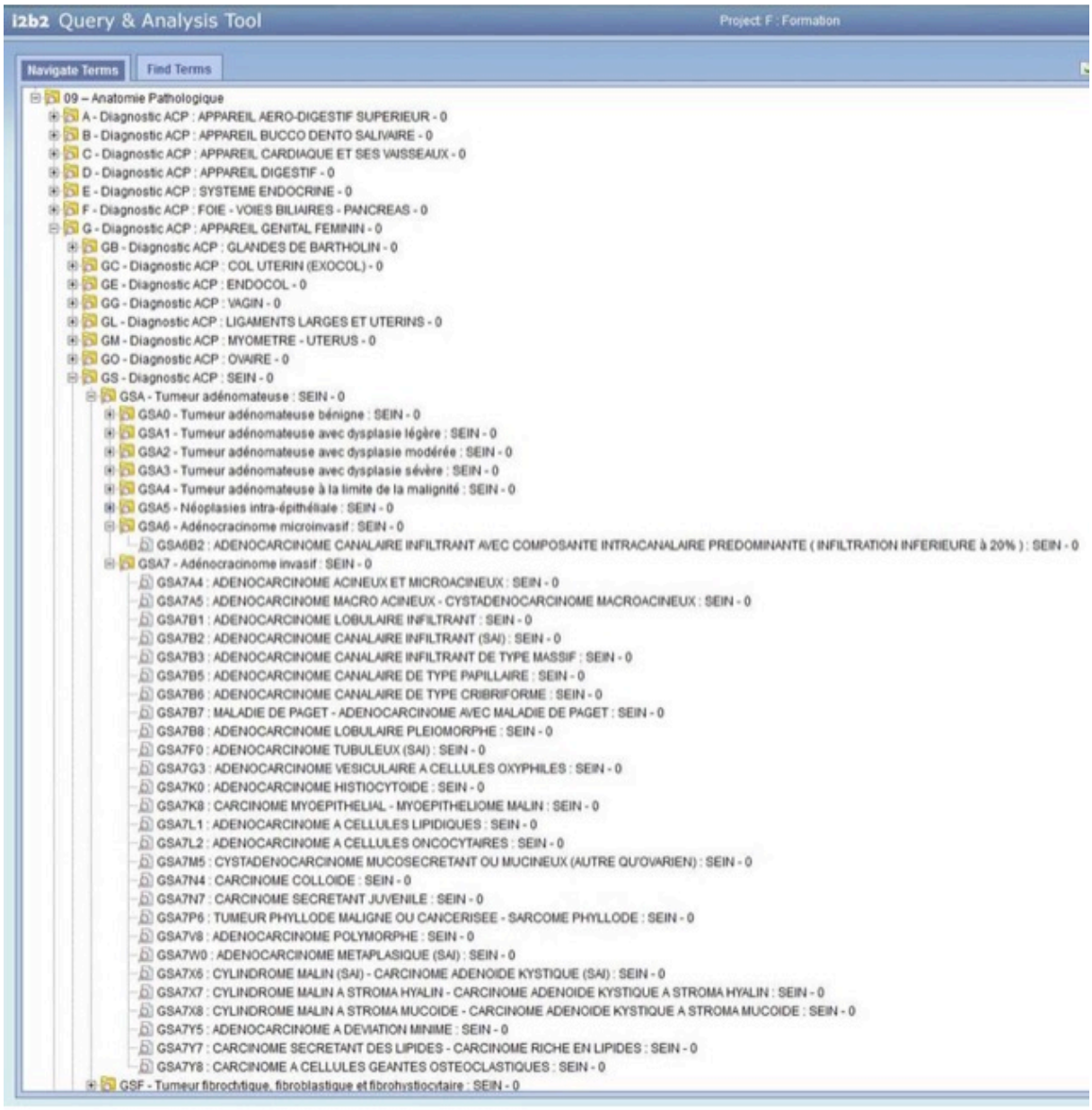


Figure 3-17: screen shot of the AP Diagnosis ontology in the AP-HP i2b2 (Informatics for Integrating Biology and the Bedside) data warehouse

3.3.2 Identification of relevant quantifiable parameters

3.3.2.1 Identified annotation corpus

The annotation corpus was constructed from 55 CAP protocols selected from 67CAP CC&P. The selection criterion is related to the relevance of their AP quantifiable prognostic observations (grade, score or treatment effect). A total of 83 "quantifiable" AP prognostic Observations were identified. The corpus consists of the observable entities; value sets and notes associated with the 83 "Quantifiable" AP prognostic Observations. Table 3-5 presents an extract of the corpus in the context of Esophagus, please refer to

Appendix 8 for the complete version.

CAP Cancer Protocols	Quantifiable observation value set (NOTE)	Expert 2 (observation quantifiable par l'humain)	Expert 2 (parametre quantifiable par la machine*)
Esophagus_16Protocol_3200_final	Histologic Grade (Note D) Not applicable Grade X Cannot be assessed Grade 1 Well differentiated Grade 2 Moderately differentiated Grade 3 Poorly differentiated Grade 4 Undifferentiated D. Histologic Grade The histologic grades for esophageal squamous cell carcinomas are as follows: Grade X Grade cannot be assessed Grade 1 Well differentiated Grade 2 Moderately differentiated Grade 3 Poorly differentiated If there are variations in the differentiation within the tumor, the highest (least favorable) grade is recorded. In general, mucoepidermoid carcinoma and adenoid cystic carcinoma of the esophagus are not amenable to grading. For adenocarcinomas, a suggested grading system based on the proportion of the tumor that is composed of glands is as follows: Grade X Grade cannot be assessed Grade 1 Well differentiated (greater than 95% of tumor composed of glands) Grade 2 Moderately differentiated (50% to 95% of tumor composed of glands) Grade 3 Poorly differentiated (49% or less of tumor composed of glands) Undifferentiated tumors cannot be categorized as squamous cell carcinoma or adenocarcinoma (or other) type. They are classified as "undifferentiated carcinomas" in the WHO classification of tumor types (see above) and may be assigned grade 4. Small cell carcinomas are not typically graded but are high-grade tumors and would correspond to grade 4. The revised TNM staging system for esophageal carcinomas incorporates tumor grade and histologic type in the stage groupings (see Note H). For purposes of staging, grade 4 carcinomas (undifferentiated carcinomas) are staged as grade 3 squamous cell carcinomas.1 Grade X tumors are grouped as grade 1 carcinomas.	Differentiation carcinoma Grade X Grade cannot be assessed Grade 1 Well differentiated Grade 2 Moderately differentiated Grade 3 Poorly differentiated Grade 4 Undifferentiated carcinomas Proportion of glands formation*	proportion of gland formation
Esophagus_16Protocol_3200_final	Treatment Effect (applicable to carcinomas treated with neoadjuvant therapy) (select all that apply) (Note G) No prior treatment Present No viable cancer cells (complete response, score 0) Single cells or rare small groups of cancer cells (near complete response, score 1) Residual cancer with evident tumor regression, but more than single cells or rare small groups of cancer cells (partial response, score 2) Extensive residual cancer with no evident tumor regression (poor or no response, score 3) Treatment history not known G. Treatment Effect Response of tumor to previous chemotherapy or radiation therapy should be reported. Several systems for tumor response have been advocated, and a modified Ryan scheme is suggested, which has been shown to provide good interobserver reproducibility provide prognostic significance in rectal cancer.11 Modified Ryan Scheme for Tumor Regression Score11 Description Tumor Regression Score	score 0 complete response score 1 near complete response score 2 partial response score 3 poor or no response Treatment history not known Count of viable cancer cells* Ryan scheme Residual tumor	Number of viable cancer cells

Table 3-5: CAP CC&P, notes and associated 83 "Quantifiable" AP prognostic Observations

3.3.2.2 Experts identification results

Table below summarizes the results of relevant terms and groups of terms identified by the 2 medical experts. Expert 1 annotated about 103 relevant group of terms, 32 of them were repeated more than once in the final list.

Total Number (#) of	Expert 1	Expert 2
Relevant terms	11	11
Relevant Group of terms	103	92
Redundant terms	04	03
Redundant Group of terms	32	25

Table 3-6: Expert 1 and 2 corpus annotation result of relevant terms and groups of terms

3.3.2.3 Inter-expert agreement analysis

The F-measure was calculated by considering all terms / groups of terms identified by the experts as shown in Table 3-7. It is estimated to be 76%, with a recall rate of 81% and an accuracy rate of 71 %.

		Expert 2 annotation	
		Positive	Negative
Expert 1 annotation	Positive	82	33
	Negative	19	d (unknown)

Table 3-7: Agreement between the two experts

3.3.2.4 Relevant terms and group of terms

The two medical experts merged the relevant terms / groups of terms identified to consensually constitute a list of 91 terms / groups of terms corresponding to quantifiable prognostic parameters and grouped them into the following 18 categories: Architecture pattern, Biomarker, Cell, Cell shape, Cell size, Cell/Architecture, Cell/Nucleus, Chromatin,

Cytoplasm, Architecture pattern/Gland, Invasion of anatomic structure, Necrosis, Nucleoli, Nucleus, Nucleus/Cytoplasm, Tumor, Tumor size, Vessel and Other. The Table below is an extract from the 91 Terms. It gives a preview of the identified quantifiable prognostic parameters (Identification column) and possible synonyms.

Category	Identification	Synonymes
Architecture pattern	Percent of solid growth (eg, cystic, solid, neurotropism)	Percent of nonsquamous solid growth pattern
Architecture pattern/Gland	Percent of glandular differenciation in the tumor	Glandular differentiation, Percent of glandular and papillary differentation of the tumor
Cell	Degree of anaplasia	Presence of anaplastic cells (solid sheets or irregular small), proportion of anaplastic cells
Cell shape	Round cell-epithelioid appearance	
Cell size	Small cell undifferentiated or rhabdoid as the predominant or sole histopathologic subtype	
Cell/Architecture	Differentiation	
Cell/Nucleus	Pleoporphism	
Chromatin	Aspect of chromatine	
Cytoplasm	Differentiation of the cytoplasme	
Invasion of anatomic structure	Vascular invasion	
Necrosis	Percent of necrosis	Percent of Extent necrosis, percent of tumor that is necrotic, tumor necrosis, extent of postchemotherapy necrosis in total specimen
Nucleoli	Nucleoli size	
Nucleus	Mitotic count	mitotic count, Mitotic rate, mitoses per HPF
Nucleus/Cytoplasm	Nuclear-to-cytoplasmic ratio	
Other	Sarcomatous stroma	
Cell	Presence of brisk tumor-Infiltrating Lymphocytes (TILs)	Number of Tumor-Infiltrating Lymphocytes
Tumor	Amount of residual cancer	Percent residual viable tumor
Tumor size	Percent of tissue involved by tumor	Proportion of prostatic tissue involved by tumor
Vessel	Basophilic staining of tumor vessels	

Table 3-8: Extract from the list of 91 terms corresponding to quantifiable prognostic parameters grouped into 18 categories

3.3.3 Validation of reference termino-ontologies

Table 3-9 summarizes the results of the identification of the 5 BioPortal ontologies offering the best coverage rate for the AP quantifiable features, based on terms / groups of terms derived from 5 reference CAP protocols.

Gold Standard		Coverage Percent (%) of the Source Ontologies				
Body Part /Cancer	**Number (#) of concepts**	**NCIT**	**SNOMED CT**	**LOINC**	**RADLEX**	**PATHLEX**
Colon & rectum	6	94%	48%	39%	38%	31%
Œsophage	20	75%	41%	51%	28%	17%
Prostate	13	85%	61%	49%	7%	27%
Breast	66	70%	52%	54%	26%	15%
Melanoma	19	66%	51%	51%	22%	9%
Average coverage per Ontology		**78%**	**50%**	**49%**	**24%**	**20%**

Table 3-9: Results of the identification of the 5 BioPortal ontologies offering the best coverage rate for the AP quantifiable features derived from 5 reference CAP protocols

3.3.4 Conceptualisation: transforming terms to relevant concepts

In total, we have 91 relevant Terms and Groups of Terms. We tried to find appropriate codes of these terms from existing semantic resources by considering the following three cases:

i) the meaning of the Term is fully represented by a single code (Concept Unique Identifier)

ii) the meaning of the Term is represented by several codes

iii) the meaning of the Term does not exist at all

The main purpose of this semantic annotation is the selection of appropriate identifier codes that cover as much as possible the meaning of the Terms and/or Group of Terms identified by experts.

For example: "acinar and papillary", BioPortal proposes to annotate "acinar pattern" apart and "papillary pattern" on the other hand so we will have two different CUI codes. Both are considered concepts that represent the meaning of the "Concept" identified by the experts.

Liste des termes (groupe) pertinents (90)	NCIT					SNOMED CT			LOINC			PathLex			RadLex
	Preferred Name	prefixIRI /Code	UMLS_CUI	Semantic Type	subClassOf	CUI	TUI	subClassOf	CUI	TUI	SubClassOf			subClassOf	SubClassOf
Acinar and papillary patterns	Acinar pattern	C35922	C1332138	Finding	Morphologic Architectural Pattern	C0332207	T082	General adjectival modifier Morphological descriptors	/	/	/	/	/	Not classified	carcinoma
	Papillary pattern	C35911	C1335326	Finding	Morphologic Architectural Pattern	C0205312	T080	General adjectival modifier Appearances	C0205312	T080	/	/	/	Not classified	ependymoma WHO grade II
ADENOCARCINOMA	Adenocarcinoma	C2852	C0001418	Neoplastic Process	Glandular Cell Neoplasm	C0001418	T191	Primary malignant adenomatous neoplasm	C0001418	T191					
ADENOID CYSTIC CARCINOMA	Adenoid Cystic Carcir	C2970	C0010606	Neoplastic Process	carcinoma	C0010606	T191	Primary malignant adenomatous neoplasm							
Aggressive growths in other parts of the neoplasm	Aggressive		C0580822	Qualitative Concept	Disease Clinical Qualifier										
	growth		C0220844	Spatial Concept	Action										
	neoplasm	C3262	C0375111	Neoplastic Process	Disease or Disorder	C0027651	T191	http://purl.bioontology.org/ontology/SNOM							prolifiration

Table 3-10: Extract of the conceptualized terms with their appropriate codes and metadata

Number (#) of Concepts issued from	Number of concepts	Percentage (%)
NCIT	203	45.11%
SNOMED CT	112	24.88%
LOINC	63	14%
RADLEX	64	14.22%
PATHLEX	8	1.77%
TOTAL	450	-

Table 3-11: AP Quantifiable features Categorization By Reference termino-ontologies

3.3.5 Concept visualisation of quantifiable parameters in the context of Breast Cancer

The semantic visual representation built with Mindmaple is illustrated in Figure 3-18, Figure 3-19 and Figure 3-20 and for the quantifiable parameters "Nuclear Pleomorphism", "Percent of glandular differentiation" and "Mitotic Count" from the Nottingham Grading System used in the evaluation Prognosis of invasive breast cancer.

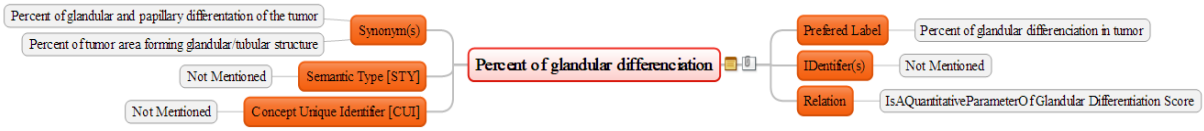


Figure 3-18: Semantic visual representation of «Percent of glandular differentiation» concept

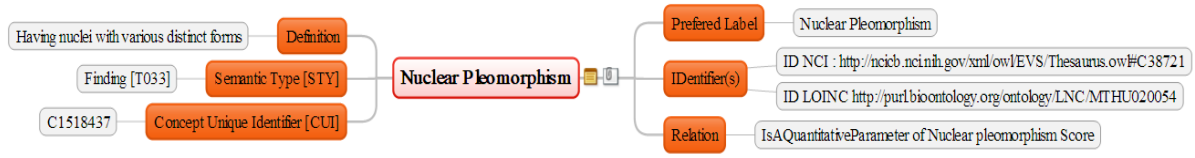


Figure 3-19: Semantic visual representation of «Nuclear Pleomorphism»concept

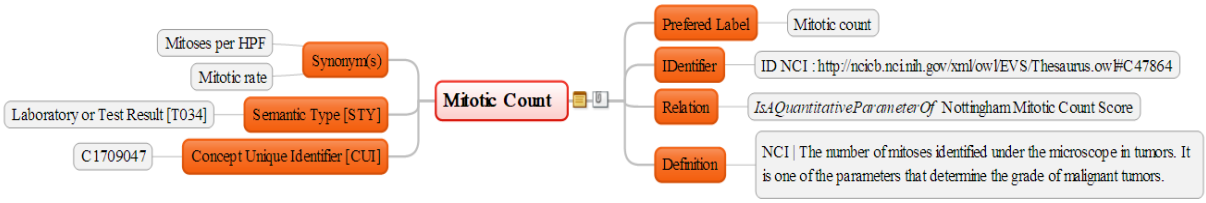


Figure 3-20: Semantic visual representation of «Mitotic Count» concept

Table 3-12 summarizes the conceptual domains of AP Diagnostics (APD), AP prognostic Observation (APO) and AP Quantifiable Features (APQF) in the prognostic assessment of breast cancer with possible relations between the different Classes.

AP Diagnostic (APD)		**AP prognostic Observations (Grades, Prognostics scores, Treatment Effect evaluation) (APO)**					**AP Quantitative Features (APQF)**	
Class	**Relation**	**Class**	**Description**	**Relation**	**Class**	**Description**	**Relation**	**Class**
Breast Invasive Carcinoma	*hasObservation*	**Nottingham Grade**	The Nottingham combined histologic grade evaluates the amount of tubule formation, the extent of nuclear pleomorphism, and the mitotic count. Grade 1 (scores of 3, 4, or 5) Grade 2 (scores of 6 or 7) Grade 3 (scores of 8 or 9)	*hasObservationComponent*	**Nottingham Glandular Differentiation Score**	Score 1 (>75% of tumor area forming glandular/tubular structures) Score 2 (10% to 75% of tumor area forming glandular/tubular structures) Score 3 (<10% of tumor area forming glandular/tubular structures)	*hasQuantitativeFeature*	Percent of glandular differentiation
					Nottingham MitoticCount Score	Score 1 (≤3 mitoses per mm) Score 2 (4-7 mitoses per mm) Score 3 (≥8 mitoses per mm)	*hasQuantitativeFeature*	Mitotic count
				hasObservationComponent	**Nottingham NuclearPleomorphism Score**	Score 1 (nuclei small with little increase in size in comparison with normal breast epithelial cells, regular outlines, uniform nuclear chromatin, little variation in size) Score 2 (cells larger than normal with open vesicular nuclei, visible nucleoli, and moderate variability in both size and shape) Score 3 (vesicular nuclei, often with prominent nucleoli, exhibiting marked variation in size and shape, occasionally with very large and bizarre forms)	*hasQuantitativeFeature*	NuclearPleomorphism
								NuclearSize
								NuclearShape
								Nuclear-to-cytoplasmic ratio
								NuclearColor

Table 3-12: Results of the formalization of concepts related to the Nottingham Grading System used in the prognostic evaluation of breast cancer

3.3.6 APQF formal representation proposal using Protégé (AP Skeleton and hierarchy of AP Quantifiable Features)

3.3.6.1 APQF in a Context Specific Approach: Breast Invasive Carcinoma

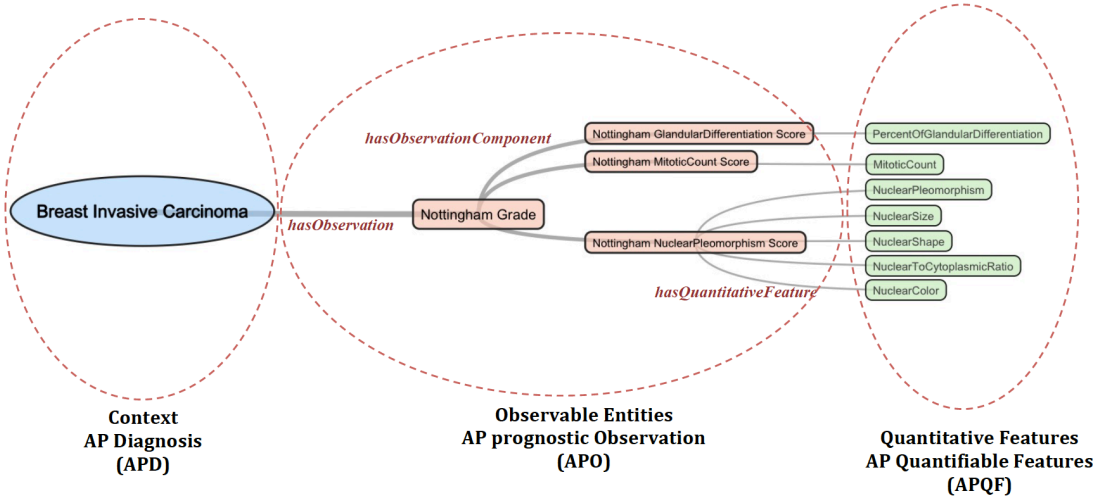


Figure 3-21:

3.3.6.2 APQF in a Context specific and Generic Approach

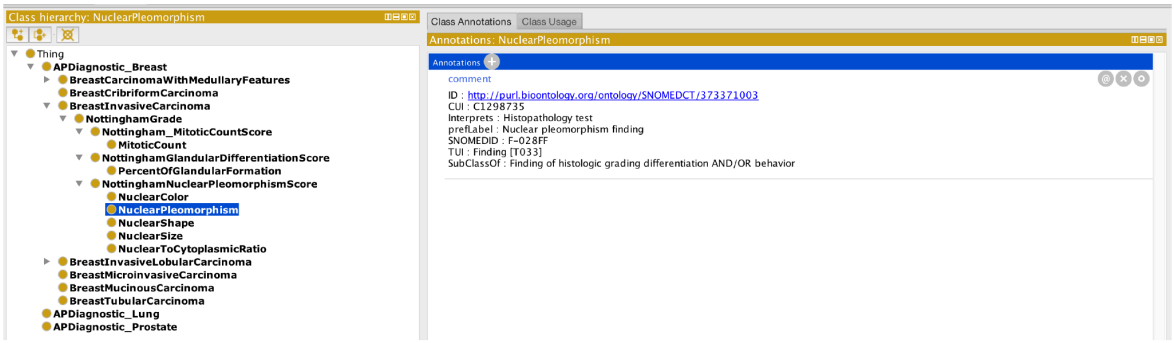


Figure 3-22: proposal of a hierarchical organization of AP Quantifiable features taking into account the Breast AP diagnostic context

We proposed a hierarchical organization of AP Quantifiable features (APQF) identified by the experts in the CAP CC&P taking into account the context i.e. AP diagnosis (tumor location and histological type) and AP Observable entity (grading/scoring system). Each APQF is associated to its corresponding definition from different sources and associated metadata (CUIs, UMLS STYs).

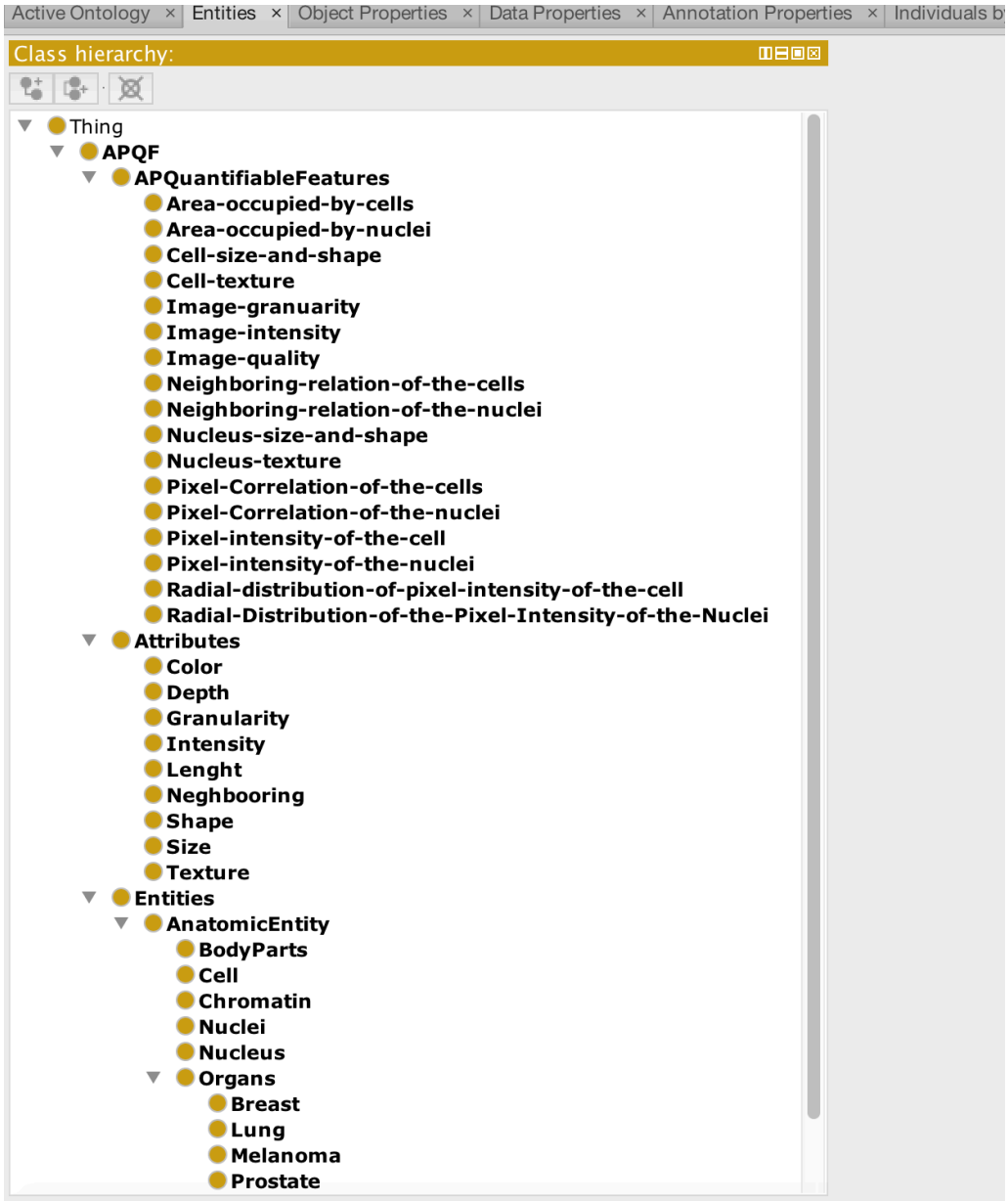


Figure 3-23: proposal of an organ independent hierarchical organization of APQF taking into account generic quantifiable features

Ontologies provide domain knowledge to drive data annotation, data integration, information retrieval, natural language processing and decision support [12] . As the number of large data sets are growing, providing a framework for data analysis and data integration using ontologies continues to be of critical importance [139], [140]

3.4 Discussion

3.4.1 Significance and comparison with related work

3.4.1.1 What was already known on the topic?

The CAP CC&P is a very valuable knowledge source about cancer grading/scoring including quantifiable observable entities of prognostic value for the most common cancers.

CAP protocols include explanatory notes describing the Quantifiable features of prognostic value that could be measured in AP images.
Standard Development Organizations (SDOs) such as HL7 or DICOM and international initiative like IHE (Integrating the Healthcare Enterprise) provide formal models representing high-level AP observations required in cancer AP reports.
Additional efforts are needed to achieve a workable standard-based formal representation of histopathological knowledge integrating both observable entities reported by humans (APO) (pathologists) and Quantifiable features (APQF) automatically computed by machines.

3.4.1.2 What this study added to our knowledge?

The CAP CC&P were used to build a formal representation of AP Quantifiable features (APQF). 167 quantifiable observable entities of prognostic value were defined within 55 out of 67 CAP CC&P. A list of 91 AP Quantifiable Features (APQF) were identified by two medical experts from the CAPCC&P explanatory notes. Inter-experts agreement, varied in the identification of terms /group of terms of Quantifiable parameters with an F-measure= 76%
We proposed a semi-automated workflow for selecting candidate ontologies/semantic sources for semantic annotation of textual documents in a given domain. This workflow was applied on the AP Quantifiable Features (APQF). Five reference ontologies/semantic sources were identified as the most relevant candidate to annotate the CAP CC&P notes and used in the process of building a formal representation of APQF. SNOMEDCT NCIT and LOINC cover about the half of the corpus. Radlex covers 14,22% of the corpus corresponding to generic terms for shape and dimensions. Pathlex, which do not include low-level morphological features, covers, not surprisingly only 1,7% of the corpus.
The proposed tool semantic visualisation tool model and the formal representation based on the CAP-CC&Ps, aim at supporting AP experts in building a standard-based representation of low-level morphological abnormalities observed in cancer that can be quantified using image analysis tools.
A formal model of Anatomic Pathology Quantifiable Features (APQF) is proposed. APQF are organized by feature categories and defined in the context of each organ specific grade/score system. The APQF model provides to the Image Analysis community a list of coded Quantifiable features associated to their context of use. These Quantifiable features are candidate parameters for building survival prediction methods based on image analysis.
This effort is complementary to the Integrating the Healthcare Enterprise (IHE) initiative building a standard-based representation of high-level AP observations required in cancer AP reports. Additional efforts are needed to achieve a workable standard-based formal representation of histopathological knowledge integrating both observable entities reported by humans (pathologists) and quantifiable entities automatically computed by machines.
Providing such unique formal representation contributes to more efficient use of computer aided diagnosis based on automatic analysis of whole slide images (WSI).

3.4.2 Limitations and perspectives

The definition of the AP Quantifiable features only covers the scope of 55 out of 67 CAP CC&P. Despite their great importance in prognostic evaluation, the data elements (observable entities) defined by CAP Cancer Biomarker Reporting Templates were not considered at this stage.
The formal model of APQF proposed in this section takes into account existing reference ontologies or semantic resources such as SNOMED CT, NCIT, RadLex, PathLex. Although a

detailed guide for semantic annotation of AP Quantifiable Features (APQF) was provided to the medical experts, they usually failed to find one to one mapping between APQF terms and concepts in the reference ontologies. In rare cases, a single code covered the meaning of the term (e.g.: Nuclear Pleomorphism Score, CUI [C1299478]). Usually, the term or group of terms corresponding to the Quantifiable feature was associated to more than one concept (e.g.; Percent [C48570] of Glandular [C0458095] differentiation [not mentioned]) or no corresponding concept could be found (E.g.; Mitotic-Karyorrhectic Index (MKI [CUI : not mentioned])

The current proposed model includes relevant terms corresponding to the various features defining the grades and scores of tumors. It provides a sustainable formal representation of the knowledge involved during the AP diagnostic process. Extending the scope of such resource would benefit from the involvement of an international consortium of pathologists provided with supportive tools enabling community members to contribute terminological content and provide feedback on existing classes and properties.

PART 3

Image Analysis Knowledge Formal representation

Image Analysis Knowledge Formal representation

Main questions

- What are the existing histopathology image analysis methods?
- What are the existing formal models for representing Practical Image Processing Tasks?
- How to build a terminological system for Practical Image Processing Task used in AP, taking into account the existing formal models?
- How to integrate the two termino-ontologies Anatomic Pathology Quantitative Features Ontology (APQFO) and Practical Image Processing Tasks (PIPTO)

In this section, we first present image analysis techniques for histopathological slide processing and computer aided diagnosis. Our description is mirrored to the general image analysis workflow for histopathological imaging in AP laboratory. We start by describing the preparation of histopathology slides for microscopic analysis. Then, we considered common image analysis methods with a focus on segmentation, feature extraction, and classification. When appropriate, we give examples from the literature in the context of cancer diagnosis and prognostic assessment (grading/scoring).
Then, we propose an approach for «Bridging the semantic gap between diagnostic histopathology and image analysis». This consists of the subsequent steps:

i) To identify effective histopathology imaging methods highlighted by recent Digital Pathology (DP) contests. Then to identify associated formalized knowledge in NCBO Bioportal and within the UMLS metathesaurus,

ii) To formalize biomedical-imaging processing knowledge sources issued from major software's (MATLAB, ITK, ImageJ) and from histopathology image analysis surveys

iii) To link relevant quantifiable observations in Anatomic Pathology Quantitative Features Ontology (APQFO) from histopathology domain to generic image analysis tasks in Practical Image Processing Tasks (PIPTO) in imaging domain).

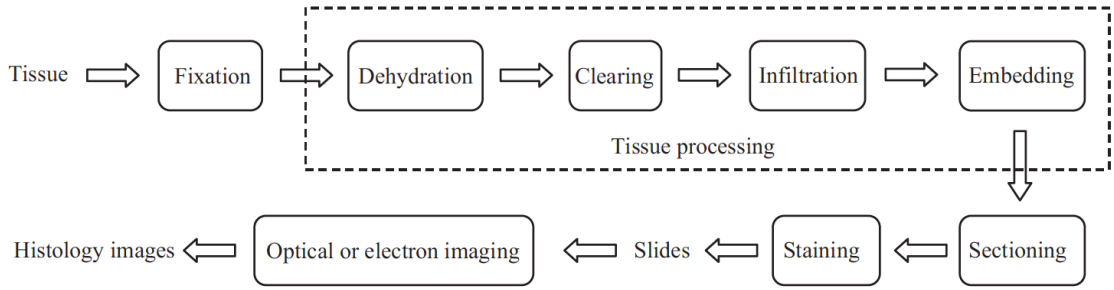
4 Image analysis in histopathology: digital pathology imaging modalities and image processing techniques

4.1 Introduction

Digital image analysis is defined by the College of American Pathologists as "*the computer-assisted detection or quantification of specific features in an image following enhancement and processing of that image, including IHC, DNA analysis, morphometric analysis and FISH*"[26][27]. Over 50 years ago JM Prewitt et al. wrote first papers on the use of computerized image analysis of cell images [141], [142]. Interestingly their most significant work, the Prewitt edge operator [143], was initially showcased in a 1965 paper on the morphological analysis of cells and chromosomes [141].
A search for "image analysis" and "digital pathology" on PubMed revealed over 1700 articles[28] just in the last decade with more than 90 review studies [34], [144]–[146]. While a number of these are application papers focusing on the use of computational image analysis tools to address specific targeted problems in digital pathology such as *quantifying specific biomarkers*, color normalization methods; "*a number of recent papers are focused on developing computer-assisted Digital Image Analysis for whole slide images (WSI)*" [147]
In the course of recent years, AP laboratories progressively follow up a change toward a digital workflow. This includes the digitization of histopathology slides and the use of computer monitors to visualize WSI, instead of virtual microscope. The use of image analysis quantification methods is considered as an answer to tackle the need of efficiency and also quality – reducing inter-observer variability during interpretation of histopathology slides: « image analysis methods have a great potential to reduce the workload in an AP laboratory and to improve the quality of interpretation. » [148], [4]

4.2 Histopathology slides preparation procedures

Histopathology is the study of a biopsy specimen by a pathologist under microscope for locating, analysing and classifying diseases like cancer [149]. To examine different architectures and components of a tissue, several preparation steps of the histopathology samples are performed. Figure 4-1 shows the preparation process.



[27] CAP Guidelines for Digital Image Analysis - 2013
[28] Pubmed search on 20/07/2017

Figure 4-1 Steps for preparation of histopathology slides, source: [149]

4.2.1 Biopsy Fixation

Samples of biological tissue are « fixed » with chemical fixation to preserve the cell or tissue.

4.2.2 Tissue processing

It consists of removing water from the gross tissue (dehydration) and replaces it with a medium, which solidifies it. This help to cut thin sections of sample. The result of embedding hardened wax blocks contains the original biological samples together with other substances in complete preparation process

4.2.3 Sectioning

Consist of producing sufficiently thin slices of sample that the detail of microstructure of the cell /tissue can be clearly observed using microscopy techniques. Then transfer the thin cut of sample on to a clean glass slide.

4.2.4 Staining

Staining is used to separate cellular components for structural as well as architectural analysis for diagnosis. Most commonly Haematoxylin and Eosin (H&E) stain is used to separate cell nuclei, cytoplasm and connective tissue. Haematoxylin stains cell nuclei blue, whereas Eosin stains cytoplasm and connective tissue pink. Other stain examples are DAB[29], immune-histochemistry stain (IHC) etc. [150], [151]

IHC is a more advanced staining technique, which makes use of antibodies to highlight specific antigens in the tissue. A useful characteristic of IHC digital slides is the determination of the percentage of pixels positively stained for a particular antigen.

In breast cancer, IHC is commonly used to highlight the presence of oestrogen (ER), progesterone (PR), and human epidermal growth factor 2 (HER2) receptors, as well as to assess the proliferation of the tumour, for example, by highlighting the Ki-67 protein, which is associated with cell proliferation.

« In contrast to H&E, most of the information that is of interest in IHC-stained sections is contained in the colour and the intensity of the staining, which makes IHC-stained samples easier to design and implement image processing algorithms on. »

4.3 Overview of conventional histopathological image analysis techniques

A typical CAD system for histology image analysis is shown in Figure 4-2. This system consists of conventional image processing and analysis tools, including pre-processing, image segmentation, feature extraction, feature dimension reduction, feature-based classification, and postprocessing.

[29] DAB (3,3'-diaminobenzidine) is an organic compound that is both chemically and thermodynamically stable. It is oxidized in the presence of peroxidase and hydrogen peroxide resulting in a dark brown réaction Product. DAB has been used in immunohistochemical staining of nucleic acids and proteins.

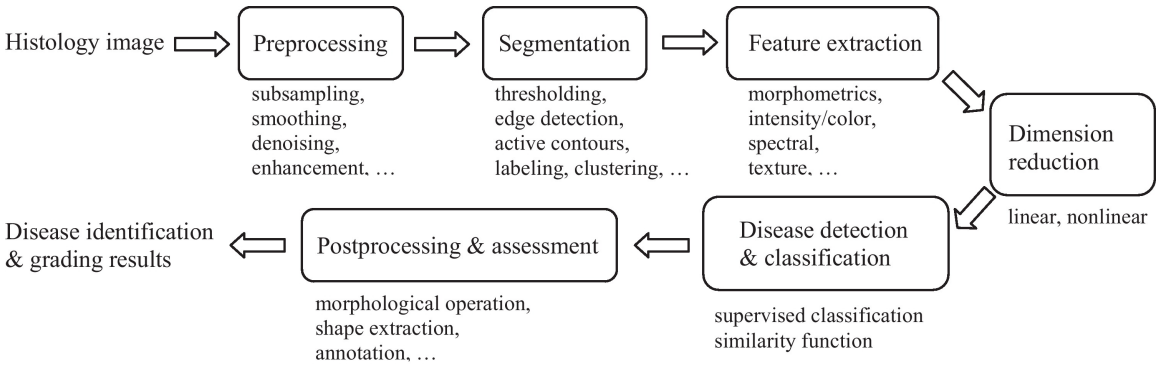


Figure 4-2 Computer assisted diagnosis flowchart source [149]

The sequential order of these functional modules may be changed in practical applications. For example, texture image segmentation requires that texture features should be computed before segmentation. Meanwhile, some modules may be omitted in particular systems, and other application specific modules not shown here, may be included.

4.3.1 Image pre-processing

Image pre-processing is the first step in an automatic histopathology analysis process.
In order to reduce visual variability and noise, raw image data is transformed. Variations of image quality can significantly affect the subsequent image segmentation and feature extraction. Appropriate pre-processing methods can contribute to reduce variations [152]. Such as colour normalization to minimize staining variations [153] spatial filtering to highlight major image structure, denoising to reduce image noise, and enhancement to optimize contrast between objects of interest and background.5 Moreover, intensity cantering and histogram equalization were presented particularly to normalize a diverse set of pathology images.

4.3.2 Image segmentation

Image segmentation consists of extracting objects or regions of interest from the background of an image. Extracted objects and regions are the focus for further disease identification and classification. Early segmentation methods still used in histopathology image analysis include *thresholding, edge detection, and region growing* [154]. For example, to separate objects or regions from background, the thresholding approaches [130,135] use a specific value (threshold) based on image intensity or its transforms such as Fourier descriptors or wavelets.

4.3.3 Feature extraction and dimension reduction

For pathologists, diagnostic criteria are inevitably described using terms such as “nucleus” and “cell.” It is thus important to develop methods capable of such object-level analysis. [155] For a CAD system, after image segmentation, image features are extracted from the regions of interest to detect morphological abnormalities that are relevant for the diagnosis or prognostic evaluation of diseases. The aim of Computed Assisted Diagnosis (CAD) of tumors is to use extracted features to support the pathologist in: i) distinguishing benignity and malignancy and defining the histopathology type of the tumour, ii) classifying different malignancy levels of the tumour (grading/scoring).[1].
This is mainly based on statistical analysis of the characteristics identified at the cellular or tissue levels. The cellular-level analysis focuses on quantifying the properties of individual cells by

considering its morphological, textural, fractal and/or intensity based features. The tissue-level features quantify the distribution of cells across the tissue based on the spatial dependency between them or the grey level dependency of the pixels.
In the literature, several types of features extraction techniques are mentioned. Traditional features [149], [156]–[160] include :
- *Morphometric with object size and shape* (e.g. compactness and regularities),
- *Topological or graph-based features* (e.g. Voronoi diagrams, Delaunay triangulation, and minimum spanning trees),
- *Intensity and colour features* (e.g. statistics in different colour spaces), and
- *Texture features* (e.g. Haralick entropy, Gabor filter, power spectrum, co-occurrence matrices, and wavelets).

	Features
Cell-level	Morphological: object size and shape (e.g. compactness and regularities)
	Texture features (e.g. Haralick entropy, Gabor filter, power spectrum, co-occurrence matrices, and wavelets).
	Fractal
	Intensity and colour features (e.g. statistics in different colour spaces),
Tissue-level	Spatial dependency of the cells
	Grey level dependency of the pixels
	Textural
	Fractal
	Topological or graph-based features (e.g. Voronoi diagrams, Delaunay triangulation, and minimum spanning trees)

Table 4-1 Major extraction features used in histopathology

Table 4-1 summarizes major extraction features used in histopathology image analysis with respect to the cell and tissue levels. In addition, besides using the image in the spatial domain, many features can also be extracted from other transformed spaces, e.g. frequency (Fourier) domain and wavelet transforms.
Another important concept in conventional histopathology image analysis is the exploration and identification of different structures at different magnifications.

→ Magnification & Resolution [152], [161], [162]

Magnification refers to increasing the proportion of biological structures, which are visible under the microscope according to the set of lenses. Conventional microscopes have a standard set of objectives 2X, 10X, 20X, 40X and 100X. It is clear that, even being the same organ, appearance of images is highly variant identifying different structures at different magnifications.
In a multi-scale framework, a set of features proven useful at a given magnification may not be relevant at another level of resolution (even within the same image): "*Feature values are related to the viewing scale or resolution*"[155].

- *At lower resolutions of histological imagery, colour or textural analysis is commonly used to capture tissue architecture, i.e. the overall pattern of glands, stroma and organ organization.*
- *At medium resolutions, architectural arrangement of individual histological structures (gland and nuclei) starts to become resolvable. Within each cancer grade, they can be described via several graph-based algorithms.*
- *At higher resolutions, morphology of specific histological structures (nuclei, margin, boundary appearance of ducts, glands) has proved to be of discriminatory importance. Many of these features can be discerned.*

On another hand, pixels classified as "non-tumour" at a lower resolution are eliminated at the subsequent higher level. This permits to reduce the number of pixels needed for analysis at higher levels. It is also important to note that "*the presence of more discriminating information at higher scales allows the classifier to better distinguish between tumour and non-tumour pixels*"[155].

4.4 Discussion & Conclusion

In this section, we presented an overview of image analysis techniques on histopathological digital images for cancer diagnosis and grading. Within the large number of image analysis methods, we focused on conventional techniques such as image pre-processing, segmentation and feature extraction by considering the general histopathology image analysis workflow. While humans have innate abilities to process and understand imagery, they do not tend to restitute how they reach their decisions.
Recent recommendations issued by the American Society of Clinical Oncology and the College of American Pathologists for testing of the ER, PR, and HER2 receptor status encourage the use of quantitative image analysis techniques to improve the consistency of the interpretation. As CAD in medical imaging domain, histopathology CAD begins to be developed for disease detection, diagnosis, and prognostic evaluation. *Large feature sets are generated in the hopes that some subset of features incorporates the information used by the human expert for analysis to ameliorate traceability and reduce redundancy*

Novel image analysis algorithms, advances in computational power, technology improvement for the storage and management of big data are promising factors to prospect a "great" development of CAD in histopathology domain to complement the opinion of the pathologist. Histopathology image analysis is a cross-disciplinary field. A continuous collaboration between researchers in imaging, computer vision, knowledge engineer and pathology is crucial to lead to new research ideas and efficient solution for both the patient and the healthcare community.

On another hand, it is important to note that "*most current CAD systems for histology image analysis are based on revising and adjusting existing image processing techniques (for radiology or cytology images) for the new applications*."[149] Such approach may not be appropriate for histopathology image analysis needs and realities. While CAD is much used in medical imaging and diagnostic radiology, the application of CAD in histopathology imaging has about 10 to 15 years delay[155], [161]. "*A histology image usually has a much more complex structure than a radiological or a cytological one, with a number of objects of interest extensively distributed in the image"*[149]. Difference in CAD approaches between

radiology and histopathology are fundamental and the questions being asked are different[155]:

- *Spatial resolution difference (limitation in spatial resolution of radiological data);*
- *Large size of data and content of histopathology images (multi-resolution framework) compared to radiology;*
- *CAD in radiology mostly deals with grey-scale while histopathology often need to process colour images*.
- *With recent advent of multi-spectral and hyper-spectral imaging, each pixel in a histopathology section could be associated with several hundred sub-bands and wavelengths*.

At last, "*Structural information about the tissue is lost when preparing the molecular assays"[86], [163]*. The semantic knowledge formalization approach can complement the promising researches to integrate imaging biomarkers from histopathology images with genomic data [164], [51], [165].

5 Image Analysis Knowledge identification and formal representation

5.1 Introduction

In this work, we focus on the image analysis domain by considering 1) descriptions of high performance histopathology imaging methods from contests and 2) concepts and functionalities found in the standard tools of three image analysis communities: Matlab (image scientists and engineers), ITK (developers) and ImageJ[30] (imaging biologists) and 3) vocabulary related to quantifiable features issued from Histopathology image analysis surveys [94], [149], [155]

Our objective is essentially:

a) To identify relevant imaging knowledge issued from contests, imaging community (Matlab, ImageJ, and ITK) and histopathology domain literature
b) To identify available formalized knowledge from the NCBO Bioportal to annotate descriptions of high performance histopathology imaging methods from contests[166]–[168]
c) To integrate the knowledge issued from a) and b) for a Practical Image Processing Ontology building proposal

5.2 Background

In this study, we continue our semantic cognitive virtual microscopy initiative[31,32] by proposing a sustainable way to bridge the content, features, performance and usability gaps[10][11] between histopathology and WSI analysis. The MICO project achieved a prototype system to perform some histopathology diagnosis related tasks on tissue slides where elementary imaging processes were combined by a logic engine, which could use formalized knowledge available as a set of rules. These rules, however, had been elaborated through local collaboration between pathologists and image scientists whereas sustainability calls for the use of publicly available knowledge gathered in standard formats from collaborative multi-centric efforts and constantly updated. The overall approach is presented in Figure 5-1.

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30 Image J : https://imagej.nih.gov/ij/

31 MICO project (COgnitive MIcroscopy) - French National Research Agency - Technologies for Health *and Autonomy (ANR TecSan): http://daniraco.free.fr/projects.htm*

FlexMIm project (Collaborative Pathology) - Consolidated Interministerial Fund (FUI - Fonds Unique Interministériel) : http://www.systematic-paris-region.org/en/projets/flexmim

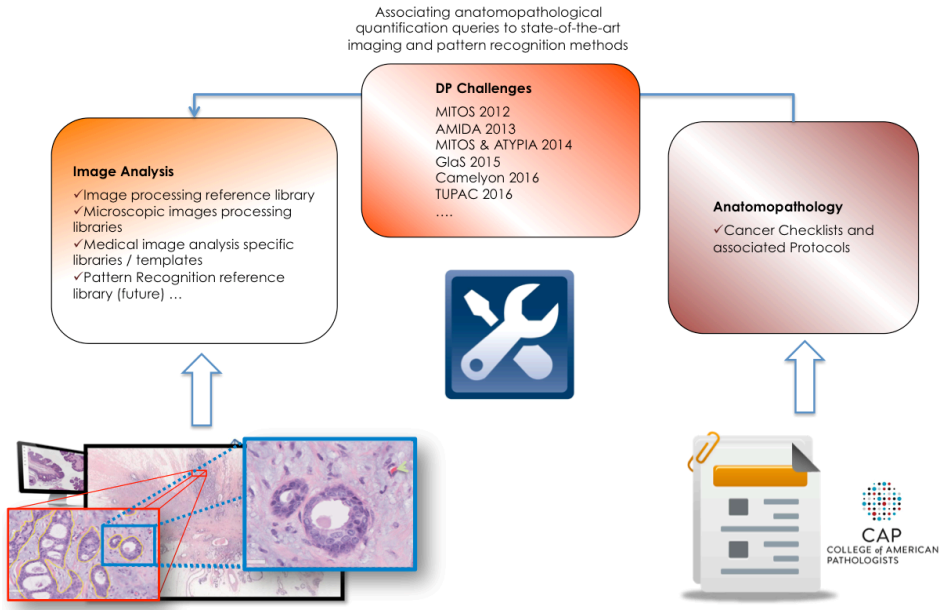

Figure 5-1: overall approach of using recent DP challenges to make an operational, instantiated link between anatomopathology and imaging.

5.3 Materials and methods

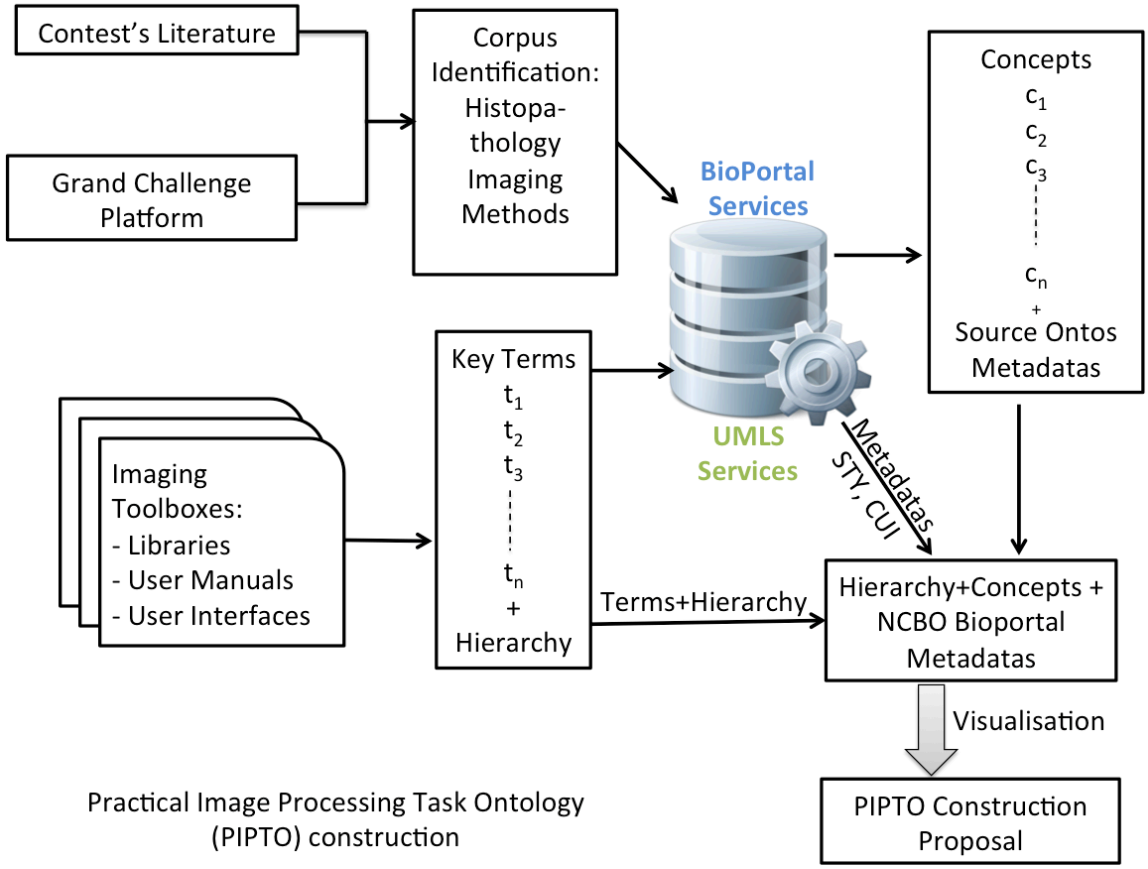

Figure 5-2: overall approach of image analysis knowledge extraction and formal representation

5.3.1 Identification of High performance histopathology imaging methods from Contests

Indeed, important milestone on the way to routine Digital Pathology, a series of international benchmarking initiatives have been launched by the team of Daniel Racoceanu for mitosis detection at MITOS 2012[33] (continued by AMIDA 2013[34], MITOS 2014[35] and TUPAC 2016[36]), as for nuclear atypia grading at ATYPIA 2014[37]. Glandular structures detection GlaS 2015[38] followed by completing some of the fundamental grading components in diagnosis and prognosis. These initiatives allow envisaging a consolidated validation referential-database for Digital Pathology in the near future. It also joins the efforts to tackle the lack of standards and ground truth as reference for algorithm validation and comparison [149].

5.3.1.1 Why Contest descriptions ***annotation corpus issued from contests?***

Contests are main Events that gathers both the histopathology (datasets, benchmarks, questions) and Imaging (algorithms, quantification support toolss, digital protocols, etc.) communities. They represent an excellent opportunity to identify new imaging methods that best answers important state-of-the-art specific histopathology questions.
Publishing a description of competing methods is a requirement for a good challenge. However, the responsibility for the content of each challenge remains with its organizers (e.i : in our work, the request of unpublished method description were demanded to respective organizers). In specific cases [169], [170] publishing or patent considerations limit the depth of the method description.

5.3.1.2 « Grand Challenge » platform initiative

Grand Challenge [166] is an online platform that provides an overview of «known» previous, on-going and upcoming challenges in biomedical image analysis. It provides tools to publish data and evaluation metrics to facilitate better comparisons between new and existing approaches. Up to date[39], there are about 149 projects (some of them are on-going) ranging in different medical and biomedical domains. We focused on histopathology imaging contests. Table 5-1 shows the corpus with the contest summary, reference papers and sources; identified methods and word count of the competing method descriptions.

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[33] Mitosis detection challenges: MITOS @ Int. Conf. Pattern Recognition (ICPR) Tsukuba, Japan, 2012: http://ludo17.free.fr/mitos_2012/ and AMIDA @ Int. Conf. Medical Image Computing and Computer Assisted Intervention (MICCAI) Osaka, Japan, 2013

[34] AMIDA 2013 : http://amida13.isi.uu.nl/

[35] MITOS & ATYPIA 2014 - Mitosis detection and nuclear atypia grading challenge, Int. Conf. Pattern Recog- nition (ICPR) Stockholm, Sweden, 2014: http://mitos-atypia-14.grand-challenge.org/

[36] TUPAC 2016 : http://tupac.tue-image.nl/

[37] MITOS & ATYPIA 2014 - Mitosis detection and nuclear atypia grading challenge, Int. Conf. Pattern Recog- nition (ICPR) Stockholm, Sweden, 2014: http://mitos-atypia-14.grand-challenge.org/

[38] GlaS 2015: Glandular structures detection challenge: GlaS @ Int. Conf. Medical Image Computing and Computer Assisted Intervention (MICCAI) Munich, Germany, 2015, http://www2.warwick.ac.uk/fac/sci/dcs/research/combi/research/bic/glascontest/

[39] Accessed 30/08/2017

5.3.1.3 Other Digital Pathology contests in the literature

Apart from contest platforms, we were interested to other histopathology imaging challenges. As demonstrated with a significant margin by the winners of contests AMIDA13 and MITOS 2012 with the Swiss team D. Ciresan and A. Guisti from the Institute of Artificial Intelligence Studies (ISDIA), the best method for detection of mitosis is by using convolutional neural networks (CNN). Papers published by this team related to AMIDA13 and MITOS 2012 contests are:

- « Mitosis Detection in Breast Cancer Histology Images with Deep Neural Networks »[40][171]
- « A Comparison Of Algorithms and Humans for Mitosis Detection »[41][172]

Other examples of active actors in histopathology image analysis from their participation to MITOS 2012 and MITOS-ATYPIA-2014 contests are :

- Tissue Image Analytics (TIA) Laboratory at Warwick University[42]
- University Medical Center Utrecht, organiser of contests AMIDA13 and TUPAC16

A significant overview paper in this domain is « Assessment of algorithms for mitosis detection in breast cancer histopathology images » of M. Veta et al., 2015[43][173]
Miscellaneous contest references are:

- TUPAC Contest during MICCAI 2016 (Tumour Proliferation Assessment Challenge)[44]
  Mitosis counting in breast cancer, M. Veta, 2016[45]
- Mitko Veta PhD Thesis on Breast cancer histopathology image analysis, 2014[46]
- Source code and presentations of summer school on deep learning in medical image analysis, 2015[47]

5.3.2 Description of the corpus issued from contests

At first, we considered the 2012-2016 period by focusing on fundamental grading components in diagnosis and prognosis respectively on mitosis detection (MITOS), nuclear atypia grading (MITOS-ATYPIA) and glandular structure detection (GlaS).
We identified 5 international benchmarking contests related to 29 top performing histopathology-imaging methods. Table 5-1 below summarizes the considered imaging methods in each of these contests.

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40 http://people.idsia.ch/~ciresan/data/miccai2013.pdf

41 http://people.idsia.ch/~ciresan/data/isbi2014.pdf

42 http://www2.warwick.ac.uk/fac/sci/dcs/research/combi/research/bic

43 https://arxiv.org/pdf/1411.5825v1.pdf

44 http://tupac.tue-image.nl/

45 https://pure.tue.nl/ws/files/29936283/asset.pdf

46 http://www.isi.uu.nl/Research/Publications/publicationview.php?id=2714

47 https://github.com/mitkovetta

Corpus Index	Challenge & Associated conference	Context/Brief Summary	Number of Identified Methods	Reference sources	Word counts
1	MITOSIS, - ICPR 2012 International Conference on Pattern Recognition	The competition consists in being able to tell what is the mitotic count on an image. Different types of images are provided. The contestants can take advantage of using the information of some of the spectral bands, which may be more discriminating for the detection of mitosis, or to concentrate only on RGB images.	4	Mitosis detection in breast cancer histological images An ICPR 2012 Contest [Roux et al. 2013]	357
2	AMIDA, MICCAI 2013	The main goal of the challenge was to evaluate and compare the performance of different (semi-)automatic mitosis detection methods that work on regions extracted from whole slide images on a large common data set. Since only the number of mitoses present in the tissue is of importance, i.e. their size and shape is not of interest, the challenge was defined as a detection problem.	11	Assessment of algorithms for mitosis detection in breast cancer histopathology images [Veta et al.201]	405
3	MITOS-ATYPIA, ICPR 2014	propose a contest using breast cancer histological images. The contest is made up of two parts: Detection of mitosis on the one hand, and evaluation of nuclear atypia score on the other hand. Mitotic count and nuclear pleomorphism are important parameters for the prognosis of breast cancer. Both tasks will be performed on images of haematoxylin and eosin (H&E) stained slides of breast cancer.	4	Detection of high-grade atypia nuclei in breast cancer imaging Mitosis Detection in Breast Cancer Histology Images via Deep Cascaded Networks	740
4	GlaS, MICCAI 2015	overview to the Gland Segmentation in Colon Histology Images Challenge Contest (GlaS) held at MICCAI'2015.	6	Gland Segmentation in Colon Histology Images: The GlaS Challenge Contest	501
5	Camelyon 16, ISBI 2016	Challenge on cancer metastasis detection in lymph node	4	https://grand-challenge.org/site/camelyon16/results/ - 29/10/2016	896

Table 5-1: Description of the corpus with the contest summary, reference sources, identified methods and word count.

5.3.3 Automatic annotation by NCBO Recommender

For each corpus, by using Recommender[114] of NCBO Bioportal we obtain the ranking of the most pertinent ontologies individually or by set of 4. The ontology-ranking algorithm used by Recommender evaluates the adequacy of each ontology to the input corpus using a combination of four evaluation criteria[48]: Coverage, Acceptance, Detail of knowledge and Specialization. For each case, we adjusted these parameters by considering default weights (Coverage=0.55, Acceptance=0.15, Knowledge Detail=0.15, Specialization=0.15) and configured weight on the coverage criterion (Coverage=1, others put to zero). We first annotated each corpus with the "imaging category ontologies" (n = 15) specified in NCBO Browse Tab. Then we redo the annotation by referring to all "ontologies available" (n = 668) in the NCBO platform. In each case, the highest ranked ontology sets (4 per set) and the first 5 single ranked ontologies were identified. Results are reported in Table 2 and Table 3.

5.4 Results

5.4.1 Annotation results

Results below summarize the corpus annotation with different weight configurations as specified in the methodology. For each corpus, we reported annotations results by referring respectively to "Imaging category Ontologies" and to "All ontologies" in NCBO Bioportal.

5.4.1.1 Automatic annotation with the 15 NCBO "imaging category" ontologies

The list of "imaging category" ontologies found in Bioportal is reported in Table 5-2. Overall 15 ontologies were found ranked with respect to their popularity (number of visits).

[48] **Coverage:** At what extent the ontology represents the input? The Recommender invokes the NCBO Annotator service to obtain all the annotations for the input and then uses those annotations to compute a coverage score for each ontology. **Acceptance:** How well-known and trusted is the ontology by the biomedical community? The number of visits to the ontology page in BioPortal and the presence or absence of the ontology in UMLS are used to compute an acceptance score for each ontology. **Detail of knowledge:** What is the level of detail provided by the ontology for the input data? It is computed using the number of definitions, synonyms and properties of the ontology classes that cover the input data. **Specialization:** How specialized is the ontology to the input data's domain? It is calculated using the number and type of the annotations done with the ontology and the position of each annotated class in the ontology hierarchy. The result is normalized by the size of the ontology, in order to identify small ontologies that are specialized to the input data.

#	NAME	CATEGORY	CLASSES
1	Radiation Oncology Ontology (ROO)	Development, Health, Human, Imaging, Vocabularies	1183
2	DICOM Controlled Terminology (DCM)	Imaging	3476
3	Information Artifact Ontology (IAO)	Biomedical Resources, Imaging, Other	180
4	Biomedical Informatics Research Network Project Lexicon (BIRNLEX)	Anatomy, Imaging	3580
5	Neural ElectroMagnetic Ontology (NEMO)	Anatomy, Biological Process, Experimental Conditions, Human, Imaging	1851
6	Biomedical Image Ontology (BIM)	Imaging	125
7	Cognitive Paradigm Ontology (COGPO)	Experimental Conditions, Human, Imaging	358
8	Biological Imaging Methods Ontology (FBbi)	Experimental Conditions, Imaging	NA
9	NIDM-Results (NIDM-RESULTS)	Imaging, Other	1
10	Magnetic Resonance Dataset Acquisition Ontology (ONL-MR-DA)	Imaging	NA
11	Dataset processing (ONL-DP)	Imaging	NA
12	Medical image simulation (OntoVIP)	Imaging	NA
13	Image and Data Quality Assessment Ontology (IDQA)	Imaging	260
14	Bioimaging Ontology (EDAM-BIOIMAGING)	Imaging	130
15	Quantitative Imaging Biomarker Ontology (QIBO)	Imaging	NA

Table 5-2 List of "imaging category" ontologies found in Bioportal with associated definitions and metrics

From NCBO "imaging category ontologies", the maximum final scores obtained with the coverage criterion (Coverage=1, others put to zero) were: 12.5% for single ranked ontology, Corpus#4 annotated with DCM and 22.6% for ontology sets, Corpus#3 annotated with the DCM, EDAM-BIOIMAGING, IAO. Table 5-3 reports the detailed annotation results.

Challenge	Corpus Number & Word Count	Selected Ontologies: Imaging (n=15), Weights Configuration: Coverage=1, Output: Ontology sets (4per set)	Selected Ontologies: Imaging (n=15), Weights Configuration: Coverage=1, Output: Ranked Ontologies (first 5)	Selected Ontologies: Imaging (n=15), Weights Configuration: Default, Output: Ontology sets (4per set)	Selected Ontologies: Imaging (n=15), Weights Configuration: Default, Output: Ranked Ontologies (first 5)
ICPR 2012 Mitosis Detection in histological images of H and E stained slides of Breast Cancer (MITOS dataset)	Corpus#1, 181 words	EDAM-BIOIMAGING, NIDM-RESULTS, NEMO, IDQA Final Score = 21.8	EDAM-BIOIMAGING (9.0) NIDM-RESULTS (6.0) NEMO (5.8) IDQA (4.9) BIRNLEX (3.9	NEMO, DCM, EDAM-BIOIMAGING, NIDM-RESULTS Final Score =21.7	NEMO (17.4) DCM (13.0) EDAM-BIOIMAGING (12.9) IAO (12.4) NIDM-RESULTS (11.6)
MICCAI 2013 Assessment of Mitosis Detection Algorithms 2013 (AMIDA13) evaluate and compare the performance of different (semi-automatic) mitosis detection methods that work on regions extracted from whole slide images on a large common data set.	Corpus#2, 405 words	BIRNLEX, EDAM-BIOIMAGING, NDM-RESULTS, NEMO Final Score =15.7	BIRNLEX (7.8) EDAM-BIOIMAGING (7.2) NIDM-RESULTS (5.5) NEMO (3.0) ROO (2.5)	NEMO, IAO, BIRNLEX, DCM Final Score =16.7	NEMO (15.8) IAO (14.0) COGPO (12.9) BIM (12.0) BIRNLEX (11.4)
ICPR 2014 is a challenge on mitosis detection and nuclear atypia scoring. The dataset of H&E (hematoxylin and eosin) stained biopsy images scanned by a Hamamatsu scanner has been used for our work. The dataset includes RGB color images at X40 magnification extracted from 16 breast cancer slides scanned by two different scanners. The ground truth was provided by senior physicians at Hospital Pitie-Salpetriere, Paris, France. The major improvements from the first edition (ICPR2012) of the contest are nuclear atypia scoring and annotations provided by two senior pathologists and three junior pathologists.	Corpus#3, 627 words Corpus#3_1 Corpus#3_2 Corpus#3_3	BIRNLEX, ROO, EDAM-BIOIMAGING, NEMO -> Final Score =16.2 EDAM-BIOIMAGING, NEMO, DCM, IDQA -> Final Score =21.3 BIRNLEX, NEMO, EDAM-BIOIMAGING, IDQA -> Final Score =10.1	BIRNLEX (5.8) ROO (5.8) EDAM-BIOIMAGING (4.6) NEMO (4.2) NIDM-RESULTS (3.3) EDAM-BIOIMAGING (9.1) NEMO (6.0) DCM (5.2) COGPO (5.0) IDQA (5.0) BIRNLEX (3.4) NEMO (3.4) EDAM-BIOIMAGING (2.5) IDQA (1.7) COGPO (1.7)	NEMO, ROO, NIDM-RESULTS, BIRNLEX Final Score =18.8 NEMO, COGPO, DCM, NIDM-RESULTS Final Score =22.4 NEMO, COGPO Final Score =16.4	NEMO (19.2) IAO (13.0) ROO (12.9) COGPO (12.4) DCM (10.8) NEMO (21.7) COGPO (15.3) IAO (14.1) DCM (13.7) BIM (12.0) NEMO (16.6) IAO (12.7) COGPO (12.3) ROO (11.1) DCM (10.5)
MICCAI 2015 Gland Segmentation Challenge (GlaS) organised at MICCAI 2015	Corpus#4, 501 words	BIRNLEX, DCM, EDAM-BIOIMAGING, NEMO Final Score =17.7	BIRNLEX (6.9) DCM (5.4) EDAM-BIOIMAGING (4.6) NEMO (3.4) NIDM-RESULTS (3.2)	NEMO, DCM, EDAM-BIOIMAGING, BIRNLEX Final Score= 17.9	NEMO (14.7) IAO (13.6) DCM (12.4) ROO (11.6) BIM (10.9)
ISBI 2016 Camelyon 16, Challenge on cancer metastasis detection in lymph node	Corpus#5, 886 words Corpus#5_1 Corpus#5_2 Corpus#5_3 Corpus#5_4	DCM, EDAM-BIOIMAGING, BIRNLEX, NIDM-RESULTS ->Final Score =18.4 NEMO, ROO, IDQA, DCM ->Final Score =20.6 EDAM-BIOIMAGING, ROO, NEMO, BIRNLEX ->Final Score =24.3 EDAM-BIOIMAGING, BIRNLEX, ROO, DCM ->Final Score =17.9	DCM (8.1) EDAM-BIOIMAGING (5.7) BIRNLEX (5.6) IAO (3.5) BIRNLEX (7.7) NEMO (7.3) ROO (4.8) IDQA (4.6) DCM (3.9) EDAM-BIOIMAGING (8.6) ROO (8.6) NEMO (7.1) BIRNLEX (5.4) NIDM-RESULTS (3.6) EDAM-BIOIMAGING (8.9) BIRNLEX (7.8) IAO (5.8) COGPO (5.8) NEMO (4.4)	DCM, IAO, EDAM-BIOIMAGING, BIRNLEX Final Score= 20.5 NEMO, DCM, ROO, IDQA Final Score= 22.6 NEMO, ROO, DCM, EDAM-BIOIMAGING Final Score= 23.2 IAO, NEMO, ROO, EDAM-BIOIMAGING Final Score= 20.4	DCM (15.7) IAO (15.6) COGPO (14.3) NEMO (12.7) EDAM-BIOIMAGING (12.0) NEMO (21.3) IAO (16.4) COGPO (15.2) DCM (12.7) BIRNLEX (11.7) NEMO (18.2) IAO (14.3) ROO (14.1) COGPO (13.2) BIM (12.2) IAO (17.2) NEMO (16.9) COGPO (16.0) ROO (15.0) EDAM-BIOIMAGING (13.0)

Table 5-3: Annotation metrics of contest corpus with adjustable weights* of Recommender and by referring to "imaging category ontologies" (n=15)

5.4.1.2 Automatic annotation with all 668 ontologies available on the NCBO platform

Table 5-4 shows the results of the annotation with all ontologies available in NCBO Bioportal. From these results, we get the list of the most relevant biomedical ontologies to be used for the annotation of the corpus describing imaging methods in histopathology domain. They are reported in Table 5-5 with related definitions and metrics.

Challenge	Corpus Number & Word Count	Selected Ontologies: All (n=665), Weights Configuration: Coverage=1, Output: Ontology sets (4per set)	Selected Ontologies: All (n=665), Weights Configuration: Coverage=1, Output: Ranked Ontologies (first 5)	Selected Ontologies: All (n=665), Weights Configuration: Default, Output: Ontology sets (4per set)	Selected Ontologies: All (n=665), Weights Configuration: Default, Output: Ranked Ontologies (first 5)
ICPR 2012 Mitosis Detection in histological images of H and E stained slides of Breast Cancer (MITOS dataset)	Corpus#1, 181 words	NCIT, SNOMEDCT, SWEET, LOINC Final Score =75.2	NCIT (57.7) SNOMEDCT (33.0) NIFSTD (24.3) SWEET (22.4) LOINC (22.4)	NCIT, SNOMEDCT, MESH, SWEET Final Score =74.2	NCIT (70.2), SNOMEDCT (45.7), MESH (40.7), NIFSTD (33.2), LOINC (33.2)
MICCAI 2013 Assessment of Mitosis Detection Algorithms 2013 (AMIDA13) evaluate and compare the performance of different (semi-automatic) mitosis detection methods that work on regions extracted from whole slide images on a large common data set.	Corpus#2, 405 words	NA - Problem getting recommendation for Set of 4 Set of 3: NCIT, SNOMEDCT, MESH Final Score =69.8	NCIT (57.8) SNOMEDCT (37.9) NIFSTD (27.0) RCD (24.5) LOINC (22.6)	NA - Problem getting recommendation for Set of 4 Set of 3: NCIT, SNOMEDCT, MESH Final Score =74.4	NCIT (69.2), SNOMEDCT (49.7), MESH (38.8), NIFSTD (37.2), RCD (35.8)
ICPR 2014 is a challenge on mitosis detection and nuclear atypia scoring. The dataset of H&E (hematoxylin and eosin) stained biopsy images scanned by a Hamamatsu scanner has been used for our work. The dataset includes RGB color images at X40 magnification extracted from 16 breast cancer slides scanned by two different scanners. The ground truth was provided by senior physicians at Hospital Pitie-Salpetriere, Paris, France. The major improvements from the first edition (ICPR2012) of the contest are nuclear atypia scoring and annotations provided by two senior pathologists and three junior pathologists.	Corpus#3, 627 words Corpus#3_1 Corpus#3_2 Corpus#3_3	NCIT, SNOMEDCT, NIFSTD, SOPHARM ->Final Score =85.0 NCIT, SNOMEDCT, RCD, ORTH > Final Score =80.9 NA For Set 4, Set 3: NCIT, LOINC, ORTH -> Final Score =74.0	NCIT (64.1) SNOMEDCT (40.4) SWEET (30.8) RCD (30.0) LOINC (28.4) NCIT (63.6) SNOMEDCT (40.2) SWEET (31.2) MATRROCKIGNEOUS (29.2) MNR (29.2) NCIT (57.6) SNOMEDCT (38.2) RCD (28.1) NIFSTD (28.0) LOINC (24.7)	NCIT, SNOMEDCT, RCD, MESH Final Score =79.8 NCIT, SNOMEDCT, RCD, ORTH Final Score =78.7 NCIT, SNOMEDCT, LOINC, ORTH Final Score = 75.5	NCIT (73.1) SNOMEDCT (50.9) RCD (39.1) MESH (37.4) LOINC (36.3) NCIT (72.9) SNOMEDCT (49.8) NIFSTD (38.2) RCD (36.4) LOINC (33.5) NCIT (69.6) SNOMEDCT (50.5) NIFSTD (39.1) RCD (38.6) LOINC (35.0)
MICCAI 2015 Gland Segmentation Challenge (GlaS) organised at MICCAI 2015	Corpus#4, 501 words	NCIT, SNOMEDCT, SWEET -> Final Score =75.1 (NA Set 4)	NCIT (59.8) SNOMEDCT (39.0) RCD (26.9) NIFSTD (22.2) LOINC (20.4)	NCIT, SNOMEDCT, RCD -> Final Score =63.1 (NA Set 4)	NCIT (55.5) SNOMEDCT (43.6) RCD (32.9) MESH (30.4) LOINC (28.1)
ISBI 2016 Camelyon 16, Challenge on cancer metastasis detection in lymph node	Corpus#5, 886 words Corpus#5_1 Corpus#5_2 Corpus#5_3 Corpus#5_4	NCIT, SWEET, SNOMEDCT, LOINC ->Final Score =82.0 NCIT, SNOMEDCT, SWEET, LOINC ->Final Score =71.8 NCIT, SWEET, SNOMEDCT, NIFSTD NCIT, SNOMEDCT, LOINC, NIFSTD ->Final Score =75.1	NCIT (65.9) SWEET (35.6) SNOMEDCT (35.1) LOINC (29.8) AURA (28.2) NCIT (59.7) SNOMEDCT (34.8) SWEET (28.0) MATRROCKIGNEOUS (25.1) MNR (25.1) NCIT (65.1) SWEET (42.9) MATRROCKIGNEOUS (38.1) MNR (38.1) MINERAL (38.1) NCIT (60.5) SNOMEDCT (30.8) SWEET (27.2) LOINC (27.2) NIFSTD (25.0)	NCIT, SNOMEDCT, LOINC, SWEET ->Final Score =79.5 NCIT, SNOMEDCT, LOINC, SWEET -> Final Score =73.9 NCIT, SNOMEDCT, NIFSTD, SWEET -> Final Score =77.4 NCIT, SNOMEDCT, MESH, NIFSTD ->Final Score =76.2	NCIT (74.7) SNOMEDCT (49.3) MESH (41.0) LOINC (39.7) RCD (37.7) NCIT (70.9) SNOMEDCT (47.8) NIFSTD (37.6) RCD (36.2) CRISP (34.5) NCIT (73.3) SNOMEDCT (44.8) NIFSTD (40.7) SWEET (39.8) MESH (37.0) NCIT (71.3) SNOMEDCT (46.3) MESH (38.4) CRISP (37.7) NIFSTD (36.2)

Table 5-4: Annotation metrics of contest corpus with adjustable weights* of Recommender and by referring to "All ontologies" (n=665) in NCBO Bioportal.

#	NAME	CATEGORY	CLASSES
1	BioModels Ontology (BIOMODELS),	All Organisms, Biological Process, Biomedical Resources	187520
2	Computer Retrieval of Information on Scientific Projects Thesaurus (CRISP)	Health	9039
3	Foundational Model of Anatomy (FMA)	Anatomy	104258
4	Logical Observation Identifier Names and Codes (LOINC)	Health	187123
5	Material Rock Igneous (MATRROCKIGNEOUS)	Upper Level Ontology	3535
6	Medical Subject Headings (MESH)	Health	261990
7	Material Natural Resource (MNR)	Upper Level Ontology	3554
8	National Cancer Institute Thesaurus (NCIT)	Vocabularies	118941
9	Neuroscience Information Framework (NIF) Standard Ontology (NIFSTD)	All Organisms, Anatomy, Biological Process, Cell, Cellular anatomy , Dysfunction, Molecule, Neurologic Disease, Neurological Disorder, Other, Subcellular, Subcellular anatomy	124337
10	Otology Ontology (ORTH)	All Organism, Genomic and Proteomic	4663
11	Radiology Lexicon (RADLEX)	Not mentioned	46140
12	Read Codes, Clinical Terms Version 3 (CTV3) (RCD)	Not mentioned	140065
13	Systematized Nomenclature of Medicine - Clinical Terms (SNOMEDCT)	Health	324129
14	Suggested Ontology for Pharmacogenomics (SOPHARM)	Genomic and Proteomic	44956
15	Semantic Web for Earth and Environment Technology Ontology (SWEET)	Not mentioned	4550

Table 5-5: List of the most relevant biomedical ontologies in NCBO Bioportal for the annotation of corpus describing imaging methods in histopathology domain

5.5 Formalization of major biomedical-imaging knowledge sources

5.5.1 Knowledge issued from major imaging community software: Matlab, ImageJ & ITK

In the perspective of building the Practical Image Processing Task Ontology – PIPTO, we propose a visual representation of concepts issued from three image analysis communities: Matlab (image scientists and engineers), ITK (developers) and Image J (imaging biologists) by using Protégé [174]. PIPTO aims at capturing image domain knowledge in a generic way and provides a consensual understanding of concepts and functionalities identified in the standard tools in these 3 communities:

- With OWLviz plugin [134], three graphical trees were obtained by conserving the source hierarchy from content of user manual and user interface menus. We exported the outline (title, subtitles and function lists) of user manuals and user interface menus into a TXT file.

- By conserving the hierarchy levels from sources, we organized all identified concepts in Protégé by using « Create class hierarchy » from the « Tools » Tab.
- Then with « OWLviz » tab, we generate the visualization of concepts related to each source.

At this stage, we only consider the hierarchical organization of concepts and their respective definitions from different sources. The management of issues related to similarities, properties, relations, semantic conflicts, etc. will be discussed in the future works.

5.5.2 Visual representation of concepts from Matlab, ImageJ and ITK

By using Protégé and OWLviz, we obtained the visual representation of concepts issued from Matlab, ImageJ and ITK image analysis communities. Three (3) graphical tree representations were obtained. Figure 5-3 shows an overview of the imaging knowledge visualization process and the number of concepts identified. Obtained graphical trees contribute to a better understanding of the hierarchy and granularity of the information contained in each source.

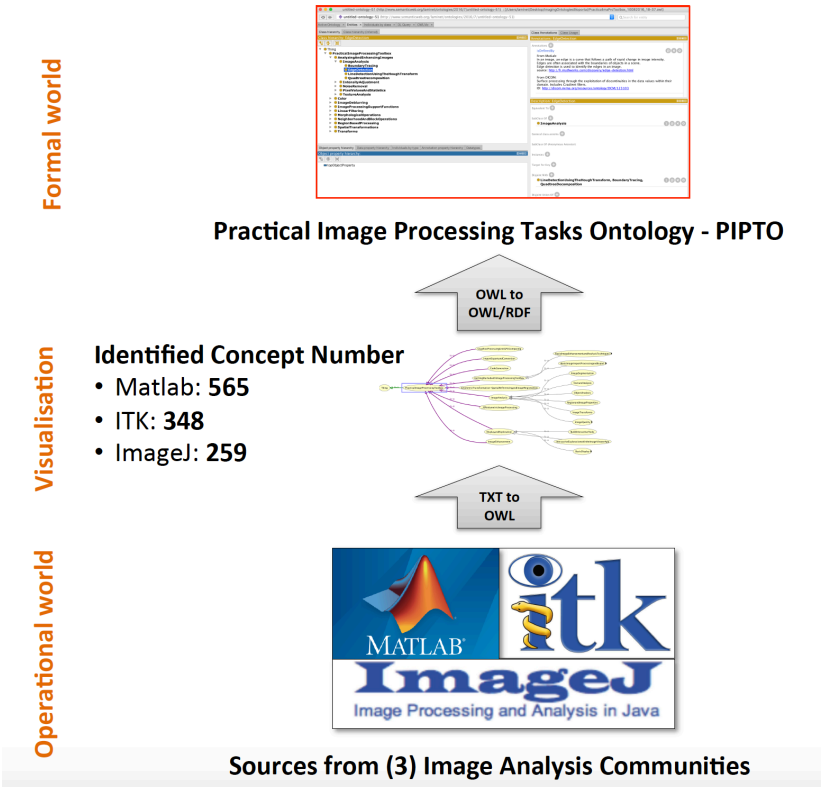


Figure 5-3: Overview of the imaging knowledge visualization process and the number of concepts identified from each source

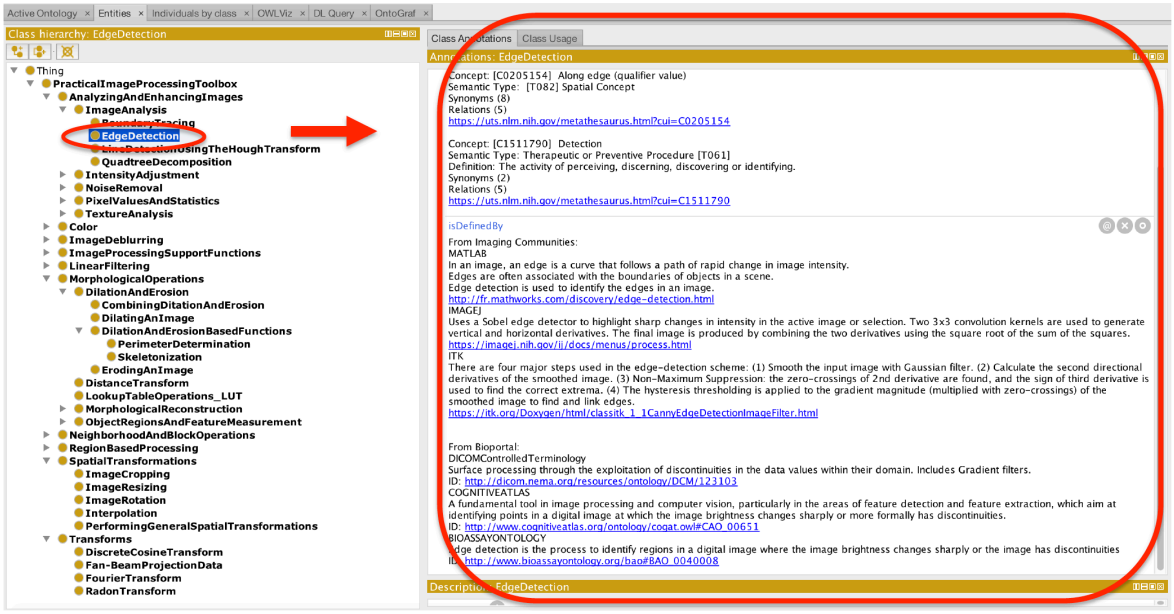


Figure 5-4: Practical Image Processing Task Ontology (PIPTO) issued from software overview

5.5.3 Generic imaging concepts identified from histopathology image analysis literature

Figure 5-5, Table 5-6, Table 5-7 and Table 5-8 are examples summarizing common histopathology image analysis methods and features related to spatial-arrangement, objectlevel and perceptive descriptor categories. These features were extracted from the literature review on histopathology image analysis. Definitions for all listed features are detailed in reference [94], [155], [167], [175]

5.5.3.1 Semi-structured diagram hierarchies

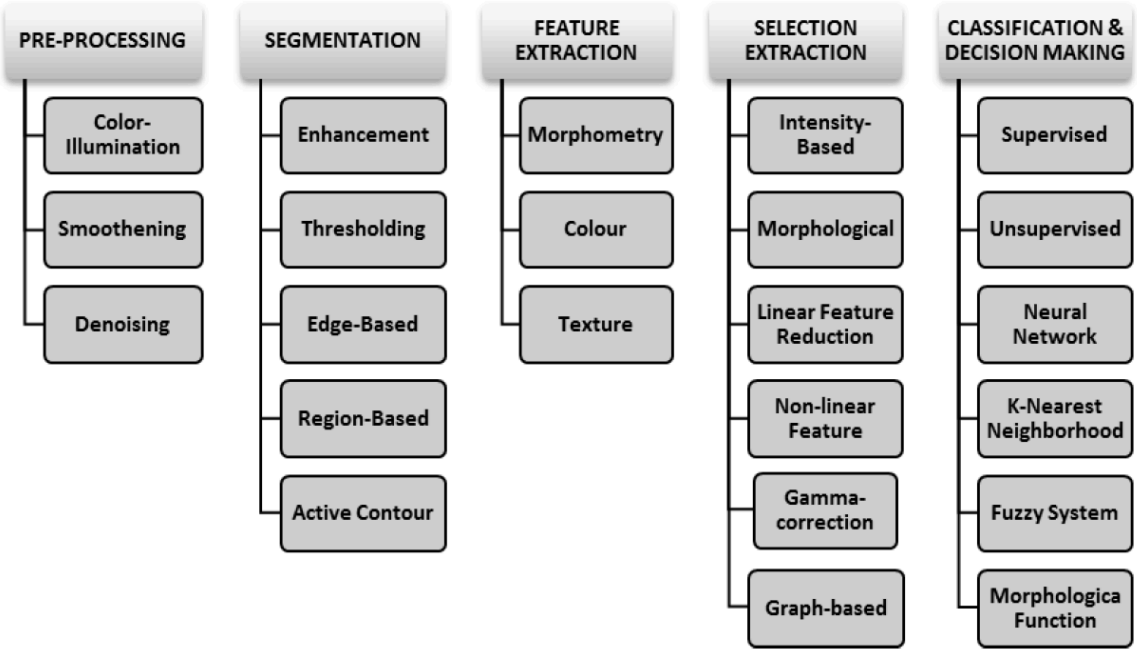


Figure 5-5 Schematic Diagram of Methods Related to Digital Microscopy, source:[176]

5.5.3.2 Table organisation hierarchies

Graph Structure	Features
Voronoi Tesselation	Number of nodes, number of edges, cyclomatic number, number of triangles, number of k-walks, spectral radius, eigenexponent, Randic index, area, roundness factor, area disorder, roundness factor homogeneity
Delaunay Triangulation	Number of nodes, edge length, degree, number of edges, cyclomatic number, number of triangles, number of k-walks, spectral radius, eigenexponent, Wiener index, eccentricity, Randic index, fractal dimension
Minimum Spanning Tree	Number of nodes, edge length, degree, number of neighbors, Wiener index, eccentricity, Randic index, Balaban index, fractal dimension
O'Callaghan Neighborhood Graph	Number of nodes, number of edges, cyclomatic number, number of neighbors, number of triangles, number of k-walks, spectral radius, eigenexponent, Randic index, fractal dimension
Connected Graph	Number of nodes, edge length, number of triangles, number of k-walks, spectral radius, eigenexponent, Wiener index, eccentricity, Randic index, fractal dimension
Relative Neighbor Graph	Number of nodes, number of edges, cyclomatic number, number of neighbors, number of triangles, number of k-walks, spectral radius, eigenexponent, Randic index, fractal dimension
k-NN Graph	Number of nodes, edge length, degree, number of triangles, number of k-walks, spectral radius, eigenexponent, Wiener index, eccentricity, Randic index, fractal dimension

Table 5-6 Summary of spatial-arrangement features used in histopathology image analysis, source: [155]

Category	Features
Size and Shape	Area
	Elliptical Features: Major and minor axis length, eccentricity, orientation, elliptical deviation
	Convex Hull Features: Convex area, convex deficiency,solidity
	Filled Image Features: Filled area, Euler number
	Bounding Box Features: Extent, aspect ratio
	Boundary Features: Perimeter, radii, perimeter FFT, perimeter curvature, bending energy, perimeter FD
	Other Shape Features: Equivalent diameter, sphericity, compactness, inertia shape
	Center of Mass (gray and OD)
	Reflection Symmetry Features (binary, gray, and OD)
Radiometric and Densitometric	Image Bands
	Intensity
	Optical Density (OD)
	Integrated Optical Density (IOD)
	Mean Optical Density (MOD)
	Hue
Texture	Co-occurrence Matrix Features (gray and OD): Inertia, energy, entropy, homogeneity, maximum probability, cluster shade, cluster prominence, correlation
	Fractal dimension (gray and OD)
	Run-length Features (gray and OD): Short runs emphasis, long runs emphasis, gray-level non-uniformity, run-length non-uniformity, runs percentage, low gray-level runs emphasis, high gray-level runs emphasis
	Wavelet Features (gray and OD): Energies of detail and low resolution images
	Entropy (gray and OD)
Chromatin-Specific	Area, IOD, MOD, number of regions compactness, distance, center of mass

Table 5-7: Summary of object-level features used in histopathology image analysis source:[155], [175]

Descriptor Category	Descriptor
Model	Model: *Blob*
Edge-Morphology	Contrast: *Numeric, Level** Shape: {curve, straight line} Status: {open, close, loop} Profile: {roof, peak, step} Straightness: *Numeric, Level**
Boundary	Nature: {edge, limit of texture, limit of homogeneous regions, no boundary} Reference: **cf. the visual primitive description: edge**
Photometry	Brightness:*Level**, *Numeric, Image* Contrast: *Level**, *Numeric, Image* Model: *Patch, Image*
Colorimetry	Hue: *Numeric, Level** Saturation: *Numeric, Level** Lightness: *Numeric, Level**
Texture	Direction: *Numeric* [deg], {horizontal, vertical, oblique} Scale: {macro, micro} Type: {no-texture, contour, dot, complex, periodic}
Topology	Number of holes: *Numeric*
Orientation	Orientation: *Numeric*, {vertical, horizontal}
Position	Center of mass: *Coordinate*
Edge-Size	Length: *Numeric* [pixel],*Level** Thickness: *Numeric* [pixel],*Level**
Region-Size	Area: *Numeric*[pixel^2],*Level** Volume: *Numeric*[pixel^3],*Level** Bounding Box: *Numeric* [pixel^2|pixel^3], *Level** Diameter: *Numeric* [pixel], *Level** Thickness: *Numeric* [pixel], *Level** Net perimeter: *Numeric* [pixel], *Level** Convex perimeter: *Numeric* [pixel], *Level**
Region-Morphology	Shape: {square, rectangular, circle, ellipsoid, parallelepiped, cubic, sphere} Compactness: *Numeric,Level** Convexity: *Numeric, Level** Elongation: *Numeric,Level** Number of angles: *Numeric* Rectangularity: *Numeric,Level** Roundness: *Numeric,Level**
InterestPoint-Morphology	Junction type: {L, T, X, Y}

**Level: {null, very low, low, medium, high, very high}*

Table 5-8 List of the perceptive descriptor category concepts with their sub-concepts for representing Image processing objectives, source [177]

5.5.4 Proposal of a second Practical Image Processing tasks Termino-Ontology – (PIPTO2)

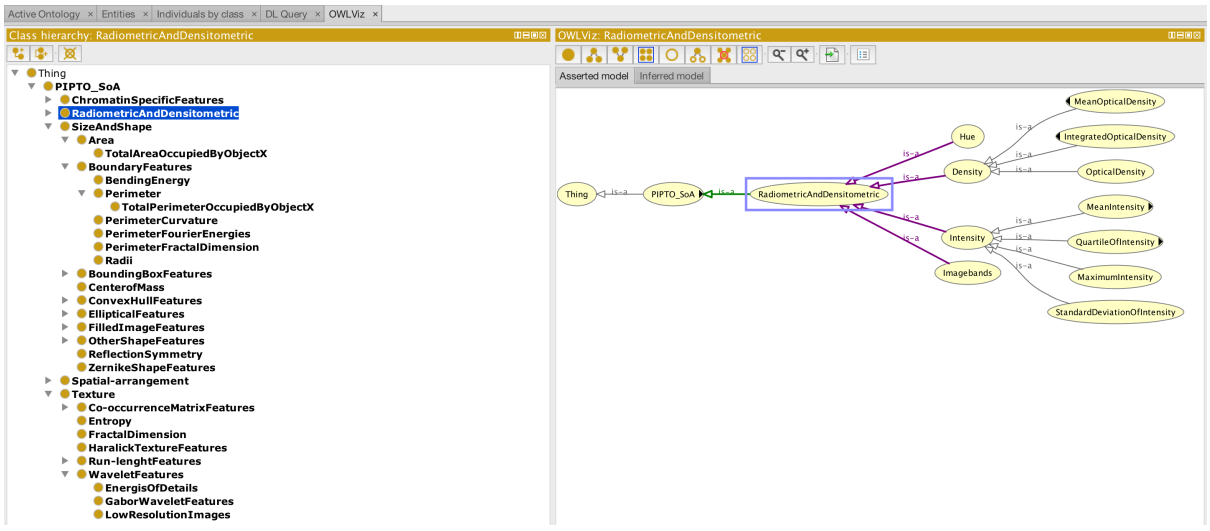


Figure 5-6 Screenshot of Practical Image Processing Task Ontology (PIPTO) issued from the State Of the Art (SoA)

5.5.5 Bridging the Semantic Gap Between Diagnostic Histopathology and Image Analysis

By considering the semantic aspect and knowledge engineering approach, building the liaison between AP and imaging requires close collaboration between the various actors.
We propose an integrative framework of histopathology and image analysis in the context of prognostic evaluation (scoring/grading) of tumours (figure 0-9). The generic integrative framework (figure 0-9) has been instantiated in the context of the Nottingham histologic score (figure 0-10) to illustrate how the different concepts of the two termino-ontologies Anatomic Pathology Quantitative Features Ontology (APQFO) and Practical Image Processing Tasks (PIPTO) are related and can be integrated according to the generic integrative framework.

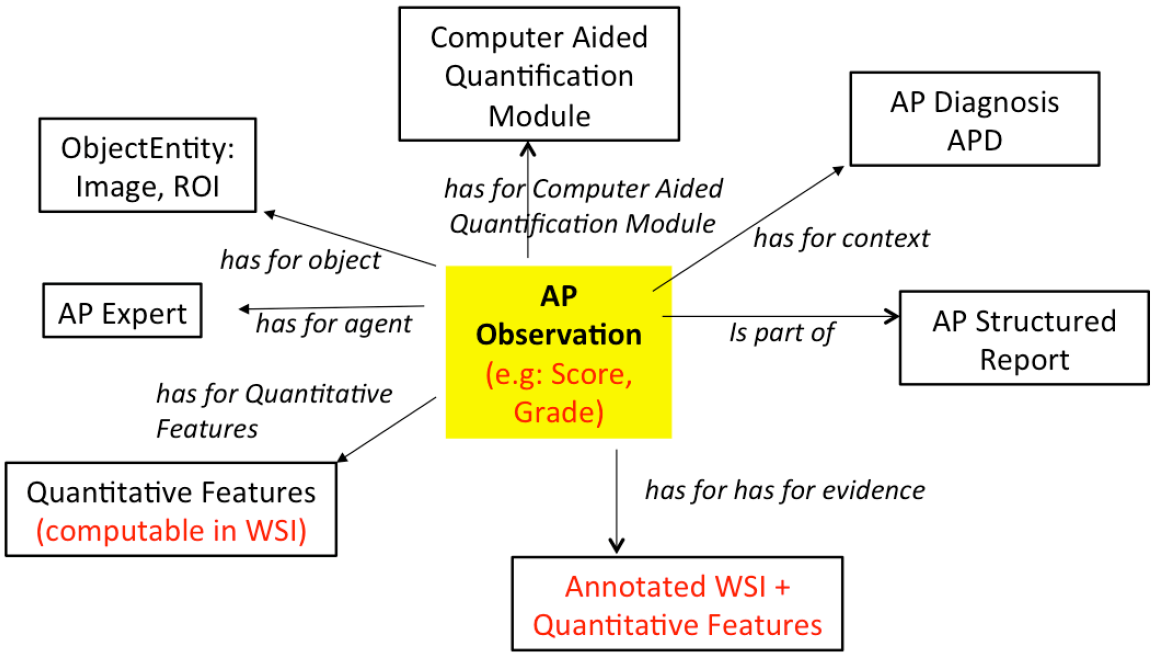


Figure 5-7 AP Observation process: prognostic evaluation (Grading/Scoring)

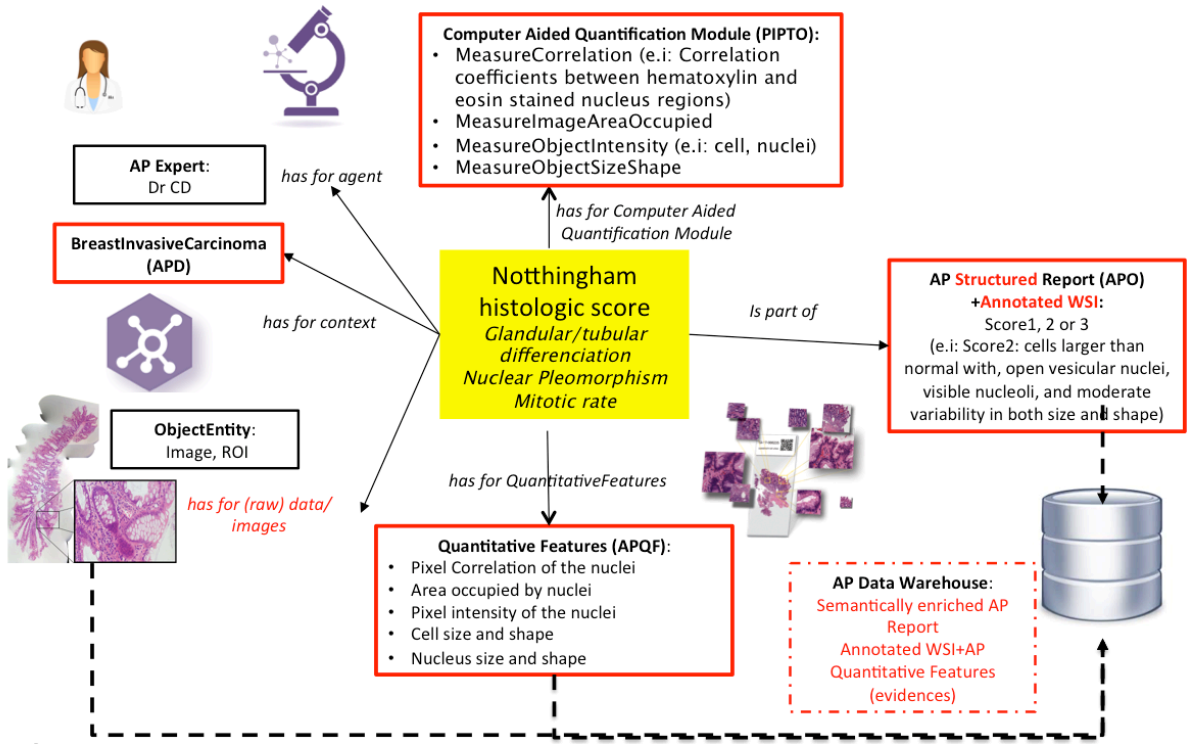


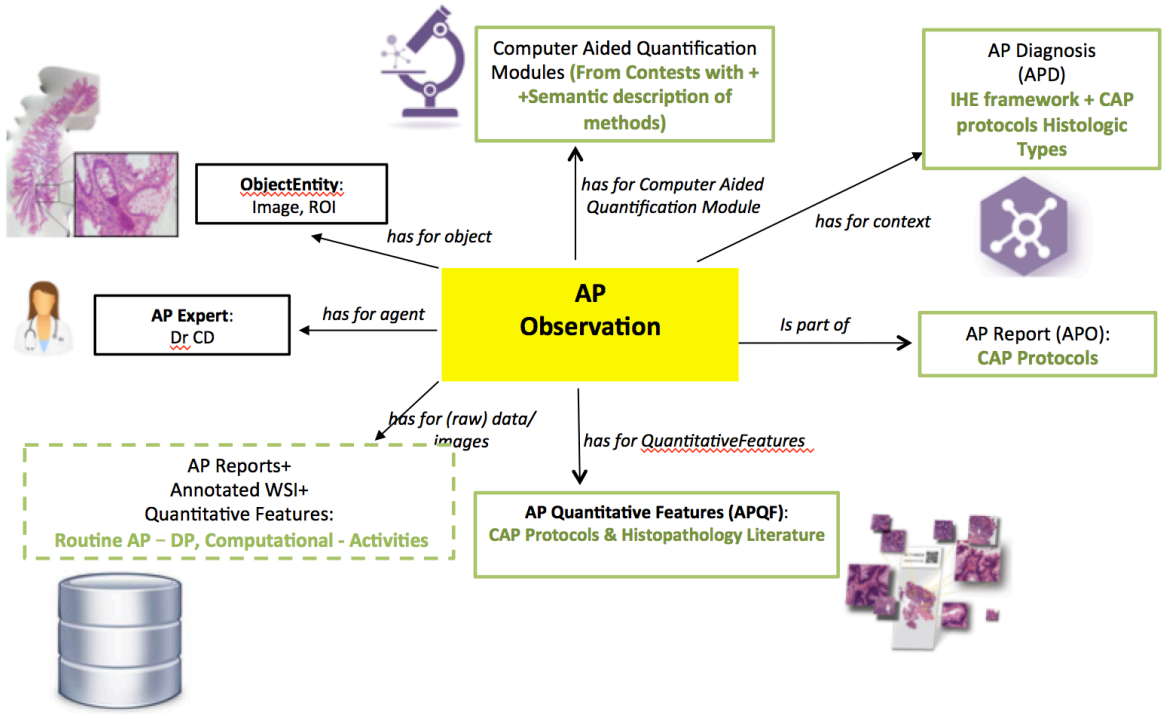
Figure 5-8 Example of Nottingham Nuclear Pleomorphism Score prognostic evaluation

In this work, we used expert-based definitions of grading/scoring systems to identify for each tumour type (breast cancer, prostate cancer, etc.) the quantifiable observations and the corresponding quantitative features that are relevant for prognostic evaluation.

APQF	PIPTO (module)
Feature Category	**CellProfiler Module**
Pixel Correlation of the cells	MeasureCorrelation
Pixel Correlation of the nuclei	MeasureCorrelation
Image granuarity	MeasureGranularity
Area occupied by cells	MeasureImageAreaOccupied
Area occupied by nuclei	MeasureImageAreaOccupied
Image intensity	MeasureImageIntensity
Image quality	MeasureImageQuality
Pixel intensity of the cell	MeasureObjectIntensity
Pixel intensity of the nuclei	MeasureObjectIntensity
Neighboring relation of the cells	MeasureObjectNeighbors
Neighboring relation of the nuclei	MeasureObjectNeighbors
Radial distribution of pixel intensity of the cell	MeasureObjectRadialDistribution
Radial Distribution of the Pixel Intensity of the Nuclei	MeasureObjectRadialDistribution
Cell size and shape	MeasureObjectSizeShape
Nucleus size and shape	MeasureObjectSizeShape
Cell texture	MeasureTexture
Nucleus texture	MeasureTexture

Table 5-9: Linking of APQF identified feature categories to PIPTO image quantification modules

This work is a step forward to build a formal representation integrating image analysis tasks with concepts related to the histopathology domain.
Providing, in each context of cancer (breast cancer, prostate cancer, etc.) such a unique formal representation contributes to consistently integrate AP observations reported by pathologists and their corresponding evidences in WSI (quantitative feature automatically extracted from WSI using image analysis tasks).
The integrative framework of APQFO and PIPTO contributes for routine use of histopathology computer assisted diagnosis (quantification aided modules) in providing for each context of cancer the list of image analysis tasks that are relevant to use according to the grading/scoring systems used by pathologists (as mentioned in the CAP P&CC).



5.6 Discussion

The proposed approach based on the annotation of contest corpus from "Grand-Challenge"[166] with NCBO Bioportal aims to evaluate available semantic resources associated to the imaging domain. From the above results, we report that there is no ontologies related to imaging domain in NCBO Bioportal to annotate efficiently the identified histopathology imaging methods. With respect to the annotation scores from Table 5-3 and the ontology list of Table 5-4, we see that the most relevant ontologies annotating imaging concepts in Bioportal are related to health, anatomy, biological process and similar categories. One should note that these huge resources are not specialized to imaging domain even if they give the highest annotation scores. This can be better understood by the position of the annotated class in the source ontology hierarchy. This also shows the need to develop an imaging domain ontology build beyond available image analysis concepts and functionalities.
Beyond NCBO Bioportal, we searched in other ontology repositories such as OBO Foundry[131]. Out of the 181 ontologies in Ontobee, we could manually identify 17 ontologies related to the imaging domain. The selection criteria were based on the "*Ontology Full name*" and "*More details*". To the best of our knowledge, there is no integrated annotating tool associated to Ontobee, we could not annotate our corpus with these semantic

resources. In future works, we plan to use these semantic resources "locally" with BioYodie[49] to annotate and evaluate the relevance of their concepts with respect to imaging methods from contests.
On another hand, we faced difficulties in getting the corpus of contest's definition methods. We could find few published papers (Table 5-1) in open access, describing newly proposed methods of considered contests. To complete this list, we sent requests to authors to obtain more descriptions. However, in some cases publishing or patent considerations limit the depth of the description related to a method. For example: our request to get descriptions of top 5 ranking methods in TUPAC16 from organizer was turn down, one of the 7 highest-ranking methods in GlaS contest was not available.
At last, concepts present in the Convolutional Neural Network methods may go beyond the imaging domain and may not be covered by the imaging ontologies. To overcome this limit, we plan to consider concepts associated to the Matlab CNN library and similar resources in the perspective of PIPTO enrichment.
The DICOM Controlled Terminology Ontology, which contains about 3384 concepts, gathers pertinent concepts and definitions related to the storage and transmission of medical imaging information relevant to our topic. Since DICOM is the main standard in medical imaging, it would be interesting to consider existing descriptions in DICOM sources to enrich the definition of concepts in PIPTO. Additional efforts are needed to achieve a workable standard-based formal representation that will be clearly understandable by humans, machine processable and sustainable.

On another hand, our perspective for enhancing our integrative framework is to use an alternative approach based on the use of real-world data and machine learning. This approach, successfully used in the context of lung cancer [Yu], will be used in the real hospital settings of the Assistance-Publique-Hopitaux de Paris (AP-HP) in collaboration with a working group of pathologists.
AP data – AP reports and AP quantitative features extracted from WSI – stored in the AP-HP Clinical Data Repository (CDR) – will be semantically annotated using the termiono-ontologies of the integrative framework. Anatomic Pathology Diagnosis Ontology (APDO), Anatomic Pathology Observation Ontology (APOO) and Anatomic Pathology Quantitative Features Ontology (APQFO) will be used to annotate the AP reports. Practical Image Processing Tasks (PIPTO) will be used to annotate the image analysis tasks and the quantitative features automatically extracted from WSI. Machine-learning techniques will be used to analyse correlation between patient outcome and quantitative features automatically extracted from WSI.

What opportunities for our integrative framework for AP research using real-world data?

- Seizing opportunities for non-interventional data research
- Improve intervention research through feasibility studies in clinical trials
- Propose an integrative Platform for the sharing and exploitation of "megadata" for the processing of massive and complex AP data
- Develop and evaluate "future" quantification and decision support algorithms.

[49] Beyond semantic annotation, Bio-Yodie manages hierarchical disambiguation between the concepts and refers to UMLS to build updated reference resources. It is based on the GATE platform (General Architecture for Text Engineering) and offers a wide range of output format for annotated concepts.

5.7 Conclusion

Overall, we could identify and evaluate relevant ontologies associated to histopathology image analysis. Then by considering concepts from main biomedical imaging tools, we could propose a formal representation of the imaging knowledge from Matlab, ImageJ, and ITK. Each of these imaging softwares includes a set of concepts, definitions, functions and relations that cover most of the imaging methods.
Future anatomopathological services need to use digital technologies in valid routine pathological diagnosis and healthcare protocols, by integrating the Whole Slide Images (WSI) observation for diagnosis purposes in a whole large specific DP case record. The goal is to describe, conceive and formalize an integrative framework of all these data, most of them already used for the diagnosis and prognostic evaluation.
This will generate an operational DP process in which the novelty relies in linking the microscopic exam of WSI to specific or generic annotations defined as micro-semiology semantic references. This enables the generation of a structured and standardized image-related report. Through Digital Pathology, the future of anatomopathology is on the way to reinforce its ethical and dynamical strengths. With the emergence of omics and integrative approaches, a traceable, semantically indexed second opinion will thus become essential for patients and healthcare professionals in personalized medicine.

5.7.1 What was already known on the topic?

- Hhistopathology contests are very valuable knowledge source about Image Processing Task used in the AP domain
- Imaging community functions and libraries invlude information describing the Image Processing Task that could be used during image analysis of AP images
- There is fundamental prognostic data embedded in pathology images and digital pathology will provide the next new source of “big data” for inform clinical research and decision making
- There is currently a lack of semantic reasoning methods to make inferences about cancerous lesions from semantic annotations.

5.7.2 What this study added to our knowledge?

- State of the Art of digital pathology imaging modalities and image processing techniques
- Identificatio (1) of histopathology image analysis Top ranking methods (n=29) within 2012-2016
- Identification (2) of relevant imaging libraries, key terms and concepts from ImageJ, Matlab and ITK
- Identification (3) of existing metadata (CUI, STY, definitions,) from Bioportal and UMLS associated with concepts issued from contests and imaging libraries.

- Proposal of a Practical Image Processing Task Termino-ontology (PIPTO) formalisation by integrating the Knowledge and semantic datas issued from results in (1), (2) and (3)
- Comparison of APQF obtained from « Experts Analysis » with « Machine Learning » results in the context of lung cancer by Yu et Al. [94] & PIPTO
- Proposing a generic integrative framework between Diagnostic Histopathology and Image Analysis

PART 4

Integration Platform and Valorisation Prospect: Smart'GRADE[50]. Concluding remarks and perspectives

[50] A knowledge driven Computer-Assisted Diagnosis (CAD) Tool for breast cancer grading

6 Concluding remarks and valorization prospect with Smart'GRADE[51] Integration Platform

This chapter summarizes the main concluding remarks and recommendations derived from this thesis. This finalizes the work carried out in this study, while providing some insights for the continuity of scientific research and some orientations of technologies to be developed and conducted within the Smart'GRADE project.

[51] : A knowledge driven Computer-Assisted Diagnosis (CAD) Tool for breast cancer grading

6.1 Concluding remarks, significance and comparison with related work

Currently, in daily AP laboratory workflow, microscopic diagnosis remains the gold standard. This process is limited by the lack of objectivity, reproducibility and considerable variability between observers [29], [136], [178]. Over the last decade, the development of "virtual slide" technology (transforming static images into dynamic images) coupled with the development of acquisition systems (motorized microscope followed by the slide scanner), networks and storage facilities, have radically stimulated the perspectives of digital / computational pathology. These advances represent a very promising solution to support the pathologist's laborious tasks during diagnosis (e.g. 95% accuracy for the identification of low-grade astrocytoma - WHO class II) [179] and prognosis [94].
Nevertheless, the adoption or practical use of these novelties and algorithms in the literature [166], [167], [180] published by peers during congress and conferences dedicated to the analysis of histopathological images, is not always used by the medical community [166] . And those used correspond only to a very precise need formulated by the laboratory in question (Ad hoc solution).

In this thesis we have mainly contributed to the development of two standard-based terminological systems in the AP domain to bridge the semantic gap between diagnostic histopathology and image analysis.
This thesis has contributed to the scientific state-of-the-art in the fields of Medical Informatics, Image analysis, Information Systems, and Biomedical Engineering. This is evidenced with the publications derived from this thesis in international conferences. The specific concluding remarks of this thesis are listed as follows:

6.1.1 Significance and comparison with related work

6.1.1.1 What was already known on the topic?

[Part 2] There do exist reference models for AP observations. The CAP CC&P is a very valuable knowledge source about cancer grading/scoring including quantifiable observable entities of prognostic value for the most common cancers as well as their corresponding explanatory notes describing the quantitative features of prognostic value that could be measured in AP images. Standard Development Organizations (SDOs) such as HL7 or DICOM and international initiative like integrating the Healthcare Enterprise provide formal models representing high-level AP observations required in cancer AP reports. These models are designed for information exchange between different modules involved in the workflow of digital pathology.
A workable standard-based formal representation of histopathological knowledge integrating both observable entities reported by humans (APO) (pathologists) and quantitative features (APQF) automatically computed by machines were still needed.

[Part 3] There do exist reference models for image analysis tasks in AP.

6.1.1.2 What this study added to our knowledge?

[Part 2] We proposed a semi-automated workflow for selecting candidate ontologies/semantic sources for semantic annotation of textual documents in a given domain. This workflow was applied on the AP Quantifiable Features (APQF).
We also proposed an approach and tool (Mental Maps) and Formal representation based on the CAP-CC&Ps, to support AP experts in building a standard-based representation of low-level morphological abnormalities.

We built a formal model of Anatomic Pathology Quantifiable Features (APQF) in which APQF are organized by feature categories and defined in the context of each organ specific grade/score system.
Additional efforts are needed to achieve a workable standard-based formal representation of histopathological knowledge integrating both observable entities reported by humans (pathologists) and quantifiable entities automatically computed by machines.
Providing such unique formal representation contributes to more efficient use of computer aided diagnosis based on automatic analysis of whole slide images (WSI).

[Part 3] We identified key imaging knowledge and concepts issued from different community sources: Matlab, ImageJ, ITK and histopathology imaging contests. We initiate a formal model PIPTO by integrating this knowledge with existing semantic resources in NCBO and UMLS.

This thesis has contributed to the scientific state-of-the-art in the fields of Medical Informatics, Image analysis, Information Systems, and Biomedical Engineering. This is evidenced with the publications derived from this thesis in international conferences.

6.1.2 Recommendations

The objectives of this thesis were motivated, first, by the background and recommendations given from the years of experience of societies of pathologists – College of American Pathologists (CAP) in US and "*Association pour le Développement de l'Informatique en Cytologie et Anatomie Pathologique*" (ADICAP) in France and standard development organizations or initiatives – IHE, HL7, DICOM – in the Anatomic Pathology (AP) domain.
And second, by the global necessity of developing innovative survival prediction methods in cancer-based knowledge driven Computer-Assisted Diagnosis (CAD) Tools as justified in the scientific state-of-the-art and Big Data tendencies.
As such, continuing with the research cycle, the developed methods and research findings in this thesis can establish the starting point of further research branches based on them, in addition to further technological developments. The following recommendations are suggested.

6.1.3 State of the art, Contribution and Innovative aspect of Smart'GRADE

Smart'GRADE offers the opportunity to consolidate and valorise this significant knowledge for better diagnostic histopathology protocols.
We propose a "Standard translation" of histopathology imaging techniques to support the interpretation of datasets from multiple modalities and at different scales. Digital pathology is a major area of application on which very few global teams have yet focused. On the market, there are solutions from the biomedical industry such as:
- TRIBVN Healthcare, which accompanies the laboratories in the implementation of the digital pathology
- DATEXIM proposes an automated screening system for cervical cancer
- IMAGIA, which detects and quantifies early changes caused by cancer.

Additional efforts are needed to achieve a workable standard-based formal representation of histopathological knowledge integrating both observable entities reported by humans (pathologists) and quantitative features and image p automatically computed by machines.

Providing such unique formal representation paves the way for more efficient use of computer aided diagnosis in AP as well as for the development of new biomarkers based on automatic analysis of whole slide images (WSI).

Not much explored up to now, the Digital Pathology field tends to become more competitive in the coming years, given the challenges in terms of health economy and patient quality care.

Following-up a first prototype realized with MICO [79] for the mitoses count in the scope of breast cancer diagnosis, we initiated the Smart'GRADE project. In this context, a preliminary work on the semantic approach for histopathologic diagnosis was recently published by our team [6] using the CAP protocols to generate a vocabulary dedicated to histopathology, capable of effectively helping the pathologist's daily work on virtual slides.

Figure 0-1, shows the expected Smart'GRADE platform with the following component modules: two (2) termino-ontologies, the semantic-core technologies », services and Imaging services. As you can see we obtain a platform combining knowledge visualization, rules for reasoning and semantic profiles of imaging tasks.

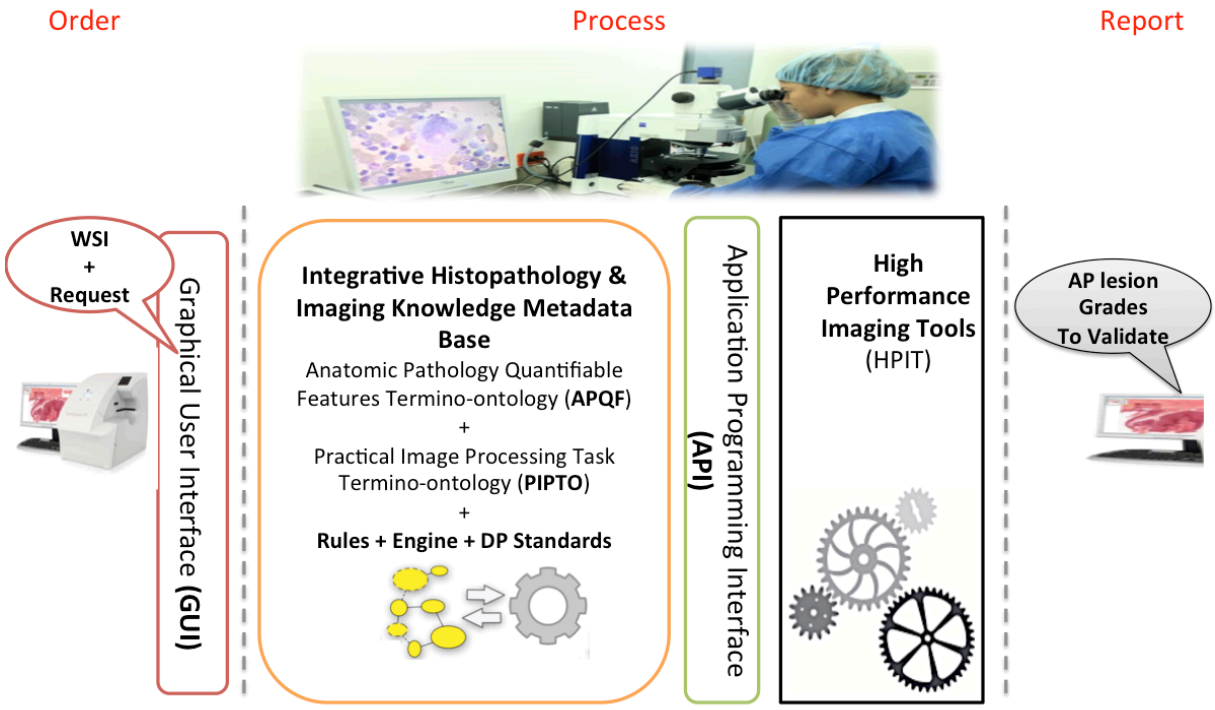


Figure 0-1: Expected workflow for integrating the Histopathology metadata base for Image tagging

Ultimately, Smart'GRADE will improve the pathologist's daily practice in terms of reliability, traceability, and performance. Smart'GRADE brings a new approach/protocol into digital pathology via:

A unique knowledge base:

- "Certified", using the protocols of the American College of Pathologists, which is a world reference for pathologists
- "Competitive", use of algorithms validated by scientific committees "Perennial", with automatic updating of the knowledge base via their semantic links with existing health resources and terminologies.

A "smart" decision support system that:

- Integrates practitioner feedback for each case. This consists of locating and presenting to the expert the alternatives used previously.
- Suggests context-appropriate approaches, explaining some practices for learning or refreshing the practitioner's memory.

It is important to note that the goal of Smart'GRADE is not to replace the pathologist, but rather to accompany and facilitate his long and cumbersome tasks, to ensure the traceability and reproducibility of the diagnosis process: the last word remains to the pathologists!

6.2 Perspectives: maturation program of Smart'GRADE

With each year about 920,000 people treated and 145,000 deaths, cancer is the leading cause of death in France[52]. Diagnoses for this scourge are increasingly in demand, with 320,000 new cases per year. However, the latter require a long and tedious process with repetitive acts, based on the experience and judgment of the physician [28], [181]. Also, opinions may vary depending on the practitioner, whose reliability decreases after hours of microscopic observation [136], [182]

The challenges of developing screening tools, improving diagnostics and aids in the therapeutic follow-up of cancers remain major in terms of medical reliability and in terms of health economics. We offer a reliable decision-making service to facilitate diagnosis via automatic scanning of the scanned slides. Thanks to Smart'GRADE, the doctor's judgment on the type and evolution of cancer detected becomes more reliable, reproducible and traceable. The Smart'GRADE technology is based on an imbrication of medical protocols with certified image analysis algorithms.

6.2.1 Smart'GRADE project: Context, Services and Process

The European Union considers cancer to be one of the main public health issues in its member countries[53]. In France, breast cancer is the first female cancer with more than 50,000 new cases estimated each year and a standardized incidence rate of + 2.1% per year on average[54].

Since 2003, the various decision-making bodies have put in place and adopted numerous recommendations and actions to support the fight against cancer. The impact of this scourge has triggered government commitment and the implementation of three "Cancer Plans" piloted by the French National Cancer Institute[55].

This is why we proposed the Smart'GRADE project, which aims to be a reliable and effective decision support tool for pathologists in the diagnosis of breast cancer precisely during the pathological examination[56]. Our technology is based on the construction of two termino-ontologies [25] by combining knowledge extracted from the medical protocols of the

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52 https://www.inserm.fr/

53 European Journal of Cancer, vol. 49, no. 6, pp. 1374–1403, Apr. 2013

54 Hospices civils de Lyon, Institut national de la santé et de la recherche médicale : www.invs.sante.fr

55 http://www.e-cancer.fr

56 Pathologic examination involves the microscopic examination of cells or tissues taken from an organ; it is also called histopathological examination.

College of American Pathologists[57] with image analysis algorithms, validated by the scientific community [166].
The objective is to establish an effective collaboration between medicine, imaging and computer science. Smart'GRADE will improve diagnostic decision support tools in terms of performance, traceability and reproducibility in the field of histopathology. Figure 0-2 summarizes the diagnosis process for breast cancer with the intervention of Smart'GRADE

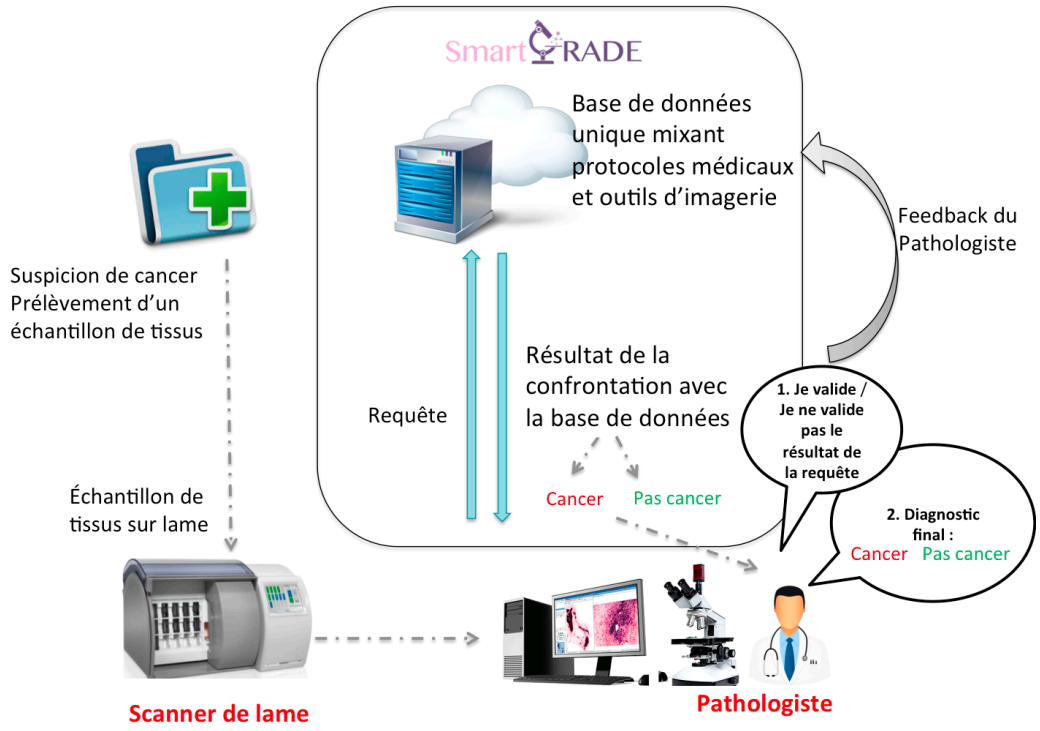

Figure 0-2 Smart'GRADE intervention in the breast cancer diagnosis process

When a person presents symptoms or abnormalities are detected during a screening test, a number of tests (see Appendix B) must be performed to make a diagnosis. This assessment also makes it possible to define the treatment proposal that will be most adapted according to the type and the aggressiveness of the cancer.
Anatomopathological examination of a biopsy tissue sample makes it possible to evaluate the type of cancer and to define its grade.
The pathologist examines the tumor under a microscope and evaluates three morphological parameters: cellular architecture[58], nucleus shape, and mitotic activity[59]. However, this process remains tedious and the reliability falls after hours of observation with the eyes focused to the microscope (***1 case out of 3*** *is not detected during the manual analyses!)* [91].

Smart'GRADE intervenes at this precise stage to help the pathologist in the evaluation of these quantifiable criteria. Instead of a subjective (educated) visual analysis of the images,

[57] With more than 18,000 member physicians, the College of American Pathologists (CAP) is the world's largest association of pathologists. Founded in 1947, CAP promotes and advocates excellence in the practice of pathology and laboratory medicine.
[58] The appearance of cancer cells
[59] The number of cells in division that reflects the rate at which cancer cells develop

Smart'GRADE proposes a formal, traceable and reproducible approach to the diagnostic approach. All the concepts manipulated during the diagnostic procedure from the image are identified with a precise definition of the morphological characteristics and their role in the final diagnosis.

6.2.2 Strengths, Weaknesses, Opportunities and Threats

The table below summarizes the SWOT analysis of Smart'GRADE project

S	• Strong scientific Board with recognized legitimacy: Laboratory of Medical Computing and e-Health Knowledge Engineering (Limics), Biomedical Imaging Laboratory (LIB) • Multidisciplinary team with R & D and Business Development profiles • A strong network already established: Research laboratories, DGRTT UPMC, SATT Lutech, PEPITE Paris Center, European network with EIT Health • Technology adapted to the understanding and practice of physicians with the use of protocols defined by the College of American Pathologists (CAP) and Ontologies designed under the supervision of pathologists. • Image analysis algorithms certified and validated by scientific committees on real medical datasets (from Contests) • Sustainability of the developed knowledge base: continuous update	O	• Major corporate issues with strong market trend and low competition • Scientific opportunities: European Congress on digital pathology organized for 13 years with a constant increase of publications in the field. • Recent advances in digital pathology (virtual slides, high-speed scanners) in the last decade • Horizon 2020 National and Regional Health Strategy Plan • Partnership and funding opportunities (Pépite, Agoranov Incubator, UPMC DGRTT, EIT Health, SATT Lutech) • Community of active digital pathology with a demanding (innovative tools) end user profile • Xerfi[60] shows that "the historical business model of the manufacturers, based on the sale of equipment and maintenance, will be gradually supplanted by new models based on the provision of services with high added value" • Great advances in artificial intelligence, computing, storage and

[60] The Global Medical Technology Industry: the market, Market Analysis – 2017-2023 Trends – Corporate Strategies

	from existing semantic platforms (Bioportal & UMLS) with the integration of feedback from the user/practitioners		transmission power (Google with DeepMind Cancer[61]) and semantic web technologies (IBM with Watson) • Increased investment in the public and private sector, the rapid growth of the aging population
W	• Need to position against the work of direct competitors and large companies: Google, IBM, Datexim, Definiens, etc. • Licensing rights for certain sources or products: (SNOMEDCT, CAP protocols, MATLAB) • Necessity of cohesion of the team and the scientific council	T	• Training and teaching of the use of Smart'GRADE, Dissemination. • Main risks associated with digital technologies: the loss of confidentiality is cited by 89% of the doctors surveyed, and this is far from unequal access to care (72%) or dehumanization of the doctor-patient relationship (71%). • Flexibility to respond to expert requests

Table 0-1 SWOT analysis of the Smart'GRADE project

6.2.3 Maturation and valorisation prospects

For the implementation of Smart'GRADE, we have established a planning in three phases:

- Maturation: Identification of image analysis algorithms, Refine knowledge base, Integration of imaging modules with medical protocols.
- Test & Regulation: Implementation, Clinical tests, Clinical regulatory standards and Validation.
- Marketing: Client early-adopters, Market penetration strategy, Marketing

[61] "Google uses artificial intelligence to diagnose breast cancer": www.deccanchronicle.com

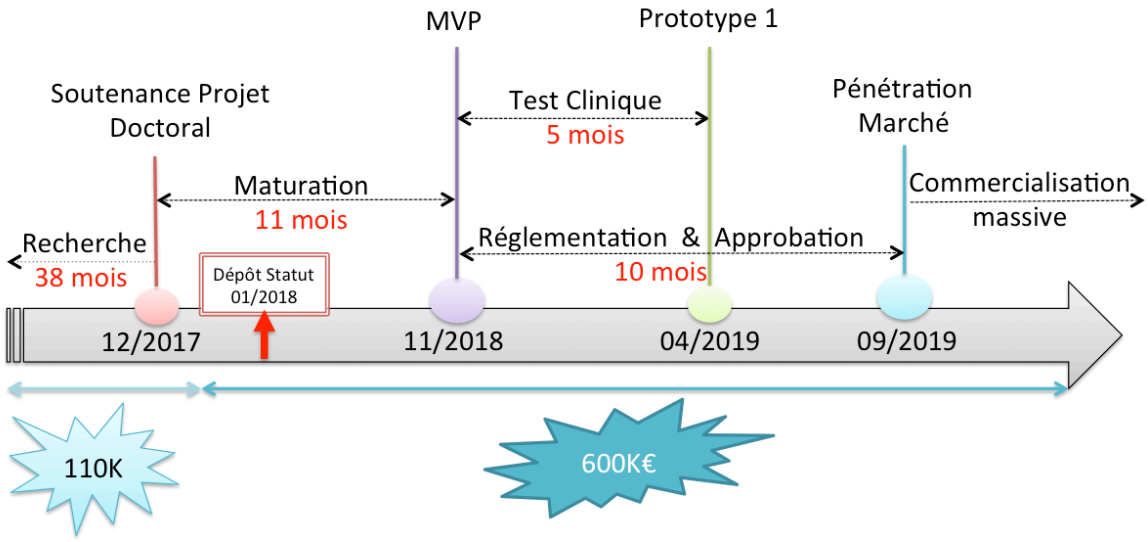

Figure 0-3: Smart'GRADE maturation and implementation planning

6.2.4 Technology Readiness Level (TRL)

Table 0-2 summarizes existing tools, concepts and models that will be brought into the Smart'GRADE project by collaborating partners. Modules will be further extended and improved during the project maturation. The TRLs (Technology Readiness Levels) for all aspects applied to the developments of Smart'GRADE modules tends to achieve a TRL level within 6 to 8 for the overall system.
After a thorough validation, Smart'GRADE can be implemented in daily routine (both for clinical diagnostics and education) at Year 2 (2019) with a TRL8 ready system. In parallel, we will work in close collaboration with the Standards Institution of the European Code of Conduct for Research Integrity, Regulations and Approvals.

Tool/Concept/Model	**Partner Concerned**	**Current TRL**	**Brief Description & Intended use in Smart'GRADE**	**Target TRL**
Knowledge database	LT	TRL4	Breast Cancer grading support modules with the two termino-ontologies	TRL6
Int Middleware	LIMICS	TRL5	Interoperability middleware conception by considering technical, semantic, security and privacy issues	TRL7
Terminology mapping Editor (TME)	AP-HP	TRL3	A semantic alignment tool developed within EHR4CR project	TRL6
Smart'GRADE	FAU	TRL7	Top ranking imaging	TRL8

Platform			algorithm catalogues	

Table 0-2 : existing tools, concepts and models contributions for Smart'GRADE

6.2.5 Scientifique Board, Team and Methodology

LTO – Senior Researcher at INSERM. Strong experience in Dynamic systems, Complexity and Computer Sciences for Health decision-making. Developed several health information platorms.

MCJ – Director of Research at INSERM. Involved in several European projects. Expert in Knowledge engineering, decision support systems and Artificial intelligence.

CD - Deputy Director of INnovation Data Web Department(WIND) of the Information Systems Division (ISD) at Assistance Publique des Hôpitax de Paris (AP-HP). Pathologist with a strong experience in eHealth, Medical informatics and standardisation.

FC- Former Head of Anatomopathology Department, Hôpital la Pitié Salpêtrière, Paris. Senior pathologist much involved in Digital Pathology, practical hospital experience, provides benchmark datasets and clinical sessions for contest.

JK – Founder and R&D responsible of Tribvn SAS, pioneer and valuable contribution in the European Digital Pathology community. Expertise in Medical image analysis, Standards (IHE & DICOM), DP industry and entrepreneurship.

6.2.6 Needed Human Ressources

	HUMAN RESOURCES	
FINANCE	M T	> Funding Strategy > Financial management > Business strategy > Marketing Strategy
RESEARCH &DEVELOPMENT	L T	> Construction of medical and imaging knowledge bases > Supervision of technical teams
ENGINEERING/TECHNICS	Semantic Web Developer (1)	> Responsible for the Semantic Web > Developing the interface between medical and imaging knowledge bases
INGENIERIE/TECHNIQUE	Back-End Developer (2)	> Web-service development > Commissioning Cloud > Transaction & Security Management
INGENIERIE/TECHNIQUE	Front-End Developer	> User interface development > Improved user experience

MARKETING/COMMERCIAL	Marketing Manager	> Ensuring a Marketing Plan > Implementation of a marketing plan (price, targeting, distribution) > Implementation of communication tools
MARKETING/COMMERCIAL	Sales manager	> Customer Prospecting > Customer loyalty
CIFRE PhD CANDIDATE	R&D	> Construction of medical and imaging knowledge bases

Table 0-3 : Human resource neeed for Smart'GRADE project

6.2.7 Targeted Market

Smart'GRADE addresses the French market but also European and North American. In the short term, the development and commercialization of Smart'GRADE will create (4) engineer positions and (3) sales positions by 2020. As part of this R & D work, Smart'GRADE will contribute to establish a collaboration for a one (1) CIFRE doctoral project and three (3) Master student training.

Smart'GRADE is an intermediate component to complement the effective implementation of the "smart" digital laboratory. It can be integrated into the Laboratory Information System (LIS), the Image Archiving and Transmission System (PACS), the sharing servers for TeleSlide to guarantee a computer aided cancer diagnosis between pathologists.

Smart'GRADE will offer a cancer diagnosis service with an online "platform" for the "semi-automatic" exploration of digital images produced in an anatomy and cytopathology (AP) laboratory. AP is a fundamental step in the diagnosis and detection of cancers. Thus, Smart'GRADE is positioned in a Business-to-Business (B2B) market composed of two client segments:
1) for companies that develop Digital Pathology software that will acquire and integrate our Technology into their Domestic Delivery (Digital Pathology software industry)
2) a cloud-based service for end-users (hospitals, diagnostic laboratories, and histopathology research centres)

6.2.8 Qualitative and quantitative market analysis

The commercialization of Smart'GRADE services will begin in France and Western Europe. This marketing strategy is due to our current scientific collaboration network. On the other hand the technological environment of digital pathology in this region is favourable. This will be followed by North America with Quebec, Canada and the United States, who are more advanced in accepting the use of CAD and machine learning methods in diagnosis.

In 2016[62], the French AP market is operated by 1,592 pathologists (2,2 anatomopathologists per 100,000 inhabitants) in regular activity with 18 million acts per year on 451 sites (322 liberal structures and 129 hospitals). Currently, 17% of the AP laboratories in hospitals are

[62] Source: Projet FlexMIm (Flexible Medical Imaging), financé par le FUI 14.

vacant and the National Council of the Medical Doctors foresees a fall of 50% by 2020[63]. Smart'GRADE proposes to facilitate the tedious task of diagnosing breast cancer in order to cope with the growth of acts, the current and future deficit.
Financially, this equipment represents:

Equipment & Service	Fees
WSI Scanner	80 to 120 k€
Software Licensing	50 k€
Fee-for-service	5€ per act
Online Storage	5 to 10k€/To

Table 0-4: Table summarizing the financial aspect of the digitization equipment and service (Source: FlexMim virtual telepathology summary document)

According to Ipsos and ASIP Santé, 63% of physicians (general practitioners and specialists) are frequent users of digital technologies and 72% expect a reduction in the redundancy of certain medical procedures or examinations.

Digital pathology is a must in the French market. The balance of virtual scanning equipment in France in June 2011 was about 40 slide scanners. This number is increased to 100 in 2016, broken down as follows:
- Approximately 50% were in Paris region (IDF) and 50% in regions,
- Approximately 50% public/private, mainly teaching and research

In the Canadian market, we identified the eastern Quebec telepathology network, which has a low density (population: 1,729,000, area: 452,600 km2). This region includes 24 hospital sites, 17 of which have an anatomopathology laboratory: 4 of the sites are without pathologists, 6 with a single pathologist. These pathologists examine an average of 24 virtual slides daily [183]. It is a favourable environment for the use of Smart'GRADE because the habits of the practitioners and the technical prerequisites are gathered. In other areas of Canada, the workload for a pathologist exceeds the Canadian Association of Pathologists' recommended limit (see figure below).

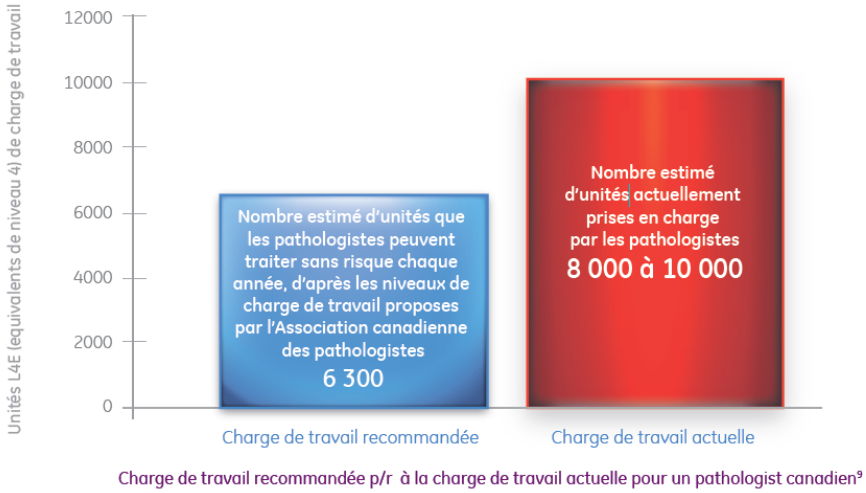

Figure 0-4: Statistics of recommended and current pathologist's workload by the Canadian Association of Pathologists

63 Source : Conseil National de l'Ordre des Médecins

Therefore, we are in two situations in favour to the use of Smart'GRADE:

- An aging population of specialists;
- The digital platforms that are increasingly used in the world and whose young generations (future pathologists or technicians) are demanding.

In addition, a Xerfi[64] study shows that "the companies historical business model, based on the sale of equipment and maintenance, will gradually be replaced by new models based on the provision of high added value services (consulting, auditing, training, research, etc.). This is an opportunity for actors who achieve higher margins on services and will have more recurring revenues through monthly or quarterly invoice. The first contracts of this type were signed between Philips and the Hospices Civils de Lyon (HCL) for the management of the imaging park, as well as between Samsung and the UniHa central purchasing office for the modernization of the Public health institutions ".

The global market for digital pathology is expected to reach $ 679.1 million in 2021, with a growth of 12.1%. Among the reasons that stimulate the demand for systems and solutions of digital pathology, in particular, we can mention:

- Increase in cancer prevalence;- Increased teleconsultations;
- Use of digital pathology for drug discovery, biomarkers, and complementary diagnosis;
- Increasing number of studies on digital pathology;
- Improved laboratory workflow efficiency

6.2.9 Possible Market and Segment Size:

For the early users and clients, we plan to rely on collaborating pathologists who accompany us in the development of Smart'GRADE with the network of AP-HP which gathers 38 hospitals, organized in 12 hospital groups.
We intend to concentrate our efforts and resources on the knowledge visualization module (to assist the training of junior pathologists) so that the added value associated with traceability and reproducibility is clearly demonstrated. Then we will move on to the clinical application that has more restriction. As mentioned above, we will start marketing on the French and European market (Western Europe in particular). The target market includes:

- The AP Hospital Laboratories
- Oncology Research Centers
- Faculties of Medicine for pathologist training
- Industrialists who may be potentially interested as a customer or partner in its development: Definiens, Tissue Gnostic, Tribun, and Datexim.

We will have a sales team that will apply a B2B approach to meeting decision makers.

6.2.10 How to build a strategic positioning?

We wish to build a Digital Pathology Smart'GRADE community through seminars, congresses, training, meetup, scientific days and press relations campaigns to promote Smart'GRADE and the services offered. This will allow us to cooperate with a learned community and communicate our services, our added value. We also want to propose a

[64] The Global Medical Technology Industry: the market, Market Analysis – 2017-2023 Trends – Corporate Strategies, Code 7XEEE02 P. FRENT., April 2017

special offer for AP Laboratories, Pathologists (targeted High-tech, which subscribe to the Digital Pathology Journal). The first 100 subscribers coming from this segment will benefit from 30% reduction in the service of Smart'GRADE.
In addition, our approach will be to identify and retain our future clients interested in DP from the Faculties of Medicine. Also by creating a coaching platform (with seniors) and offering histopathological knowledge visualization services for their training.

6.2.11 Business Model and Financial Tables

Our business model is based on a monthly service subscription. The fees and charges are determined with respect to the size of the institution as shown in the figure below. For example the monthly subscription of a hospitals with less than 200 beds is 199,90 euros. For institutions within 200 and 500 beds it is 399,90 euros and for bigger centres with more than 500 beds the fee is 599,90 euros per month. For each subscription Training for users is proposed at 499,90 per participant.

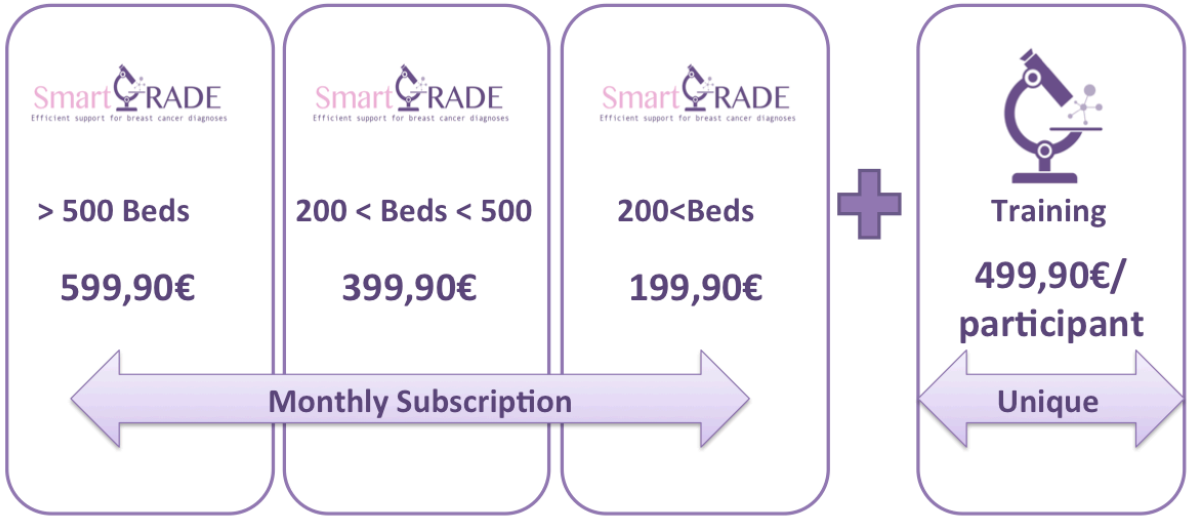

Figure 0-5: Smart'GRADE business model based on a monthly service subscription and Training fees

6.2.12 Impact and Sustainable growth

6.2.12.1 Evaluation of direct and indirect employment creation within a period 5 years

In the short term, the development and commercialization of Smart'GRADE will create (4) engineer positions and (3) sales positions by 2020. As part of its R&D work, Smart'GRADE will propose a CIFRE agreement PhD project in collaboration with a DP industrial partner and 3 proposals for Master 2 internship training.

6.2.12.2 Security & Privacy

Smart'GRADE respects the ethical principles of protection of personal data and confidentiality according to the rules of the European Union: security, confidentiality and legal framework surrounding health data. The required approval and regulatory issues are assessed with respect to the standards of the European Code of Conduct. Breast cancer is a public health problem in our modern society. This can have an impact on patients physically and psychologically. Especially for the woman, for whom the disease can affect her societal relationships: family, friends and the community in which she lives. At the organizational level and in all the key areas of expertise of the Smart'GRADE project, the gender dimension

is respected. We have expert women (5) and men (7) specializing in medical, imaging and knowledge engineering.

6.2.12.3 Healthcare Impact

Smart'GRADE addresses « Improving healthcare systems » which is a key public health and business issue.
The developments of Smart'GRADE will bring the technological innovations from ideas, to prototype, then to the widespread use in a daily clinical workflow.
Smart'GRADE project aim is to sustainably advance the "foundations" of Digital Pathology with a focus on Breast Cancer Diagnosis. Thus we propose reliability and performance in Pathology process to enhance productivity of the clinician for better care of Breast
Cancer Patients.

6.2.12.4 Social Impact

Smart'GRADE proposes an organizational innovation by helping the pathologist to respond to the growing demand by a collaborative technology of decision support to the diagnosis. Smart'GRADE also initiates an ease of updating the knowledge base derived from standard protocols, linked to the existing semantic resources and to the terminology of the domain. Beyond the diagnosis of cancer, this innovative approach can be applied to cytology, biology, molecular imaging and a wide range of approaches for the semi-automatic exploration of high-content images.

6.2.13 Legal status of the Company

Smart'GRADE will be registered under the legal status of « Société par Action Simplifiée » (SAS). We made this choice in order to facilitate future entries to the capital and employee management. We intend to submit the status of the company in January 2018 under the mentoring of Pepite Paris Centre and Agoranov incubation support.

6.2.14 Discussion and Conclusion

Smart'GRADE aims to create an efficient knowledge driven decision support system for breast cancer diagnosis. Our technology is based on the construction and integration of two termino-ontologies with relevant concepts issued from medical protocols and imaging communities. The objective is to establish an efficient collaboration between involved actors and to improve traceability, reproducibility and performance of computer aided diagnosis tools in histopathology domain particularly in Breast Cancer Grading.
Smart'GRADE provides an insight on existing Image Analysis tools for Computer Aided Diagnosis (CAD). It showcases High performance algorithms for an operational use in clinical problem solving. This dynamics trend permits to initiate activities and jobs that can be valorised. From contests and hackaton, we expect to host Top 5 imaging algorithms in Smart'GRADE System. These identified imaging tools will be remunerated with respect to their usage in the system.
According to a new report by Grand View Research Inc., CAD is most widely used for breast cancer imaging and the Market Worth $1.9 Billion By 2022.
Smart'GRADE will also help to preserve the sustainability of Digital Pathology healthcare system by providing flexible Knowledge Database issued from standard protocols and linked to existing semantic resources and terminology.
Beyond breast cancer grading, this innovative approach can be applied to cytology, biology, molecular imaging and to a large set of high-content automatic screening approaches.
Smart'GRADE aims to improve existing process for the assessment of Breast Cancer (BC) histopathologic grading. BC causes more than 460,000 new cases and 130,000 deaths death

per year in EU. Currently, analysis of breast cancer slides largely remains the work of human experts. For pathologists, this consists of hundreds of slides examined daily, a complex and time-consuming repetitive work.

Our approach is expected to yield an effective, traceable and reliable process associated to high performance image analysis identified from contests and validated with benchmarks from pathology departments. This project should therefore ease pathologist's daily work, benefit to patient care in general by ameliorating BC grading process and associated cost.

Furthermore, Smart'GRADE initiates an organizational innovation by assisting the pathologist to respond increasing demand for collaborative second opinion technology and personalized medicine by patients. Research indicates that there is growing need for integrated modules of digital pathology, which enables better interoperability reliable outcome to pathologists.

Our ambition is to improve pathologist's diagnostic performance (by not replacing them but assisting them in their daily repetitive tasks) and working conditions by enabling existing Computer Aided Quantification Tool to "understand", "assess" and better respond complex clinical requests based on their meaning. This type of "understanding" requires that the relevant information sources be semantically structured.

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Appendices

Appendix 1: Overall Protocol for the Examination of Specimens From Patients with Invasive Carcinoma of the Breast (pdf)

Protocol for the Examination of Specimens From Patients With Invasive Carcinoma of the Breast

Protocol applies to all invasive carcinomas of the breast, including microinvasive carcinoma with or without ductal carcinoma in situ (DCIS)

Based on AJCC/UICC TNM, 7th edition
Protocol web posting date: January 2016

Procedures
- Complete Excision (Less Than Total Mastectomy, Including Specimens Designated Biopsy, Lumpectomy, Quadrantectomy, and Partial Mastectomy) With or Without Axillary Contents
- Mastectomy (Total, With or Without Axillary Contents; Modified Radical; Radical)

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Full (37 pages) CAP protocol for the Examination of Specimens From Patients With Invasive Carcinoma of the Breast accessible on :
http://www.cap.org/ShowProperty?nodePath=/UCMCon/Contribution%20Folders/WebContent/pdf/cp-breast-invasive-13protocol-3200.pdf

Appendix 2: Subset of five quantifiable AP observations notes extracted from Breast DCIS and IC CAP-CC&Ps

```
Note#1- Glandular (Acinar)/Tubular Differentiation
___ Score 1: >75% of tumor area forming glandular/tubular structures
___ Score 2: 10% to 75% of tumor area forming glandular/tubular structures
___ Score 3: <10% of tumor area forming glandular/tubular structures
___ Only microinvasion present (not graded)
___ No residual invasive carcinoma after presurgical (neoadjuvant) therapy
___ Score cannot be determined
F.  Histologic Grade
The Nottingham combined histologic grade evaluates the amount of tubule formation, (…).


        Note#2- Mitotic Rate
___ Score 1 (≤3 mitoses per mm2) (see Table 1)
___ Score 2 (4-7 mitoses per mm2) (see Table 1)
___ Score 3 (≥8 mitoses per mm2) (see Table 1)
___ Only microinvasion present (not graded)
___ No residual invasive carcinoma after presurgical (neoadjuvant) therapy
___ Score cannot be determined
+ Number of mitoses per 10 high-power fields: ___
+ Diameter of microscope field: ___ mm
F.  Histologic Grade
All invasive breast carcinomas should be graded.12  The Nottingham combined histologic grade
(Elston-Ellis modification of Scarff-Bloom-Richardson grading system) should be used for
reporting.  Within each stage grouping there is a relation between histologic grade and outcome.

The Nottingham combined histologic grade evaluates the amount of tubule formation, the extent of
nuclear pleomorphism, and the mitotic count (or mitotic rate). Each variable is given a score of
1, 2, or 3, and the scores are added to produce a grade. The mitotic score is determined by the
number of mitotic figures found in 10 consecutive high-power fields (HPF) in the most mitotically
active part of the tumor. Only clearly identifiable mitotic figures should be counted;
hyperchromatic, karyorrhectic, or apoptotic nuclei are excluded. Because of variations in field
size, the HPF size must be determined for each microscope and the appropriate point score
determined accordingly. It is recommended that the size be measured by using a micrometer.
However, the diameter of an HPF can also be calculated by using the method below.

Measuring the Size of a High-Power Field (HPF) With a Ruler

Use a clear ruler to measure the diameter of a low-power field.  This number can be used to
calculate a constant based on the following formula:

Eyepiece Magnification x Objective Magnification x Microscopic Field Diameter
 = A Constant

When the value of the constant is known, the diameter of an HPF can be calculated for other
objectives by using the following formula:
Unknown Field Diameter = Constant/(Eyepiece Magnification x Objective Magnification)

Half of the field diameter is the radius of the field (r), which can then be used to calculate the
area of the HPF:

3.1415 x r 2 = Area of Microscopic Field

If the microscopic field diameter or the area of the field is known, Table 1 can be used to
determine the number of mitoses corresponding to different scores.

Table 1.  Score Categories According to Field Diameter and Mitotic Count
Scoring Categories of Mitotic Counts

Field diameter (mm)
Area (mm2)
Number of mitoses per 10 fields corresponding to:
```

```
        Note#3- Nuclear Pleomorphism
___ Score 1: Nuclei small with little increase in size in comparison with normal breast epithelial
cells, regular outlines, uniform nuclear chromatin, little variation in size
___ Score 2: Cells larger than normal with open vesicular nuclei, visible nucleoli, and moderate
variability in both size and shape
___ Score 3: Vesicular nuclei, often with prominent nucleoli, exhibiting marked variation in size
and shape, occasionally with very large and bizarre forms
___ Only microinvasion present (not graded)
___ No residual invasive carcinoma after presurgical (neoadjuvant) therapy
___ Score cannot be determined
F.  Histologic Grade
The Nottingham combined histologic grade evaluates the (…) extent of nuclear pleomorphism (…)


        Note#4- Overall Grade
___ Grade 1: scores of 3, 4, or 5
___ Grade 2: scores of 6 or 7
___ Grade 3: scores of 8 or 9
___ Only microinvasion present (not graded)
___ No residual invasive carcinoma after presurgical (neoadjuvant) therapy
___ Score cannot be determined
Note: The grade corresponds to the largest area of invasion. If there are smaller foci of invasion
of a different grade, this information should be included under “Additional Pathologic Findings.”
F.  Histologic Grade
All invasive breast carcinomas should be graded.12  The Nottingham combined histologic grade
(Elston-Ellis modification of Scarff-Bloom-Richardson grading system) should be used for
reporting.  Within each stage grouping there is a relation between histologic grade and outcome.
The Nottingham combined histologic grade evaluates the amount of tubule formation, the extent of
nuclear pleomorphism, and the mitotic count (or mitotic rate). Each variable is given a score of
1, 2, or 3, and the scores are added to produce a grade.

        Note#5- Nuclear Grade (Note F)
___ Grade I (low)
___ Grade II (intermediate)
___ Grade III (high)

F.  Nuclear Grade
The nuclear grade of DCIS is determined using 6 morphologic features (Table 2).4,30
Table 2. Nuclear Grade of Ductal Carcinoma In Situ
Feature
Grade I
(Low)
Grade II (Intermediate)
Grade III
(High)
Pleomorphism
Monotonous (monomorphic)
Intermediate
Markedly pleomorphic
Size
1.5X to 2X the size of a normal RBC or a normal duct epithelial cell nucleus
Intermediate
>2X the size of a normal RBC or a normal duct epithelial cell nucleus
Chromatin
Usually diffuse, finely dispersed chromatin
Intermediate
Usually vesicular with irregular chromatin distribution
Nucleoli
Only occasional
Prominent, often multiple
Mitoses
Only occasional
Intermediate
May be frequent
Orientation
Polarized toward luminal spaces
Intermediate
Usually not polarized toward the luminal space
Definition: RBC, red blood cell.
```

Appendix 3: Summary for the 67 CAP-CC&Ps Required_Element (xls)

Refer to «Appendix3_Summary67CAPCCP_RequiredElements.xls»

Appendix 4: "gold standard" list identified in the 5 notes - Breast CAP-CC&Ps - by the two senior pathologists

Note#1- Glandular (Acinar)/Tubular Differentiation
tubule formation
tumor area
Note#2- Mitotic Rate
mitotic count (mitotic rate,mitotic score, mitoses per mm2)
number of mitotic figures
consecutive high-power fields (HPF)
most mitotically active part of the tumor
clearly identifiable mitotic figures
hyperchromatic nuclei
karyorrhectic nuclei
apoptotic nuclei
high-power fields (HPF) size (mitoses per mm2)
Note#3- Nuclear Pleomorphism
nuclear pleomorphism
nuclei small with little increase in size in comparison with normal breast epithelial cells
normal breast epithelial cells
regular outlines
uniform nuclear chromatin
little variation in size
Cells larger than normal
open vesicular nuclei
visible nucleoli
moderate variability in both size and shape
vesicular nuclei
(often with) prominent nucleoli
marked variation in size and shape
(occasionally with) very large and bizarre forms
Note#4- Overall Grade
Histologic Grade
Nottingham combined histologic grade (Elston-Ellis modification of Scarff-Bloom-Richardson grading system)
tubule formation
nuclear pleomorphism
mitotic count (mitotic rate)"

Note#5- Nuclear Grade (Note F)
Ductal Carcinoma In Situ
Nuclear Grade
Feature
(Nuclear) Grade I (Low)
(Nuclear) Grade II (Intermediate)
(Nuclear) Grade III (High)
(Nuclear) pleomorphism
Monotonous (monomorphic) (nuclei)
Markedly pleomorphic (nuclei)
(Nuclear) size
(Nuclear) size of a normal red blood cell
(Nuclear) size of a normal duct epithelial cell nucleus
normal duct epithelial cell nucleus
red blood cell
Chromatin
Usually diffuse, finely dispersed chromatin
Usually vesicular chromatin with irregular chromatin distribution
Nucleoli
Only occasional nucleoli
Prominent, often multiple nucleoli
Mitoses
Only occasional mitoses
May be frequent mitoses
(Nuclear) orientation
Polarized toward luminal spaces
Usually not polarized toward the luminal space

Appendix 5:Copy of Invasive breast carcinoma "work aid document" with related filled examples

Recognizing that there is significant variability in format from institution to institution, the CAP has established a specific format for 'synoptic reporting' within a surgical pathology report on cancer specimens. This format minimizes the variability between institutions and is presented in such a way that clinicians can easily and quickly find it in the surgical pathology report, and ensures that the appropriate information needed for patient care is provided. Figure 6 shows a copy of Invasive breast carcinoma work aid documents. Figure 7 and Figure 8 represent respectively "Acceptable synoptic Report Example" for "Invasive Breast Carcinoma" and "Ductal Carcinoma In Situ" of the Breast. At last Figure 9 is an example of "Unacceptable synoptic Report Example for Colon, right hemi-colectomy".

INVASIVE BREAST CANCER WORK AID

Advancing Excellence

Patient Name:	Hospital #:	Case/Accession #:
Diagnosis:		

Specimen Laterality: ☐ Right ☐ Left ☐ Not specified

Specimen: ☐ Partial breast ☐ Total breast

☐ Other: _____

Procedure:

☐ Excision w/o wire-guided loc ☐ Total mastectomy

☐ Excision w/ wire-guided loc. ☐ Other: _____

Specimen Integrity:

☐ Single intact specimen ☐ Fragmented

☐ Multiple designated specimens ☐ Other: _____

Lymph Node Sampling:

☐ No lymph nodes ☐ Axillary dissection

☐ Sentinel lymph node(s) ☐ Lymph node(s) present w/in breast

☐ Other: _____

Specimen Size: _____ cm ☐Cannot be determined

Invasive Carcinoma Size: _____ cm ☐Cannot be determined

Tumor Focality: (referring to invasive carcinoma)

☐ Single ☐ Indeterminate

☐ Multiple* # of foci _____ *size of largest focus: _____

Extent of Tumor:

☐ Skin not present ☐ Skin invasion w/o ulceration ☐ Skin ulceration

☐ Skin Satellite nodules ☐ No skin invasion

☐ No skeletal muscle present ☐ Skeletal muscle present, no invasion

☐ Pectoralis muscle invasion ☐ Chest wall beyond pectoralis invasion

☐ Nipple Paget disease ☐ DCIS does not involve nipple epidermis

In Situ Component:

☐ None Present ☐ LCIS

☐ DCIS ☐ Mixed DCIS and LCIS

☐ DCIS, EIC positive

☐ DCIS, EIC negative

Histologic Type:

☐ Ductal ☐ Lobular ☐ Only microinvasion present

☐ Mixed ductal and lobular

☐ Other (specify): _____

Histologic Grade (SBR/Nottingham):

☐ 1 (3-5 points) ☐ 2 (6-7 points) ☐ 3 (8-9 points)

Glandular/Tubular Differentiation:

☐ 1 (>75% tubules) ☐ 2 (10 to 75% tubules) ☐ 3 (<10% tubules)

Nuclear Pleomorphism:			**Mitotic Count:**		
☐ 1	☐ 2	☐ 3	☐ 1	☐ 2	☐ 3

Margins:

☐ **Margins cannot be assessed**

☐ **Margins uninvolved by invasive carcinoma**

Distance from closest margin: _____ mm *Specify margin_____

☐ **Margins involved by invasive carcinoma** (*Specify margins and extent)

☐ *Superior _____ ☐ *Medial _____ ☐ *Anterior _____

☐ *Inferior _____ ☐ *Lateral _____ ☐ *Posterior _____

☐ **Margins uninvolved by DCIS**

Distance from closest margin: _____ mm *Specify margin_____

☐ **Margins involved by DCIS** (*Specify margins and extent)

☐ *Superior _____ ☐ *Medial _____ ☐ *Anterior _____

☐ *Inferior _____ ☐ *Lateral _____ ☐ *Posterior _____

Lymph nodes (if present):

_____ # SLNs Examined	_____ # LNs w/ Micromets
_____ # Total LNs examined	_____ # LNs w/ ITCs
_____ # LNs w/ Macromets	_____ Size of largest metastatic deposit (if present)

Pathologic Staging: (Descriptors: m-multiple, r-recurrent, y-post tx)

☐ m	☐ r	☐ y	pT _____	pN _____	pM (if present) _____

Estrogen (ER) Receptor: ☐ Not performed ☐ Pending

☐ Percentage of cells with nuclear immunoreactivity: _____ %

Intensity of staining: ☐ Absent ☐ Weak ☐ Medium ☐ Strong

☐ Positive ☐ Negative ☐ Other: _____

Progesterone (PgR) Receptor: ☐ Not performed ☐ Pending

☐ Percentage of cells with nuclear immunoreactivity: _____ %

Intensity of staining: ☐ Absent ☐ Weak ☐ Medium ☐ Strong

☐ Positive ☐ Negative ☐ Other: _____

Her-2/Neu IHC: ☐ Not performed ☐ Pending

☐ Negative (Score 0) ☐ Negative (Score 1+) ☐ Equivocal (Score 2+)

☐ Positive (Score 3+) ☐ Other: _____ ☐ Unknown

Her-2/Neu FISH: ☐ Not performed ☐ Pending

☐ Not Amplified ☐ Equivocal

☐ Amplified

*Average number of HER2 gene copies per cell: _____

*Average number of chromosome 17 per cell: _____

*Ratio: _____

☐ Unknown ☐ Other: _____

Comments:

If the patient has received neo-adjuvant therapy, or if only microinvasion is present, refer to the Invasive Breast Carcinoma protocol.

* Data elements with asterisks are not required.

Breast Work Aid 3.0.0.0

Figure 6: Work Aid document format for Invasive Breast Carcinoma

Specimen laterality:	Right		
Specimen:	Partial breast		
Procedure:	Excision with wire guided localization		
Specimen integrity:	Single intact specimen		
Lymph node sampling:	Sentinel lymph node(s), Axillary dissection		
Specimen size:	6 cm		
Invasive carcinoma size:	1.5 cm		
Tumor focality:	Single		
Extent of tumor			
Skin	No skin present		
Chest wall	No skeletal muscle present		
In situ component:	DCIS present, EIC positive		
Histologic type:	Ductal		
Histologic grade:	2 (6 -7 points)		
Nuclear score:	2		
Tubule score:	2		
Mitosis score:	2		
Margins			
Invasive carcinoma	Margins uninvolved by invasive carcinoma		
	Distance from closest margin: 4 mm		
DCIS	Margin(s) positive for DCIS		
Lymph nodes			
# of Sentinel LNs	4		
# of Total LNs	9		
# of LNs with macrometastases	3		
# of LNs with micrometastases	1		
# of LNs with ITCs	1		
Size of largest metastasis	8 mm		
Pathologic TNM classification:	pT1c	pN2a	
Estrogen and Progesterone Receptors			
ER	Percentage of cells with nuclear immunoreactivity:		30%
	Intensity of staining:		Medium
	Interpretation:		Positive
PgR	Percentage of cells with nuclear immunoreactivity:		20%
	Intensity of staining:		Medium
	Interpretation:		Positive
HER2/neu Immunohistochemistry Result: Negative (score 0)			
HER2/neu FISHResult: Not amplified			

Pathologic Staging [AJCC 7th edition TNM Staging System] Key:

Primary Tumor (Invasive Carcinoma) (pT)

pTX: Primary tumor cannot be assessed **pT0:** No evidence of primary tumor **pTis** (DCIS): Ductal carcinoma in situ **pTis** (LCIS): Lobular carcinoma in situ **pTis** (Paget): Paget disease of the nipple *not* associated with invasive carcinoma and/or carcinoma in situ (DCIS and/or LCIS) in the underlying breast parenchyma **pT1:** Tumor ≤20 mm in greatest dimension **pT1mi:** Tumor ≤1 mm in greatest dimension (microinvasion) **pT1a:** Tumor >1 mm but ≤5 mm in greatest dimension **pT1b:** Tumor >5 mm but ≤10 mm in greatest dimension **pT1c:** Tumor >10 mm but ≤20 mm in greatest dimension **pT2:** Tumor >20 mm but ≤50 mm in greatest dimension **pT3:** Tumor >50 mm in greatest dimension **pT4:** Tumor of any size with direct extension to the chest wall and/or to the skin (ulceration or skin nodules). *Note:* Invasion of the dermis alone does not qualify as pT4. **pT4a:** Extension to chest wall, not including only pectoralis muscle adherence/invasion **pT4b:** Ulceration and/or ipsilateral satellite nodules and/or edema (including peau d'orange) of the skin which do not meet the criteria for inflammatory carcinoma **pT4c:** Both T4a and T4b **pT4d:** Inflammatory carcinoma

Regional Lymph Nodes (pN)

pNX: Regional lymph nodes cannot be assessed (eg, previously removed, or not removed for pathologic study) **pN0:** No regional lymph node metastasis identified histologically **pN0 (i-):** No regional lymph node metastases histologically, negative IHC **pN0 (i+):** Malignant cells in regional lymph node(s) no greater than 0.2 mm and no more than 200 cells (detected by H&E or IHC including ITC) **pN0 (mol-):** No regional lymph node metastases histologically, negative molecular findings (reverse transcriptase polymerase chain reaction [RT-PCR]) **pN0 (mol+):** Positive molecular findings (RT-PCR), but no regional lymph node metastases detected by histology or IHC **pN1mi:** Micrometastases (greater than 0.2 mm and/or more than 200 cells, but none greater than 2.0 mm). **pN1a:** Metastases in 1 to 3 axillary lymph nodes, at least 1 metastasis greater than 2.0 mm **pN2a:** Metastases in 4 to 9 axillary lymph nodes (at least 1 tumor deposit greater than 2.0 mm) **pN3a:** Metastases in 10 or more axillary lymph nodes (at least 1 tumor deposit greater than 2.0 mm)

Distant Metastasis (M)

cM0(i+): No clinical or radiographic evidence of distant metastasis, but deposits of molecularly or microscopically detected tumor cells in circulating blood, bone marrow, or other nonregional nodal tissue that are ≤0.2 mm in a patient without symptoms or signs of metastasis **pM1:** Distant detectable metastasis as determined by classic clinical and radiographic means and/or histologically proven >0.2 mm

Breast Work Aid 3.0.0.0

Figure 7: Acceptable Synoptic report example for Invasive Breast Carcinoma

Specimen, Laterality, Procedure: Partial breast, right, excision without wire-guided localization

Specimen Integrity: single intact specimen

Specimen Size (for excisions less than total mastectomy): 8.2 cm in greatest dimension

Lymph Node Sampling: No lymph nodes present

***Tumor Site**: Not specified

Estimated size (extent) of DCIS (greatest dimension using gross and microscopic evaluation): at least 3.8 cm

Histologic Type: Ductal carcinoma in situ.

***Architectural Patterns:** Solid

Nuclear Grade: Grade II (intermediate)

Necrosis: Present, focal (small foci or single cell necrosis)

Margins: Margin(s) uninvolved by DCIS
Distance from closest margin: 4 mm
*Specify margins:
*Distance from superior margin: 4 mm
*Distance from inferior margin: >10 mm
*Distance from medial margin: 6 mm
*Distance from lateral margin: >10 mm
*Distance from anterior margin: >10 mm
*Distance from posterior margin: >10 mm

Pathologic Staging (pTNM)
Primary Tumor (pT): pTis (DCIS):Ductal carcinoma in situ

Regional Lymph Nodes (pN): pNX (Cannot be assessed (not removed for pathologic study)

Distant Metastasis (pM): Not applicable

December 13, 2011 v2.0

Figure 8: Acceptable synoptic Report Example for Ductal Carcinoma In Situ of the Breast (This example combines specimen, laterality, and procedure on one line, as allowed)

Diagnosis:
Colon, right hemicolectomy:
Invasive adenocarcinoma, 3.4 x 3.0 cm involving muscularis propria
All margins negative
No lymphatic invasion
No metastatic tumor identified
NOT ACCEPTABLE AS SYNOPTIC STYLE REPORTING:
NOT ALL ELEMENTS ARE PRESENT AND DIAGNOSTIC PARAMETER PAIR IS ABSENT

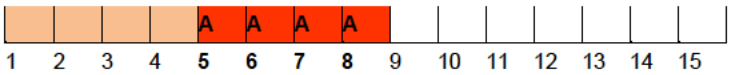
Figure 9: Unacceptable synoptic Report Example for Colon, right hemi-colectomy

Appendix 6: List of organ-specific AP observations derived from the CAP-CC&Ps of the 20 most frequent cancers (xls)

Refer to «Appendix6_ APObservationsDerivedFromCAPCCP-20MostFrequentCancers.xls»

Appendice 7: Example of inconsistencies correction in the classification of the histogénétic types of the ADICAP coding system (pdf)

La pathologie tumorale est codée au niveau des 5e, 6e, 7e et 8e caractère du code obligatoire par 4 caractères alphanumériques. Le 1e caractère est une lettre identifiant les grandes familles histogénétiques des tumeurs par leurs initiales. La classification histogénétique des tumeurs, basée sur le type cellulaire/tissulaire d'origine a peu changé depuis un siècle, bien qu'elle soit imparfaite et parfois difficile à appliquer.



L'une des limitations du système de codification ADICAP est liée à une incohérence du dictionnaire D5 en ce qui concerne la classification histogénétique (1er caractère). Par exemple: sous le Caractère B « Tumeur Basocellulaire » et « Tumeur Blastemateuse » sont classées ensemble alors qu'il s'agit de classes histogénétique différentes. Ce type d'incohérence est retrouvé dans les sous classes B, D, M, R, S et T illustrés dans la figure ci-dessous.

A Tumeur Adénomateuse - Adénocarcinome
B Tumeur Basocellulaire - tumeur Blastémateuse
C Tumeur Cartilagineuse
D Tumeur Dento-maxillaire - tumeur Dysgénétique
E Tumeur Epidermoide
F Tumeur Fibrocytique, Fibroblastique et Fibrohistiocytaire
G Tumeur Germinative ou épithéliale - tumeur germinale, tumeur gonadique
H,J,K Tumeur et maladie de système des tissus Hématopoïétiques
– H Tumeur et maladie de système des tissus Hématopoïétiques
– J Lymphomes (Classification de Kiel actualisée - 1988) puis OMS2000
– K Lymphomes hodgkinien et non hodgkinien (WForm ne plus utiliser)
L Tumeur Léiomyomateuse - tumeur Lipomateuse
M Tumeur Mélanique - tumeur du Mésenchyme pluripotentiel, tumeur Mésoth
N Tumeur Nerveuse
P Paragangliome - tumeur pilaire- tumeur pinéale
Q Tumeur ostéogénique
R Tumeur du revêtement Respiratoire - tumeur rhabdomyomateuse
S Tumeur Sébacée, sudoripare, synoviale et du S.E.D.
T Tumeur Thymique - tumeur Trophoblastique
U Tumeur Urothéliale
V Tumeur Vasculaire
X Tumeur d'histogenèse indéterminée ou particulière

Tumeur Basocellulaire: Le carcinome basocellulaire (CBC) est une tumeur épithéliale développée aux dépens du tissu épidermique, survenant le plus souvent de novo, localisée uniquement à la peau, jamais sur les muqueuses, et de malignité locale.
Tumeur Blastemateuse: Tumeur du rein. Histologiquement, ces tumeurs reproduisent l'aspect d'un blastème, et associent en général :i) des zones blastémateuses indifférenciées, faites de nappes de « petites cellules rondes et bleues » ; ii) des zones blastémateuses différenciées : la différenciation varie selon le type de blastème : elle peut être épithéliale

(tubes rénaux primitifs dans un néphroblastome), neuronale (dans un neuroblastome) ou mésenchymateuse (différenciation musculaire dans certains néphroblastomes ou médulloblastomes).

La figure ci-après montre l'organisation détaillée des « Tumeur Basocellulaire » et « Tumeur Blastemateuse » dans la classification ADICAP actuelle.

B - TUMEUR BASOCELLULAIRE - TUMEUR BLASTEMATEUSE

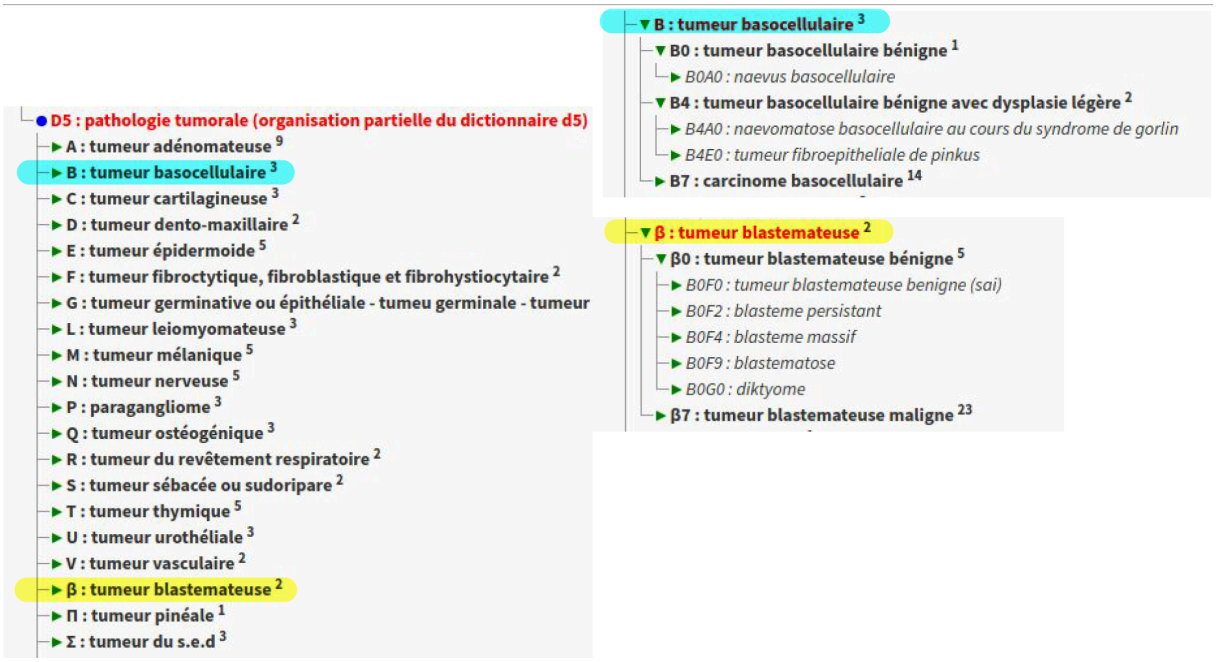
TUMEUR BASOCELLULAIRE

B0A0			**NAEVUS BASOCELLULAIRE**
B4A0	1999	MODIF	NAEVOMATOSE BASOCELLULAIRE AU COURS DU SYNDROME DE GORLIN
B4E0			TUMEUR FIBROEPITHELIALE DE PINKUS
B7A0			**CARCINOME BASOCELLULAIRE (SAI)**
B7A2			CARCINOME BASOCELLULAIRE ADAMANTINOIDE
B7A4			CARCINOME BASOCELLULAIRE ADENOIDE
B7A6			CARCINOME BASOCELLULAIRE BOWENOIDE OU A CELLULES ATYPIQUES
B7A7			CARCINOME BASOCELLULAIRE CYLINDROMATEUX
B7A8			CARCINOME BASOCELLULAIRE A DIFFERENCIATION SEBACEE
B7B2			CARCINOME BASOCELLULAIRE KERATINISANT OU PILAIRE
B7B4			CARCINOME BASOCELLULAIRE KYSTIQUE
B7B6			CARCINOME BASOCELLULAIRE PAGETOIDE OU SUPERFICIEL
B7B8			CARCINOME BASOCELLULAIRE PIGMENTE
B7D2			CARCINOME BASOCELLULAIRE SARCOMATOIDE (A CELLULES FUSIFORMES)
B7D4			CARCINOME BASOCELLULAIRE SCLERODERMIFORME
B7D6			CARCINOME BASOCELLULAIRE DE TYPE INTERMEDIAIRE
B7D8			CARCINOME BASOSPINOCELLULAIRE - CARCINOME MIXTE JUXTAPOSE BASC ET SPINOCELLULAIRE

TUMEUR BLASTEMATEUSE

B0F0			**TUMEUR BLASTEMATEUSE BENIGNE (SAI)**
B0F2			BLASTEME PERSISTANT
B0F4			BLASTEME MASSIF
B0F9			BLASTEMATOSE
B0G0	1992	AJOUT	DIKTYOME
B7F0			**TUMEUR BLASTEMATEUSE MALIGNE (SAI)**
B7F2			ESTHESIONEUROBLASTOME
B7F4			ESTHESIONEUROCYTOME
B7F6			HEPATOBLASTOME
B7F8			PANCREATOBLASTOME
B7G2			PNEUMOBLASTOME
B7G4			RETINOBLASTOME
B7G6			RETINOCYTOME
B7G8	1992	AJOUT	EPENDYMOBLASTOME
B7K0			**NEPHROBLASTOME (SAI)**
B7K1	1992	AJOUT	NEPHROBLASTOME AU SEIN D'UN RELIQUAT NEPHROGENIQUE
B7K2			NEPHROBLASTOME EPITHELIAL
B7K3	1992	AJOUT	NEPHROBLASTOME KYSTIQUE PARTIELLEMENT DIFFERENCIE
B7K4			NEPHROBLASTOME MESENCHYMATEUX
B7K5	1992	AJOUT	NEPHROBLASTOME RHABDOMYOMATEUX FOETAL
B7K6			NEPHROBLASTOME MIXTE OU COMPLEXE
B7K7	1992	AJOUT	NEPHROBLASTOME MODIFIE PAR LA THERAPEUTIQUE
B7K8	1992	AJOUT	NEPHROBLASTOME AVEC ANAPLASIE
B7K9	1992	AJOUT	NEPHROBLASTOME SANS ANAPLASIE (HISTOLOGIE FAVORABLE)
B7L0	1992	AJOUT	NEPHROBLASTOMATOSE (SAI)
B7L2	1992	AJOUT	NEPHROBLASTOMATOSE CORTICALE SUPERFICIELLE
B7L5	1992	AJOUT	NEPHROBLASTOMATOSE CORTICALE PROFONDE
B7N0	1998	MODIF	**SARCOME EMBRYONNAIRE INDIFFERENCIE - SARCOME BOTRYOÏDE**

La figure ci-après montre un exemple de notre proposition de correction de cette incohérence pour faciliter la construction de requêtes portant sur les données ACP.



Appendix 8 : CAP Cancer Protocol Annotation Results

Refer to « Appendix8_FormalismeAnnotation.xlsx » File

Appendix 9 :NCBO_REST_TextAnnotation program codes (txt)

Refer to «AppendixI_NCBO-UMLS_RestExples» Folder

```
##############################################################################################################
# This program examines text input and generate annotations from NCBO Bioportal by returning relevant classes.
# It matches words in the text to terms in specified reference ontologies
# source code: https://github.com/ncbo/ncbo_rest_sample_code
# Adapted for json output by LT laminet@gmail.com
# usage:python NCBOAnnotate_text.py
##############################################################################################################


import urllib2
import json
import os
from pprint import pprint
import requests

REST_URL = "http://data.bioontology.org"
API_KEY = "541e08cb-cd96-4db8-88b9-22f19c385edf"

def get_json(url):
    opener = urllib2.build_opener()
    opener.addheaders = [('Authorization', 'apikey token=' + API_KEY)]
    return json.loads(opener.open(url).read())

def print_annotations(annotations, get_class=True):
    for result in annotations:
        class_details = get_json(result["annotatedClass"]["links"]["self"]) if get_class else result["annotatedClass"]
        print "Class details"
        print "\tid: " + class_details["@id"]
        print "\tprefLabel: " + class_details["prefLabel"]
        print "\tontology: " + class_details["links"]["ontology"]

        print "Annotation details"
        for annotation in result["annotations"]:
            print "\tfrom: " + str(annotation["from"])
            print "\tto: " + str(annotation["to"])
            print "\tmatch type: " + annotation["matchType"]

        if result["hierarchy"]:
            print "\n\tHierarchy annotations"
            for annotation in result["hierarchy"]:
                class_details = get_json(annotation["annotatedClass"]["links"]["self"])
                pref_label = class_details["prefLabel"] or "no label"
                print "\t\tClass details"
                print "\t\t\tid: " + class_details["@id"]
                print "\t\t\tprefLabel: " + class_details["prefLabel"]
                print "\t\t\tontology: " + class_details["links"]["ontology"]
                print "\t\t\tdistance from originally annotated class: " + str(annotation["distance"])

        print "\n\n"

text_to_annotate = "Glandular (Acinar)/Tubular Differentiation. Score 1: >75% of tumor area forming glandular/tubular structures. Score 2: 10% to 75% of tumor area forming glandular/tubular structures. Score 3: <10% of tumor area forming glandular/tubular structures. Only microinvasion present (not graded). No residual invasive carcinoma after presurgical (neoadjuvant) therapy. Score cannot be determined. Histologic Grade. The Nottingham combined histologic grade evaluates the amount of tubule formation."
```

Glossary

- Adenocarcinoma : A carcinoma originating in glandular tissue.

- Atypia : Cells or tissue displaying some characteristics of a malignancy, but not considered either malignant or benign. The diagnosis of atypia generally requires a more comprehensive (and possibly invasive) follow-up to determine the true diagnosis.

- Benign : A condition which will not metastasize and is not harmful in and of itself.

- Carcinoma : A cancer of the epithelium.

- Chromatin : Nuclear material that is readily stained, consisting of the nucleic acids and associated proteins.

- Confocal : Confocal microscopy images different focal planes through the specimen.

- Counterstain : A stain used as contrast to another, generally more specific, stain.

- Cytology : The study of cells at a microscopic level, generally via a light microscopy technique.

- Cytopathology : The study of diseased cells at the microscopic level.

- Densitometry : Measurements related to the optical density of a sample.

- Ductal carcinoma : Carcinoma originating in ductal structures.

- Eosin : A pink-staining acidic dye that stains membranes and fibres.

- Epithelium : The internal and external lining of cavities within the body; also the external covering (skin).

- Fine needle aspiration : A procedure using a small needle inserted into the lesion and drawing a small amount of cellular material into a syringe; a form of aspirative cytology.

- Fluorescence imagery : Fluorescent dyes are attached to antibodies specific to some feature of interest (e.g., certain proteins) and imaged by exciting the fluorescence of the dyes with appropriate incident light. This method can very specifically target certain molecular attributes of a biological specimen.

- Gleason grading : A grading for prostate cancer, characterizing the tumor into one of 5 categories based on tumor differentiation.

- Hematoxylin : A blue-staining basic dye that stains genetic material; this is mainly seen in nuclear material, although some components of cytoplasmic and extracellular material is also stained.

- Histology : The study of tissue at a microscopic level, generally via a light microscopy technique.

- Histopathology : The study of diseased tissue at the microscopic level.

- Hyperchromasia : An overall increase in staining intensity.

- Hyperplasia : Abnormalities in the characteristics of cells and tissues, generally including an increase in cellularity and/or mitosis; often used interchangeably with dysplasia.

- Immunostain : Immunostains use antibodies to specifically target molecules of interest, similar to fluorescence imaging, but use standard dyes for viewing with light microscopy.

- in situ : Within normal boundaries, not invading surrounding tissues.

- Karyometry : Nuclear characteristics, generally texture.

- Lobular carcinoma : A type of adenocarcinoma.

- Malignant : A condition which will eventually lead to death if untreated. Malignant conditions tend to metastasize, grow uncontrollably, and lack proper tissue differentiation.

- Metastasis : The spread of cancer from the originating tissue to other parts of the body.

- Microarray : Tissue microarrays align many (hundreds or thousands) of tissue core samples on a single slide; this allows for simultaneous analysis of all samples and is commonly used in
- high-throughput operations.

- Nucleolus : A small, round sub-organelle within the cell nucleus.

- Pathology : The study of disease, with emphasis on disease structure and the effects on the body as a whole.

- Pleomorphic : Containing more than one stage of the life cycle.

- Premalignancy : A diseased state that, while not considered cancerous, will progress to cancer if left untreated.

- Stroma : Connective tissue.

- Thesauri: A thesaurus serves to minimise semantic ambiguity by ensuring uniformity and consistency in the storage and retrieval of the manifestations of content objects. It is composed by at least three elements: 1-a list of words (or terms), 2-the relationship amongst the words (or terms), indicated by their hierarchical relative position (e.g.

parent/broader term; child/narrower term, synonym, etc.), 3-a set of rules on how to use the thesaurus.

- Taxonomy: a formal list of concepts, denoted by controlled words or phrases, arranged from abstract to specific, related by subtype-supertype relations or by superset-subset relations.

- Thesaurus: a collection of categorized concepts, denoted by words or phrases, that are related to each other by narrower term, wider term and related term relations.

- Data model: an arrangement of concepts (entity types), denoted by words or phrases, that have various kinds of relationships. Typically, but not necessarily, representing requirements and capabilities for a specific scope (application area).

- Network (mathematics): an arrangement of objects in a random graph.

- Ontology: an arrangement of concepts that are related by various well-defined kinds of relations. The arrangement can be visualized in a directed acyclic graph.

- Simple Knowledge Organization System (SKOS): The Simple Knowledge Organization System is a common data model for knowledge organization systems such as thesauri, classification schemes, subject heading systems and taxonomies. Using SKOS, a knowledge organization system can be expressed as machine-readable data. It can then be exchanged between computer applications and published in a machine-readable format in the Web.